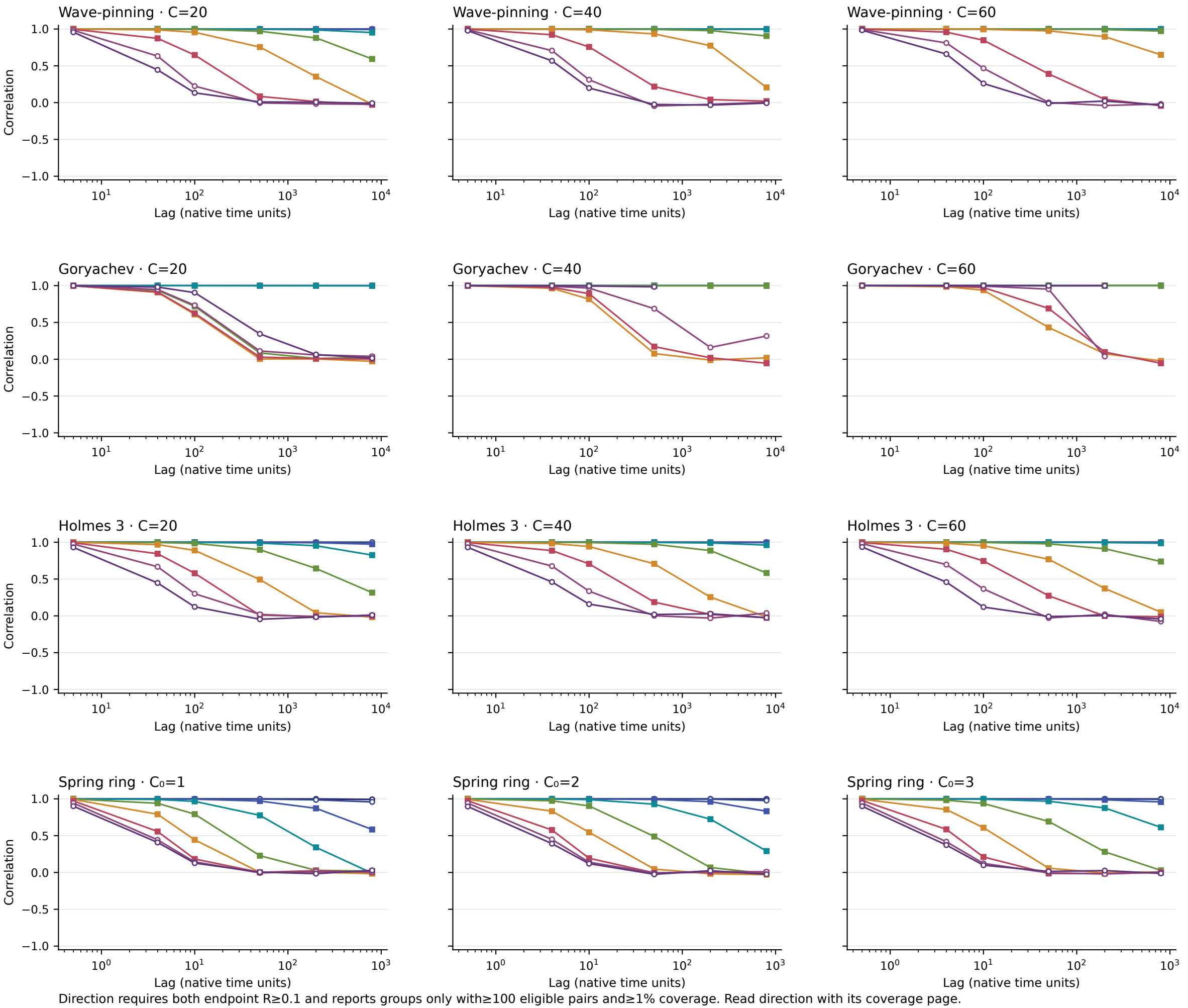
Conditional direction memory: all nine strengths

Whole observed run · chemical0–32000; Spring0–3200

f=0.0001 f=0.0003 f=0.001 f=0.003 f=0.01 f=0.03 f=0.1 f=0.3 f=1

Filled squares: n=12. Open circles: n=2 exploratory levels. The table retains exact n, pairs and contributing seeds.



Direction requires both endpoint R≥0.1 and reports groups only with≥100 eligible pairs and≥1% coverage. Read direction with its coverage page.

Uncentered vector correlation uses all raw first-moment pairs and lag-energy normalization; it also reflects magnitude/forced adaptation. It is descriptive here.

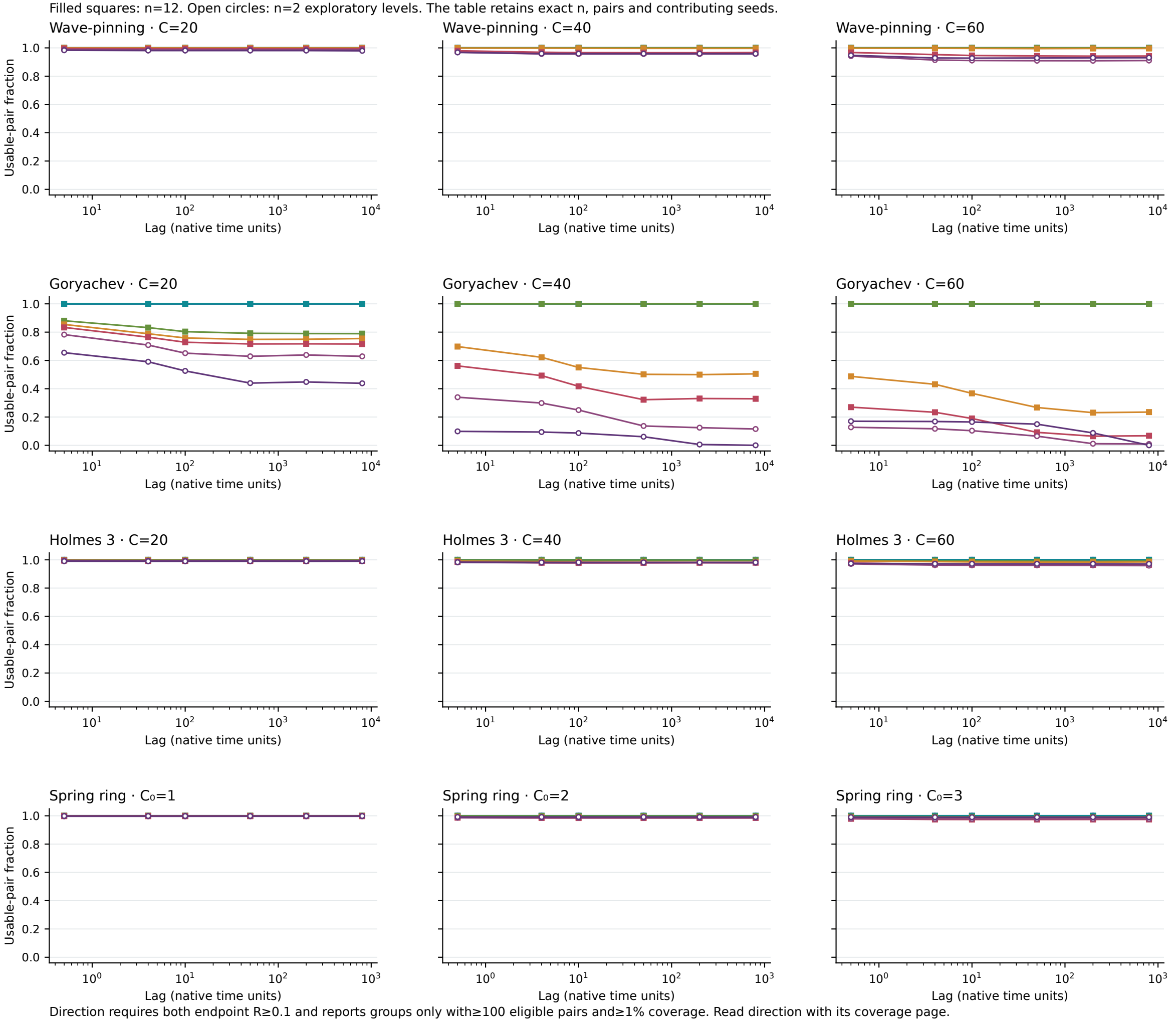
No stationarity or migration-velocity law is assumed. Chemical and Spring clocks/noise laws differ. All six lag checkpoints are retained; lines guide the eye.

Usable direction-pair coverage: all nine strengths

Whole observed run · chemical0–32000; Spring0–3200

f=0.0001 f=0.0003 f=0.001 f=0.003 f=0.01 f=0.03 f=0.1 f=0.3 f=1

Filled squares: n=12. Open circles: n=2 exploratory levels. The table retains exact n, pairs and contributing seeds.



Direction requires both endpoint R≥0.1 and reports groups only with≥100 eligible pairs and≥1% coverage. Read direction with its coverage page.

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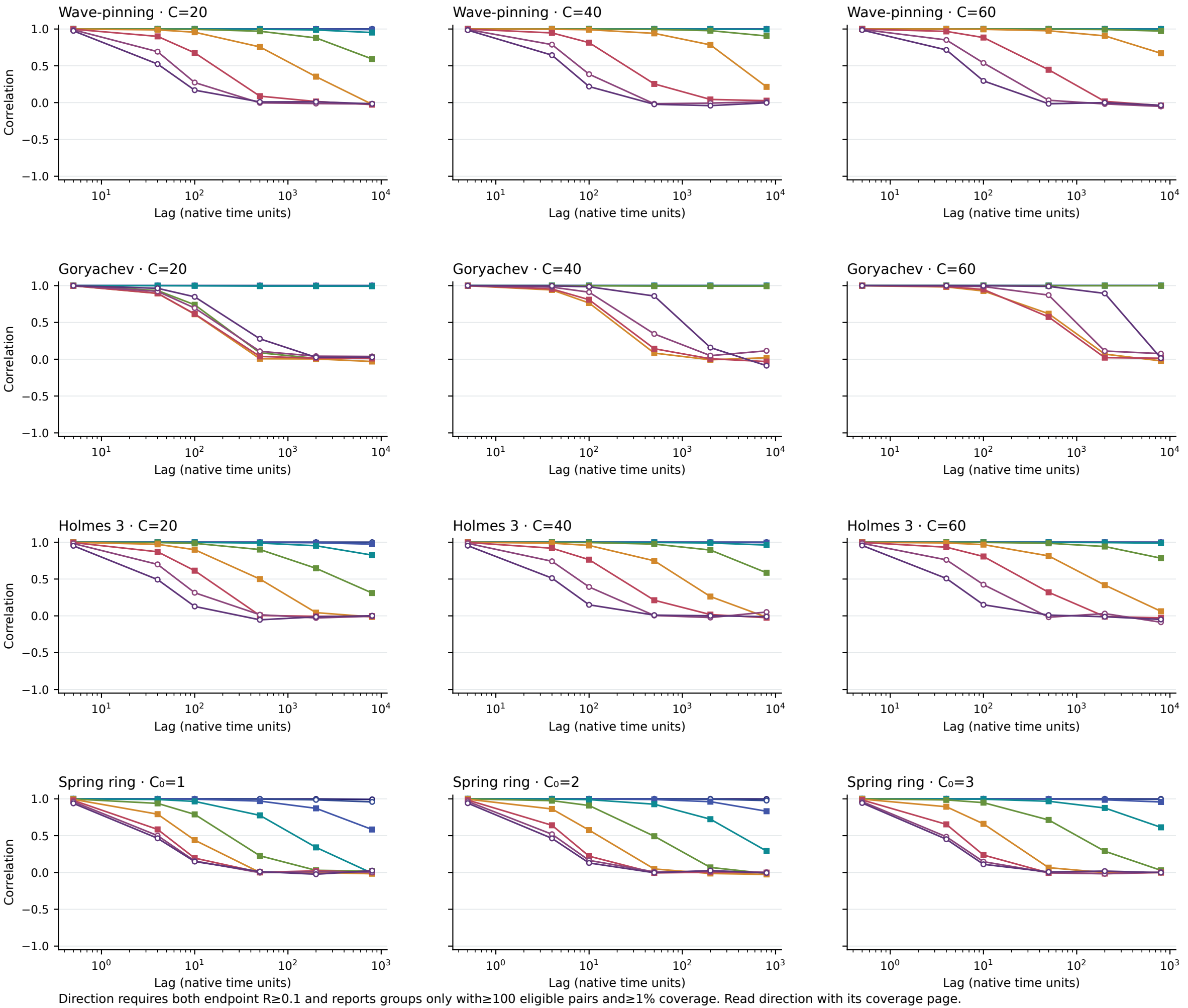
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Uncentered first-moment correlation: all nine strengths

Whole observed run · chemical0–32000; Spring0–3200

f=0.0001 f=0.0003 f=0.001 f=0.003 f=0.01 f=0.03 f=0.1 f=0.3 f=1

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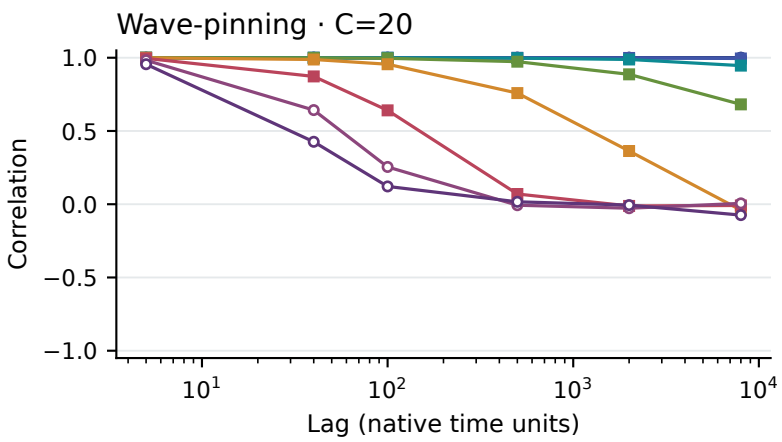
No stationarity or migration-velocity law is assumed. Chemical and Spring clocks/noise laws differ. All six lag checkpoints are retained; lines guide the eye.

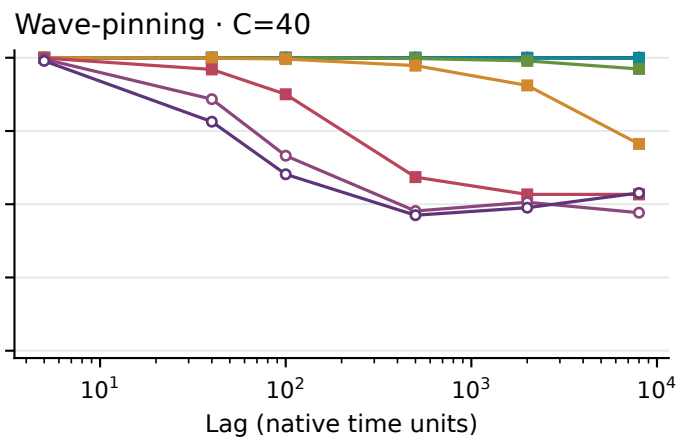
Conditional direction memory: all nine strengths

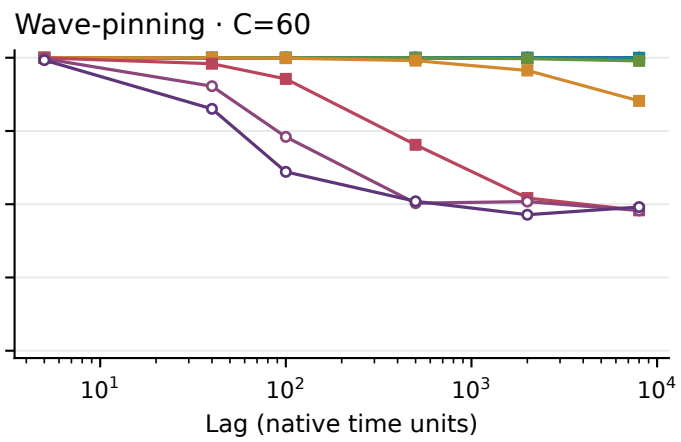
First half · chemical0–16000; Spring0–1600

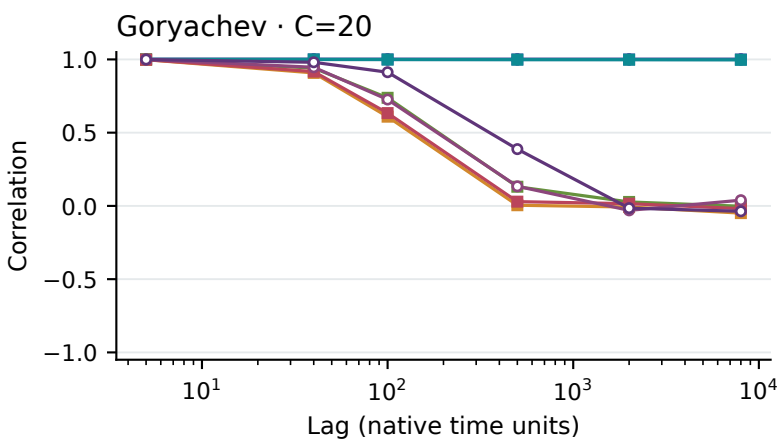
f=0.0001 f=0.0003 f=0.001 f=0.003 f=0.01 f=0.03 f=0.1 f=0.3 f=1

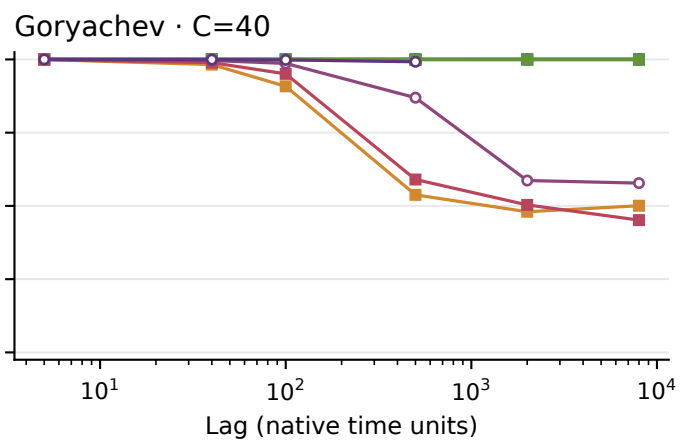
Filled squares: n=12. Open circles: n=2 exploratory levels. The table retains exact n, pairs and contributing seeds.

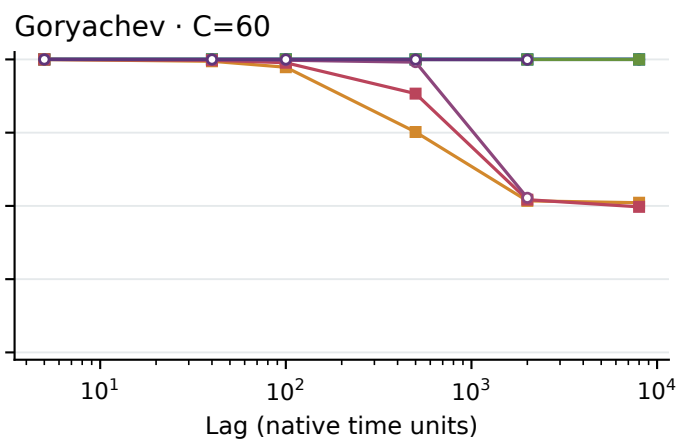


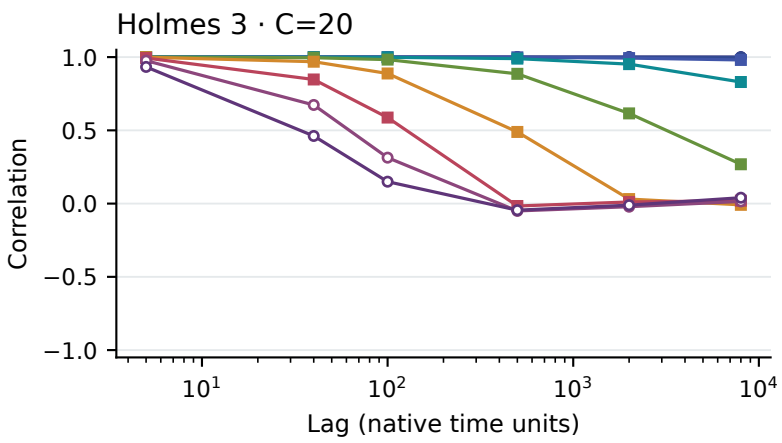


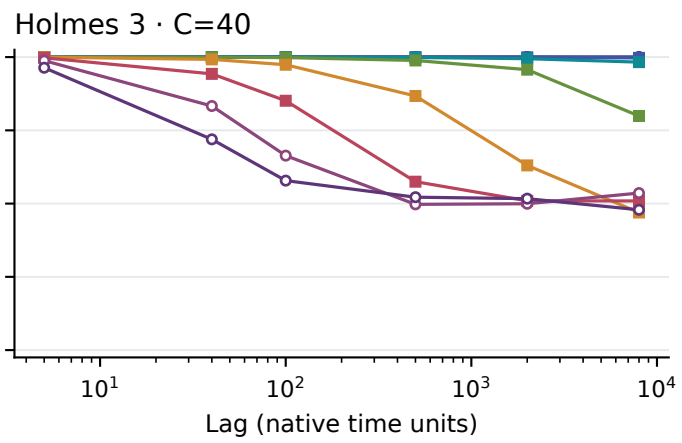


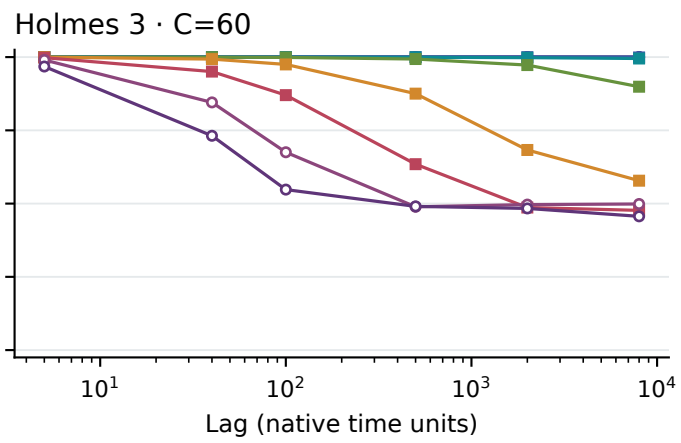


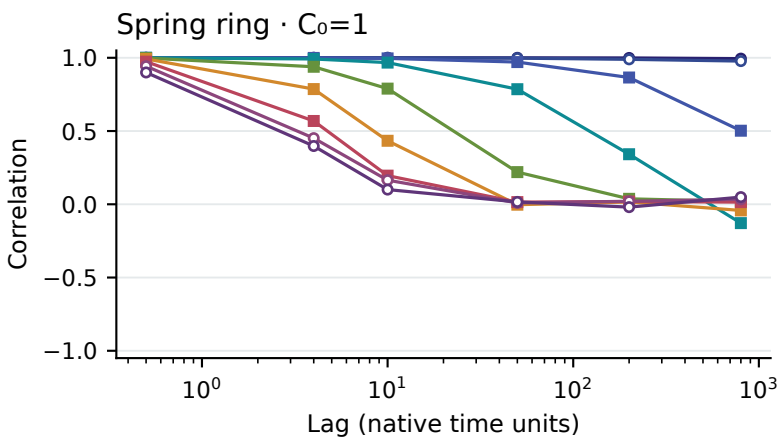


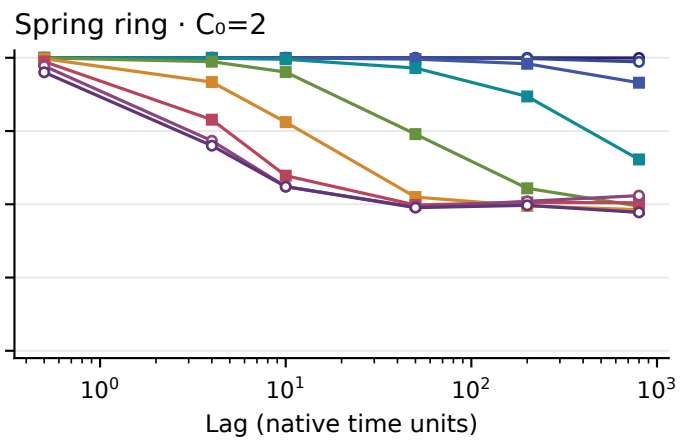


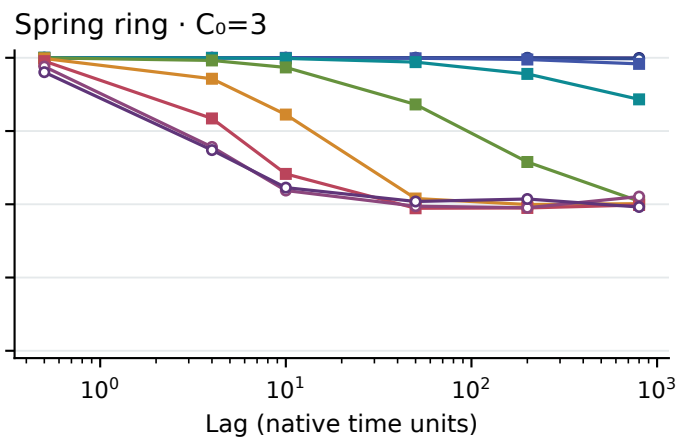












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Uncentered vector correlation uses all raw first-moment pairs and lag-energy normalization; it also reflects magnitude/forced adaptation. It is descriptive here.

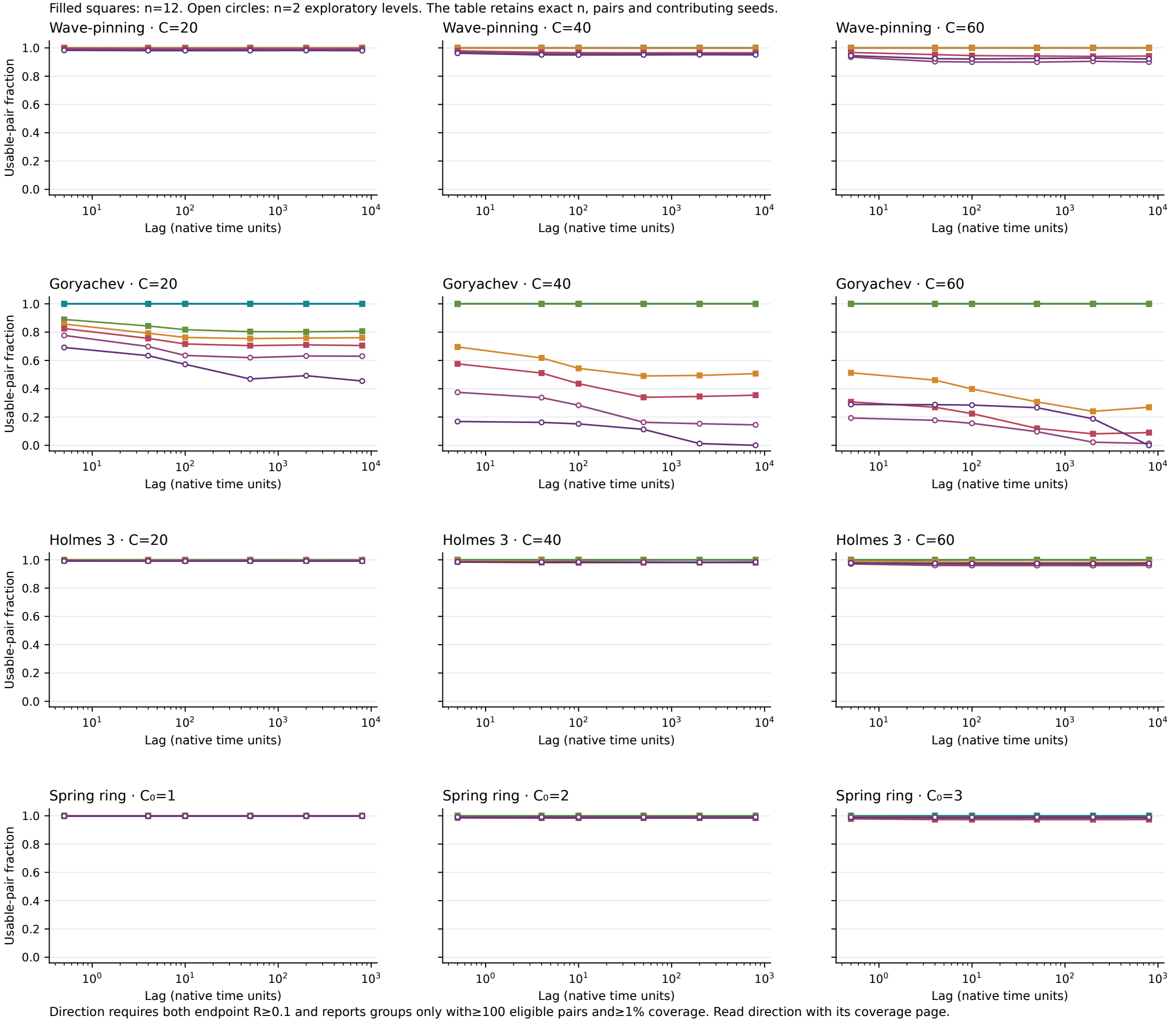
No stationarity or migration-velocity law is assumed. Chemical and Spring clocks/noise laws differ. All six lag checkpoints are retained; lines guide the eye.

Usable direction-pair coverage: all nine strengths

First half · chemical0–16000; Spring0–1600

f=0.0001 f=0.0003 f=0.001 f=0.003 f=0.01 f=0.03 f=0.1 f=0.3 f=1

Filled squares: n=12. Open circles: n=2 exploratory levels. The table retains exact n, pairs and contributing seeds.



Direction requires both endpoint R≥0.1 and reports groups only with≥100 eligible pairs and≥1% coverage. Read direction with its coverage page.

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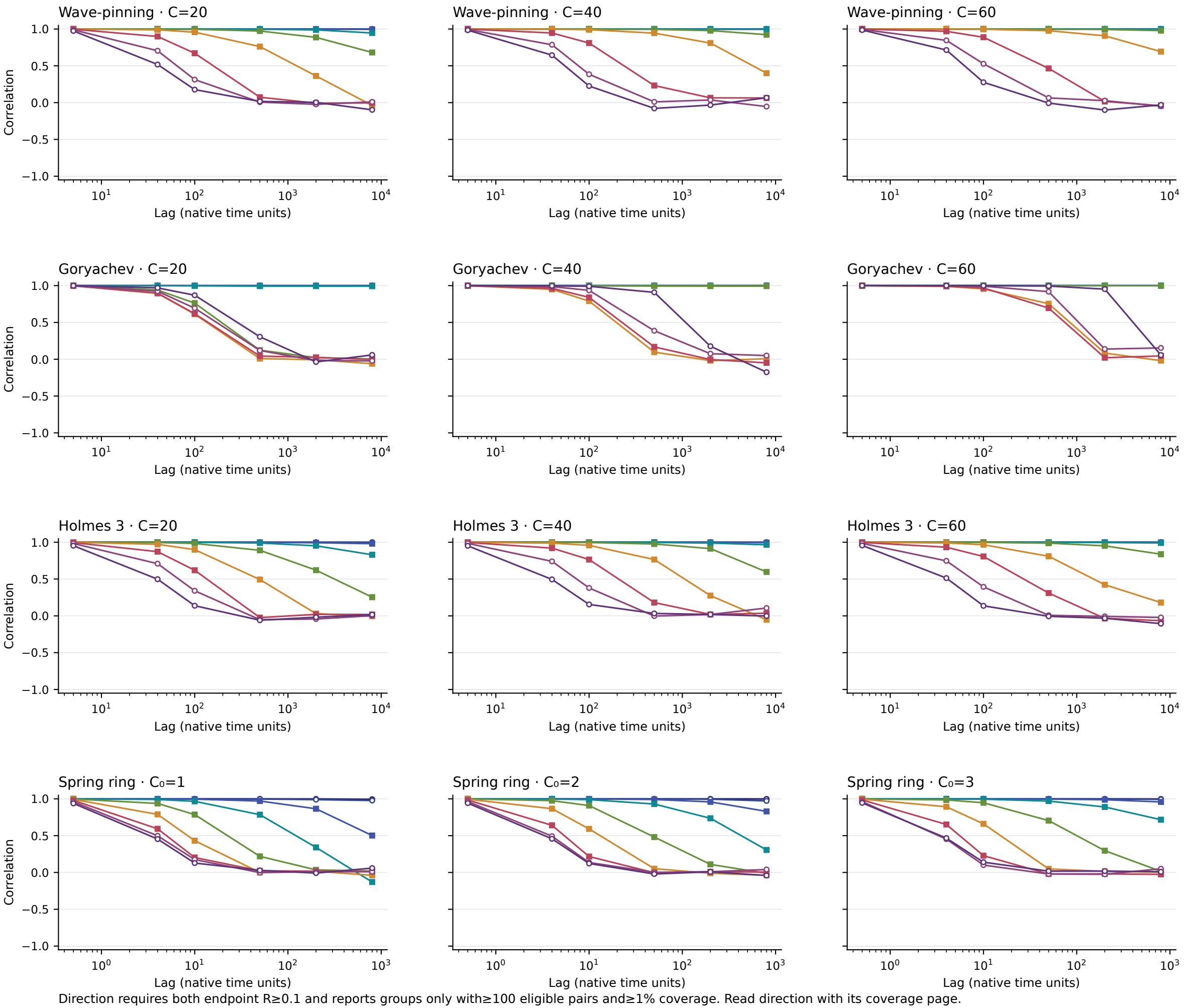
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Uncentered first-moment correlation: all nine strengths

First half · chemical0–16000; Spring0–1600

f=0.0001 f=0.0003 f=0.001 f=0.003 f=0.01 f=0.03 f=0.1 f=0.3 f=1

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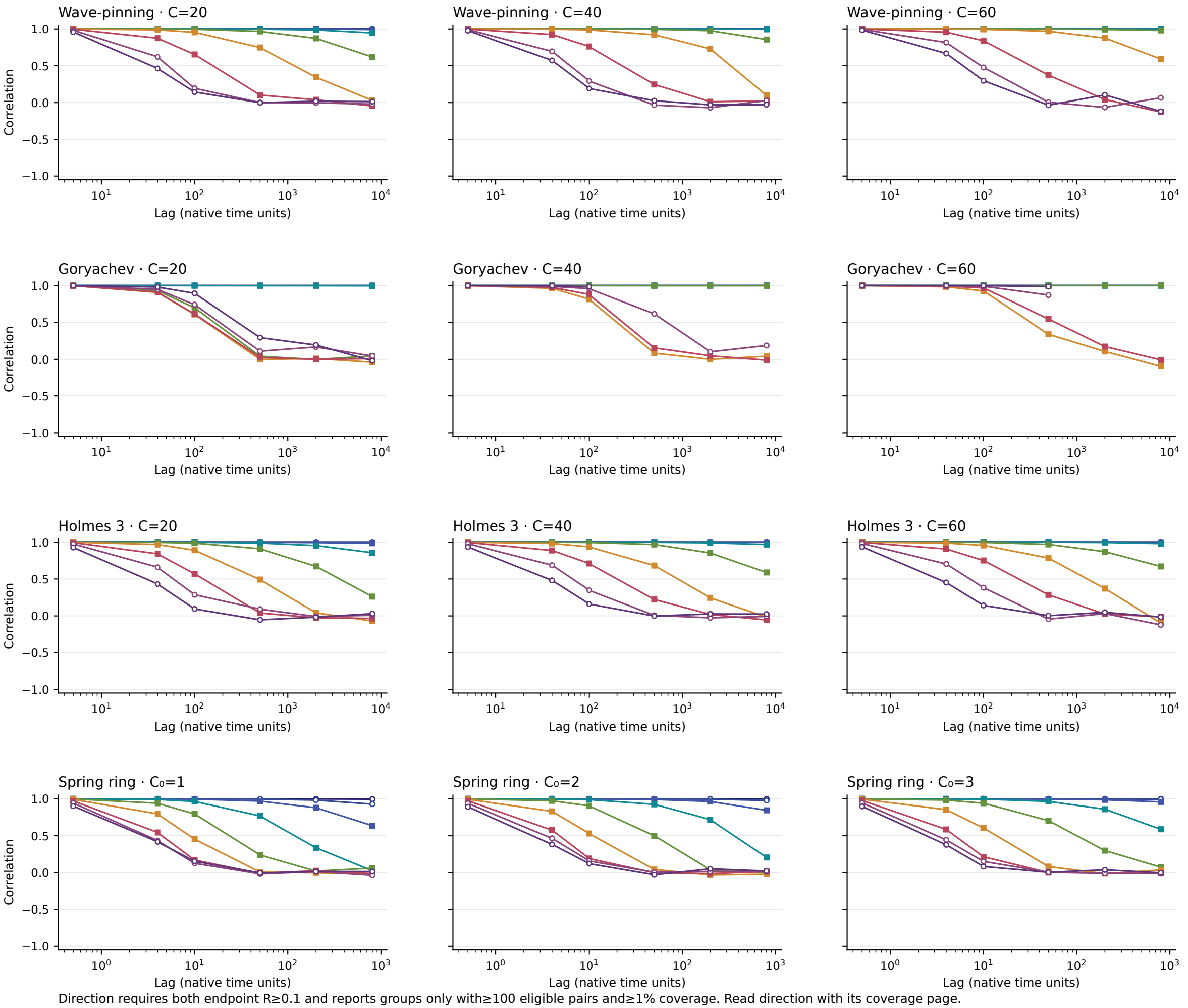
No stationarity or migration-velocity law is assumed. Chemical and Spring clocks/noise laws differ. All six lag checkpoints are retained; lines guide the eye.

Conditional direction memory: all nine strengths

Second half after discarding the first half · chemical16000–32000; Spring1600–3200

f=0.0001 f=0.0003 f=0.001 f=0.003 f=0.01 f=0.03 f=0.1 f=0.3 f=1

Filled squares: n=12. Open circles: n=2 exploratory levels. The table retains exact n, pairs and contributing seeds.



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Uncentered vector correlation uses all raw first-moment pairs and lag-energy normalization; it also reflects magnitude/forced adaptation. It is descriptive here.

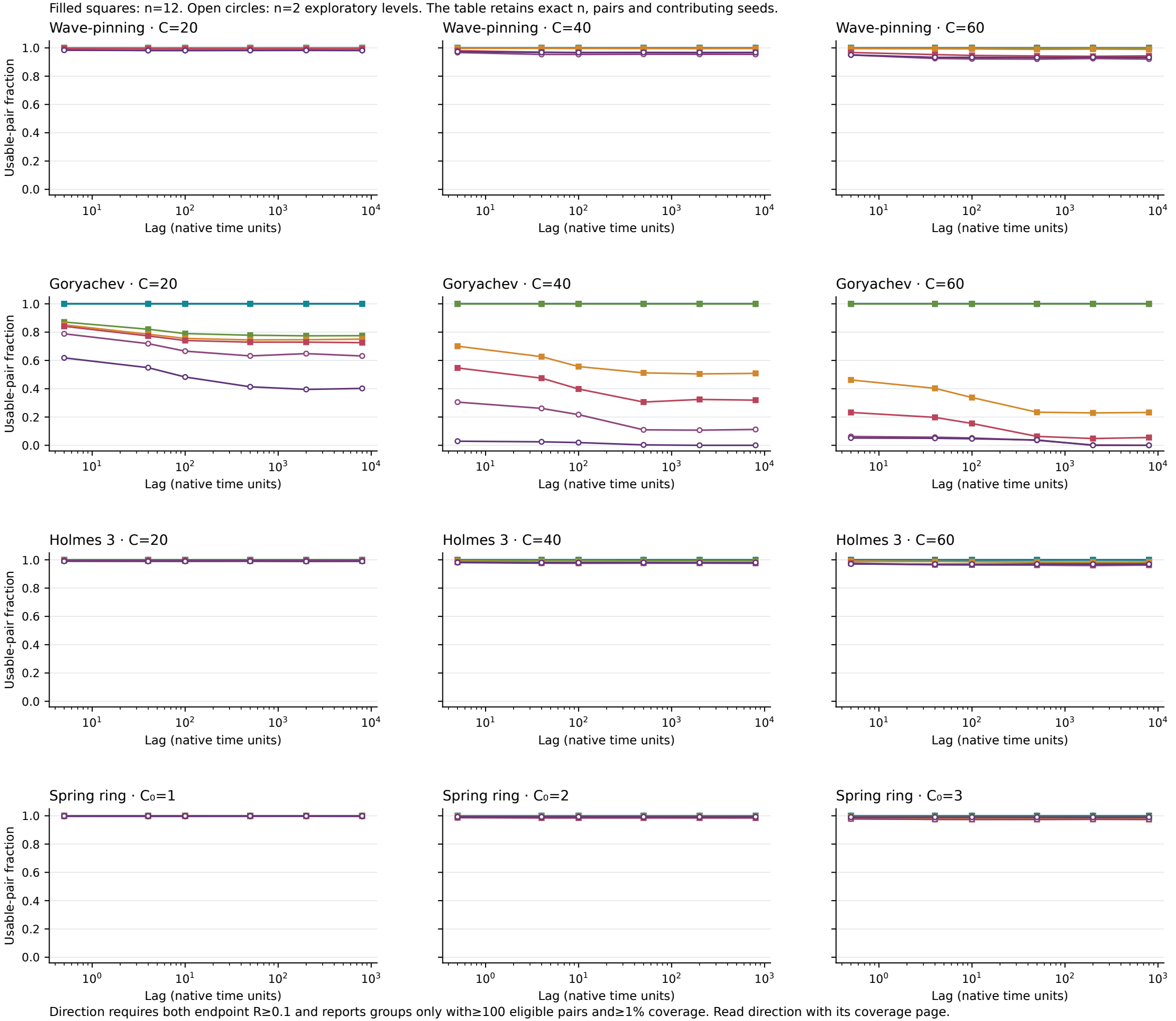
No stationarity or migration-velocity law is assumed. Chemical and Spring clocks/noise laws differ. All six lag checkpoints are retained; lines guide the eye.

Usable direction-pair coverage: all nine strengths

Second half after discarding the first half · chemical16000–32000; Spring1600–3200

f=0.0001 f=0.0003 f=0.001 f=0.003 f=0.01 f=0.03 f=0.1 f=0.3 f=1

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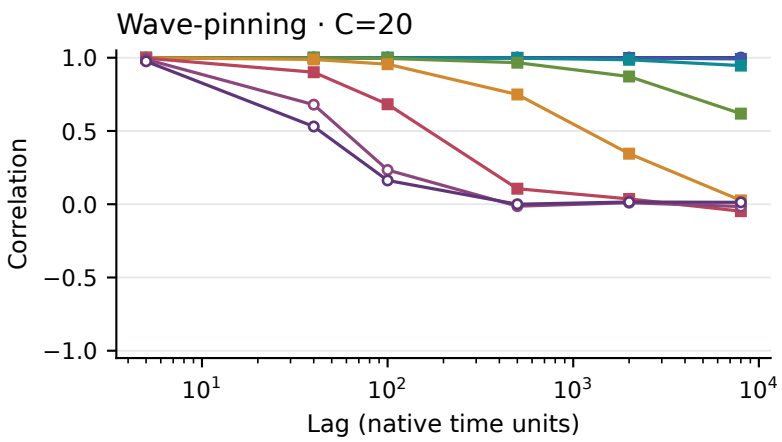
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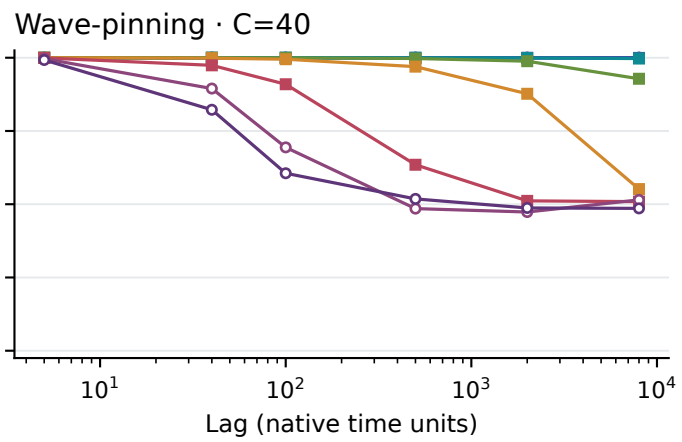
Uncentered first-moment correlation: all nine strengths

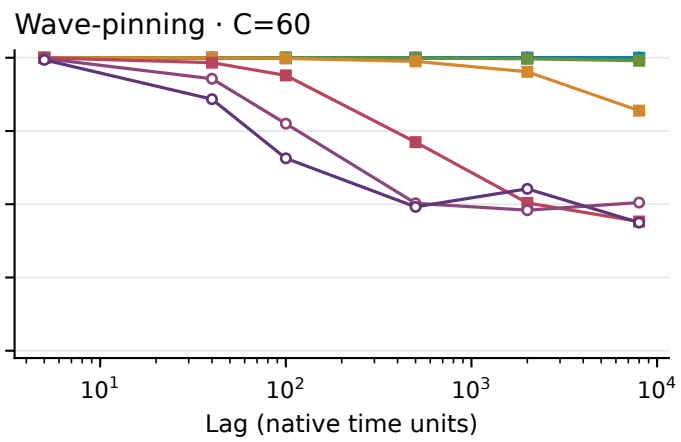
Second half after discarding the first half · chemical16000–32000; Spring1600–3200

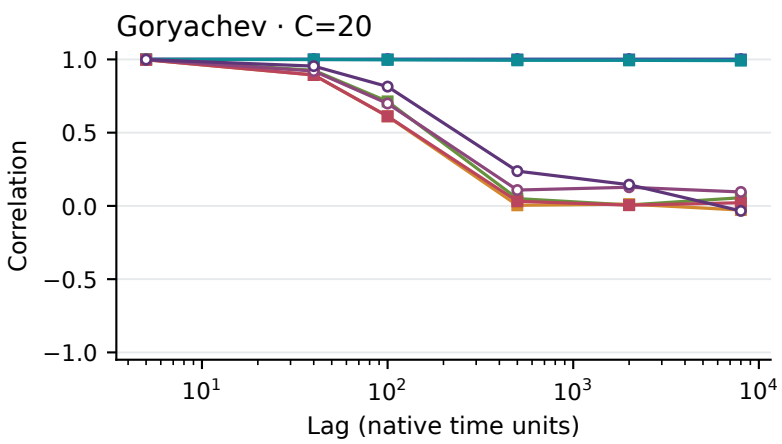
f=0.0001 f=0.0003 f=0.001 f=0.003 f=0.01 f=0.03 f=0.1 f=0.3 f=1

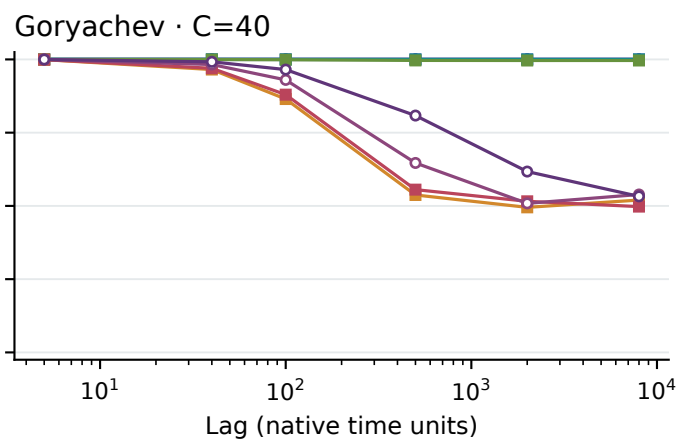
Filled squares: n=12. Open circles: n=2 exploratory levels. The table retains exact n, pairs and contributing seeds.

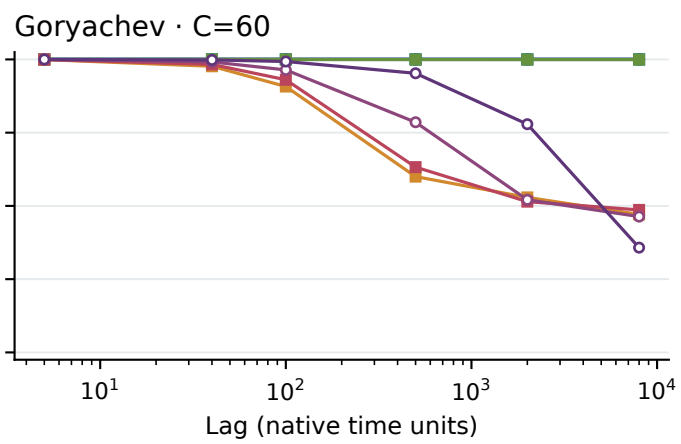


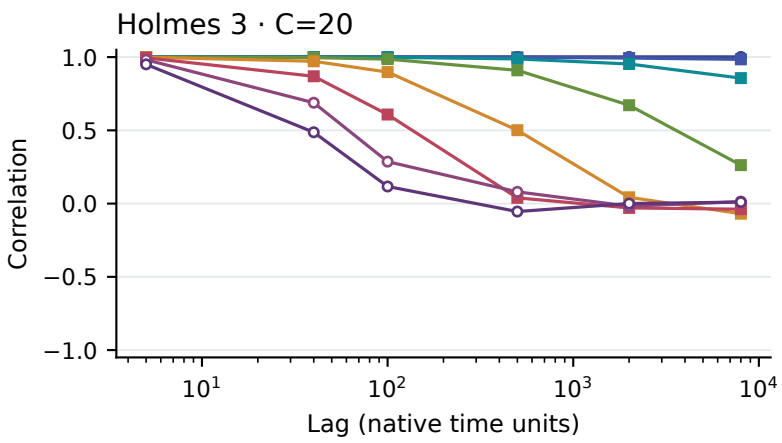


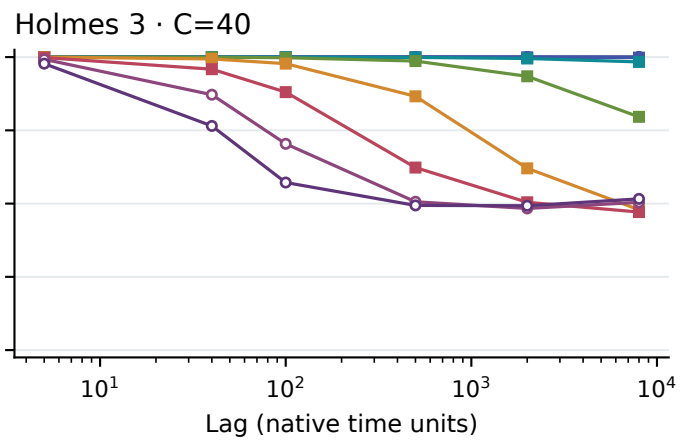


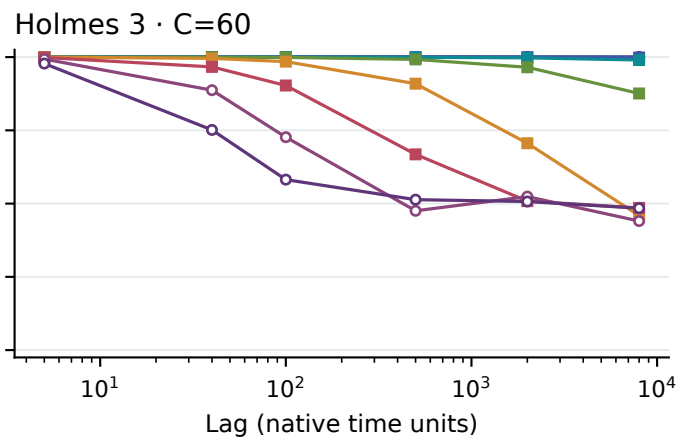


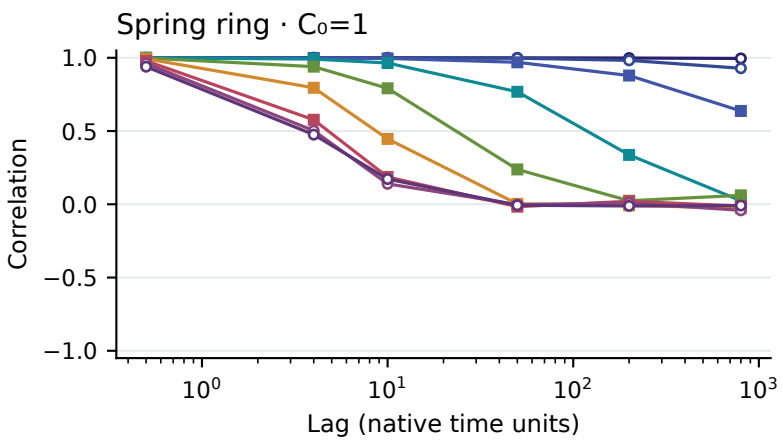


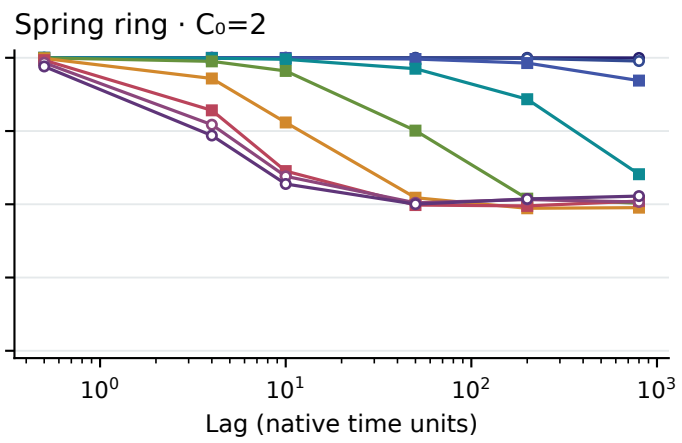


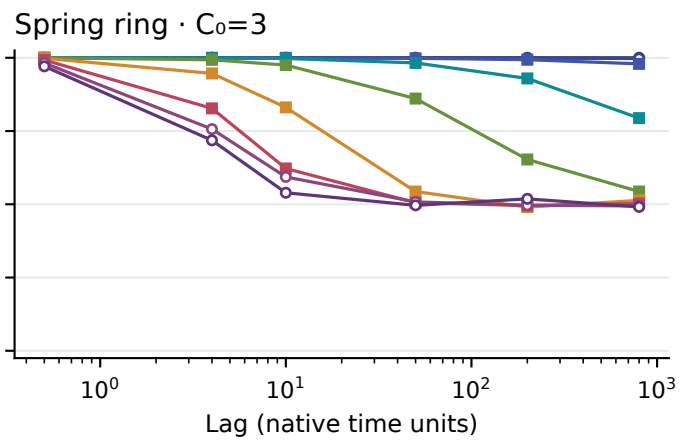












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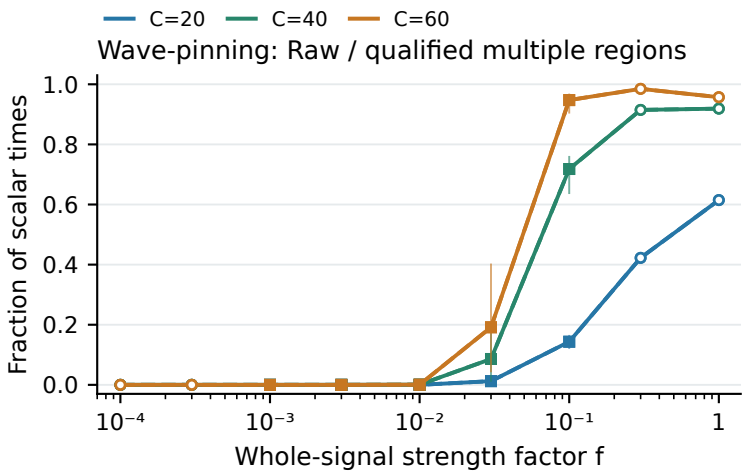
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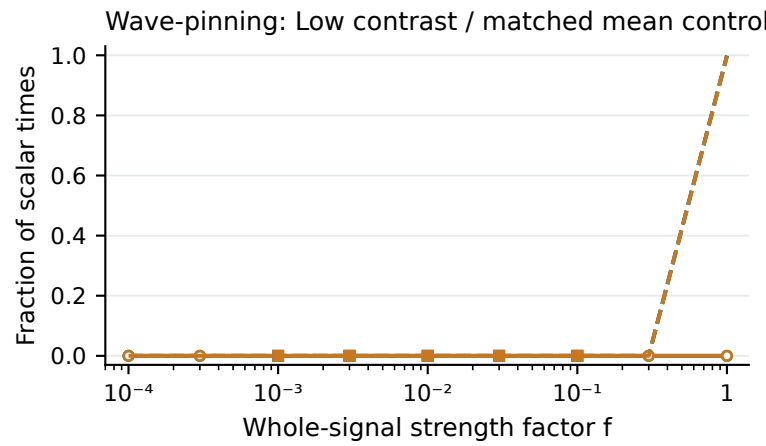
Organization, contrast and global magnitude: all sizes and strengths

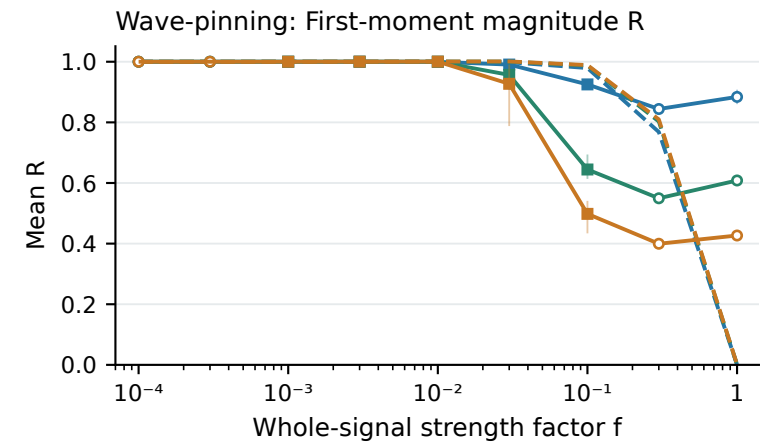
Whole observed run · chemicals0–32000; Spring0–3200. Half-window midpoint shared.

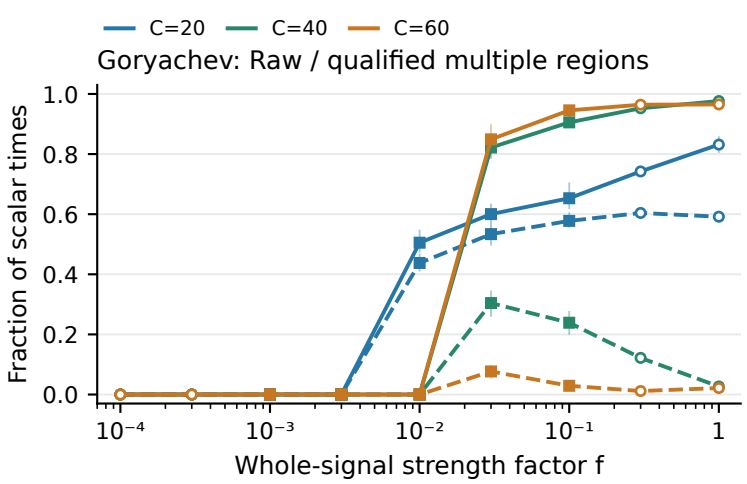
Size colors: blue smallest, teal middle, orange largest. Squares n=12; circles n=2. Whiskers: observed seed range, not confidence intervals.

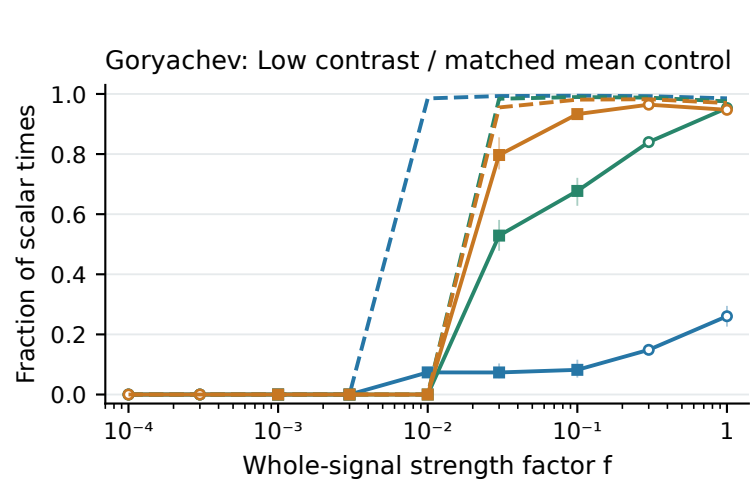
solid: raw multi / noisy low contrast / noisy R dashed: contrast-qualified multi / matched mean-control low contrast / mean-control R

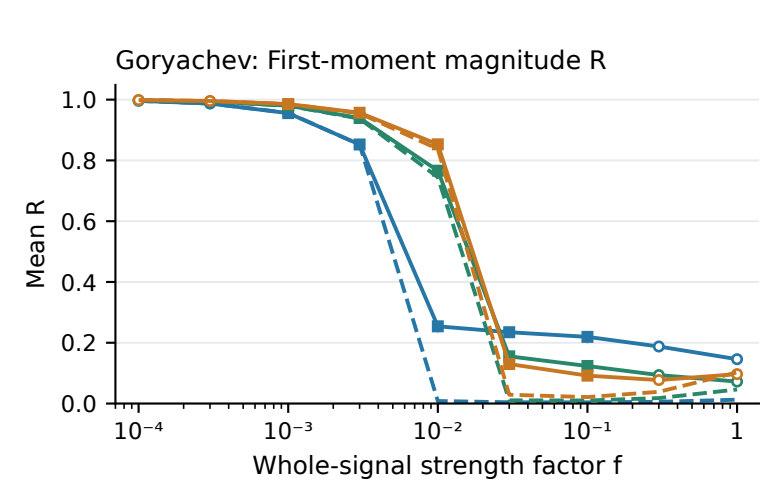


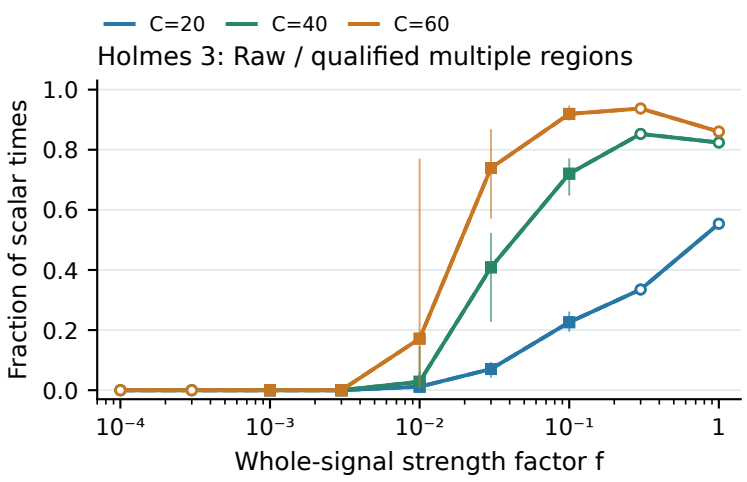


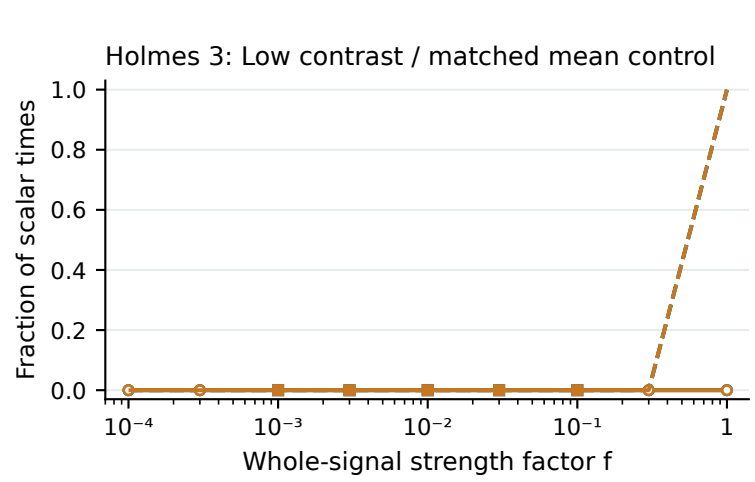


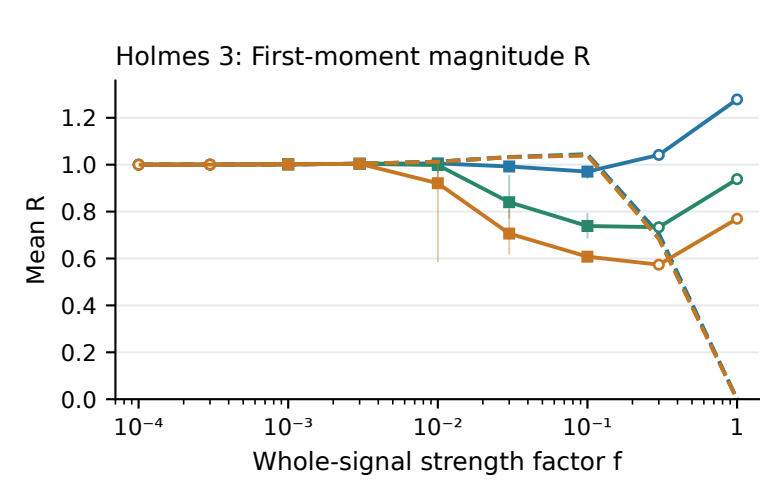


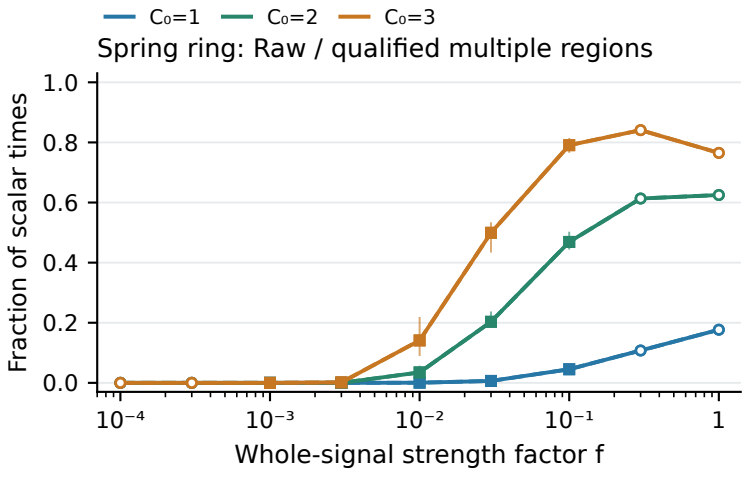


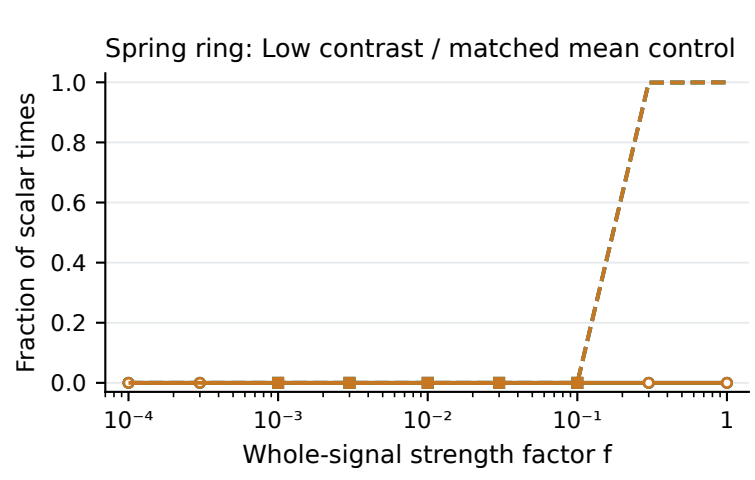


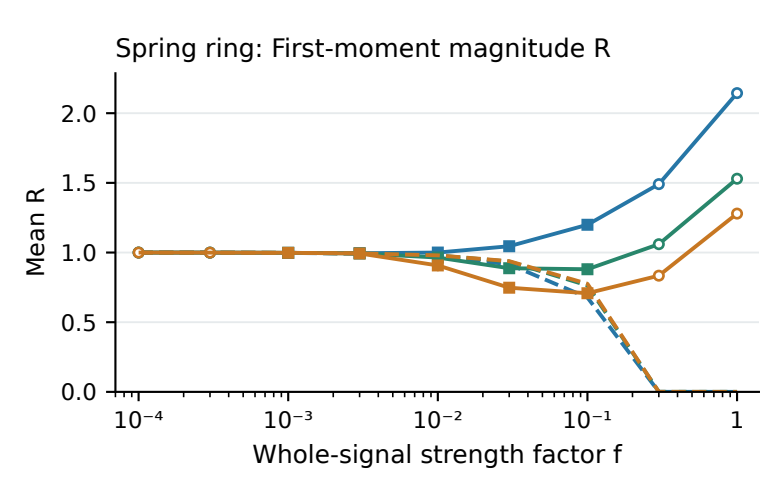












All scalar times remain in occupancy denominators; cadence5 chemicals/0.5 Spring. Threshold regions may coexist with weak or strong resultants; these are not exclusive phases.

R=AQ exactly per frame; mean R is mean of the paired product, not product of means. $q_1=|M_1|/M_0$ and $q_2=|M_2|/M_0$ retain basal coherence differences.

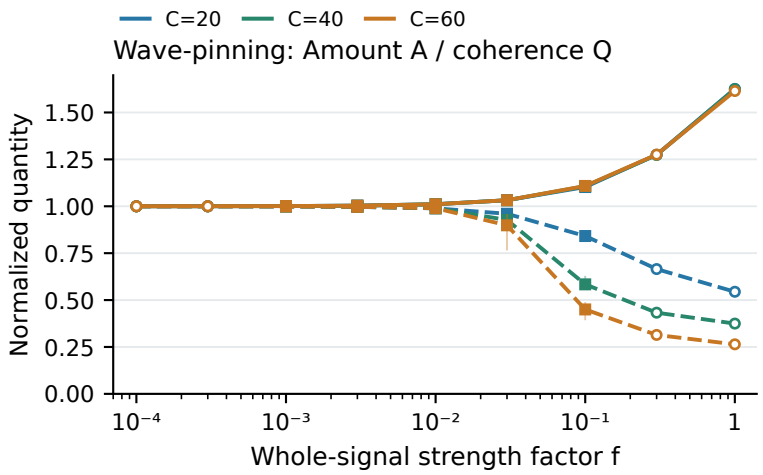
Mean controls match declared ensemble input means, without per-path normalization. Native clocks/laws/units differ; full/half differences do not certify stationarity.

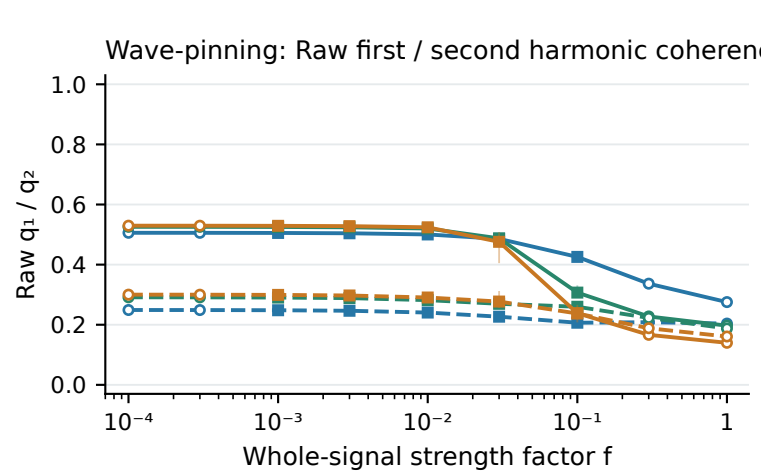
Active amount, coherence and spatial harmonics: all sizes and strengths

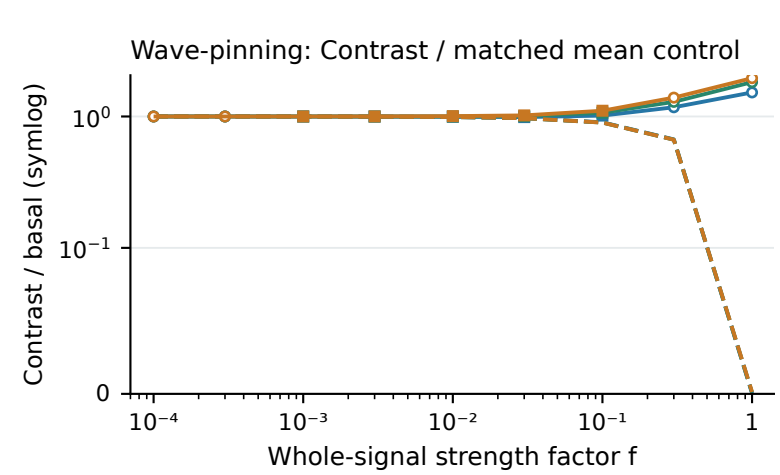
Whole observed run · chemicals0–32000; Spring0–3200. Half-window midpoint shared.

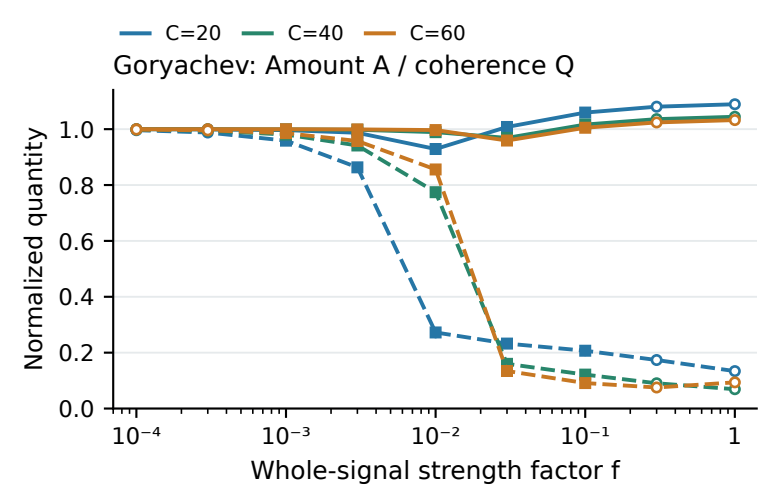
Size colors: blue smallest, teal middle, orange largest. Squares n=12; circles n=2. Whiskers: observed seed range, not confidence intervals.

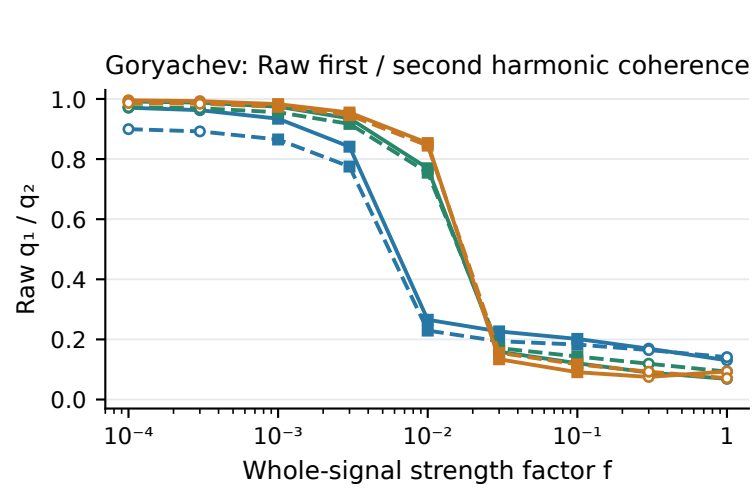
solid: active amount A / raw q_1 / noisy contrast dashed: coherence Q / q_2 / matched mean-control contrast

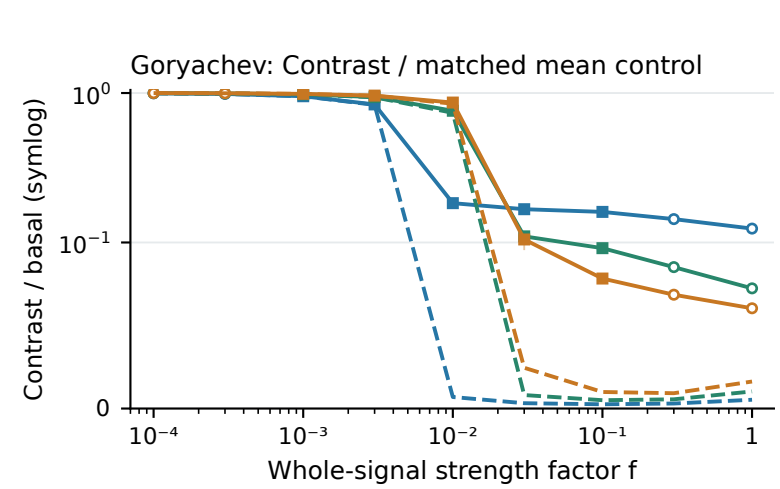


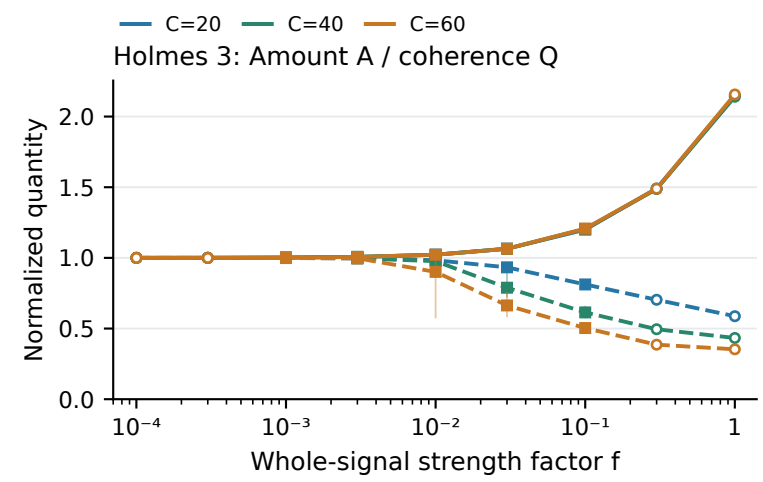


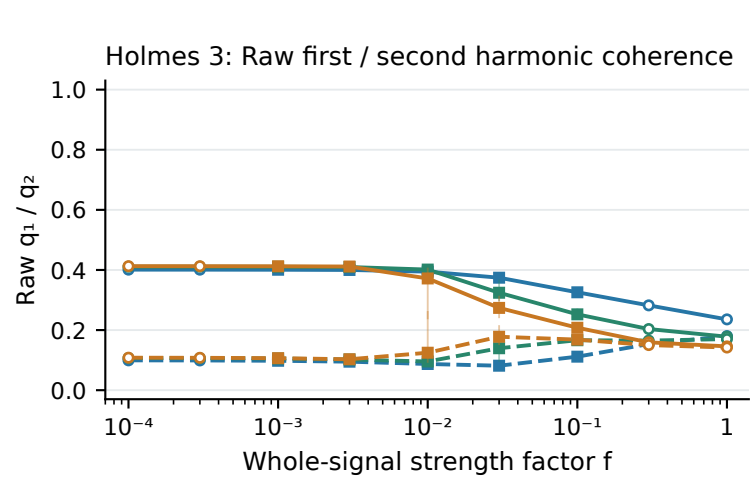


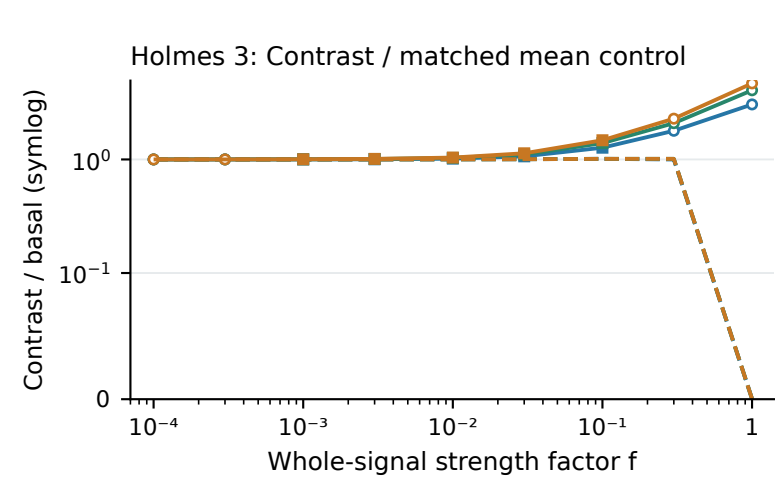


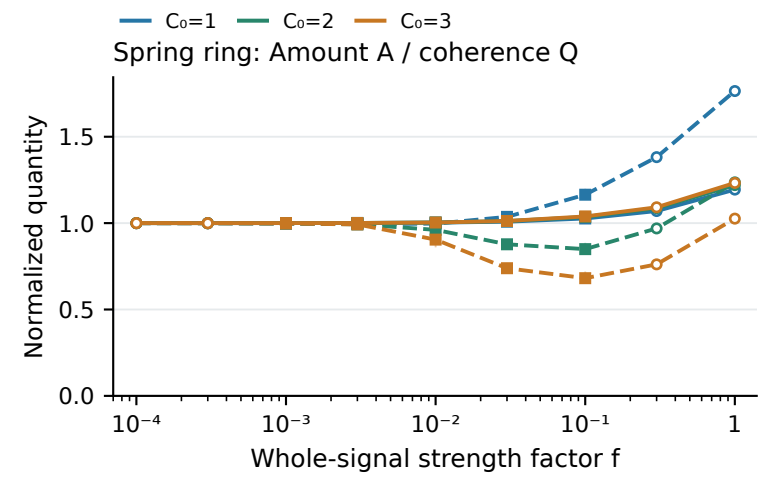


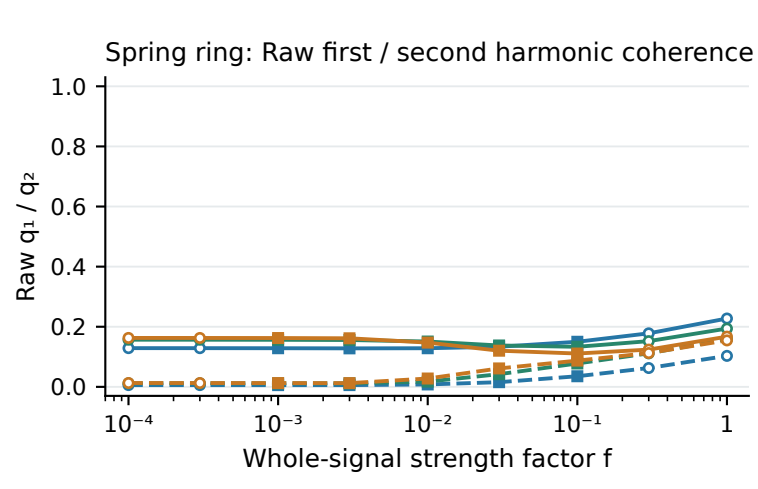


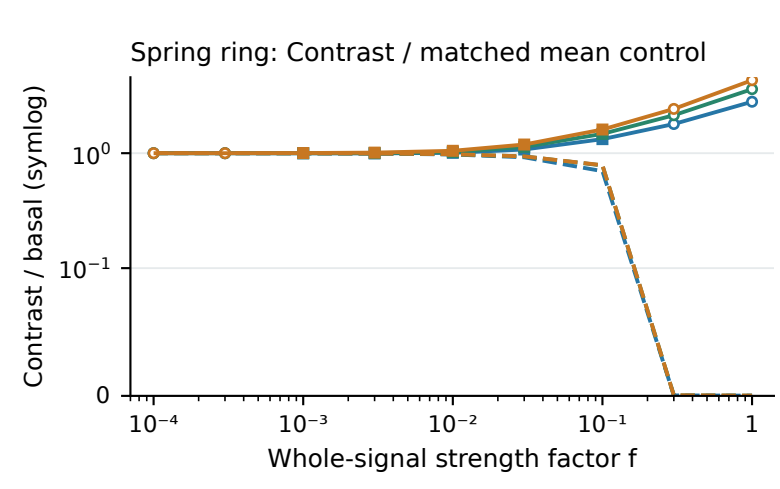












All scalar times remain in occupancy denominators; cadence5 chemicals/0.5 Spring. Threshold regions may coexist with weak or strong resultants; these are not exclusive phases.

R=AQ exactly per frame; mean R is mean of the paired product, not product of means. $q_1=|M_1|/M_0$ and $q_2=|M_2|/M_0$ retain basal coherence differences.

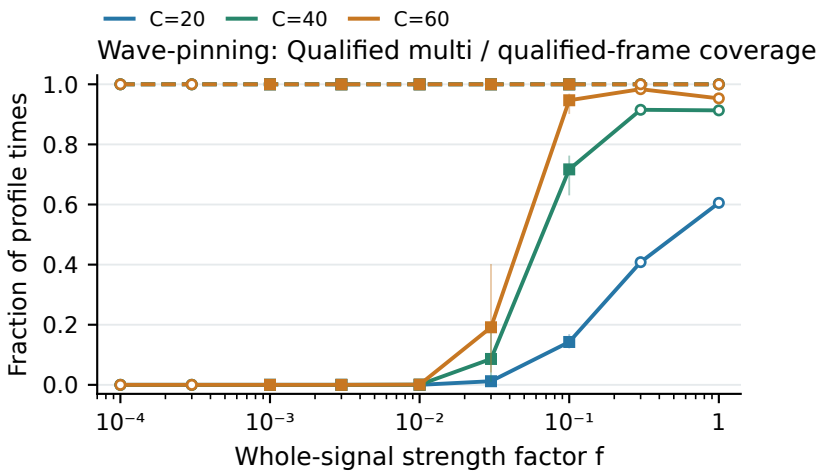
Mean controls match declared ensemble input means, without per-path normalization. Native clocks/laws/units differ; full/half differences do not certify stationarity.

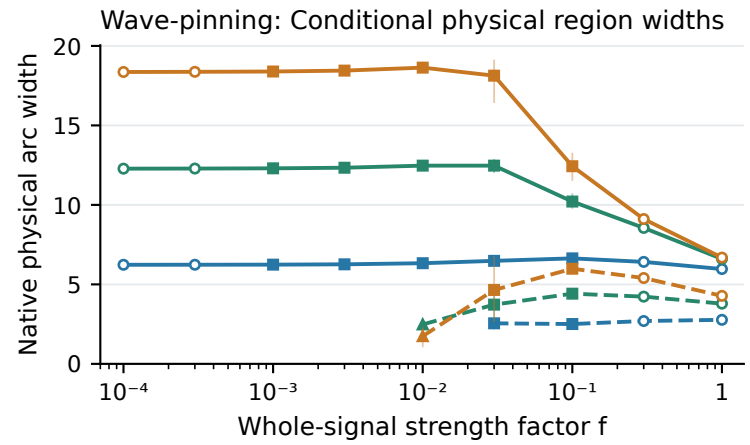
Instantaneous region widths with qualification coverage: all sizes and strengths

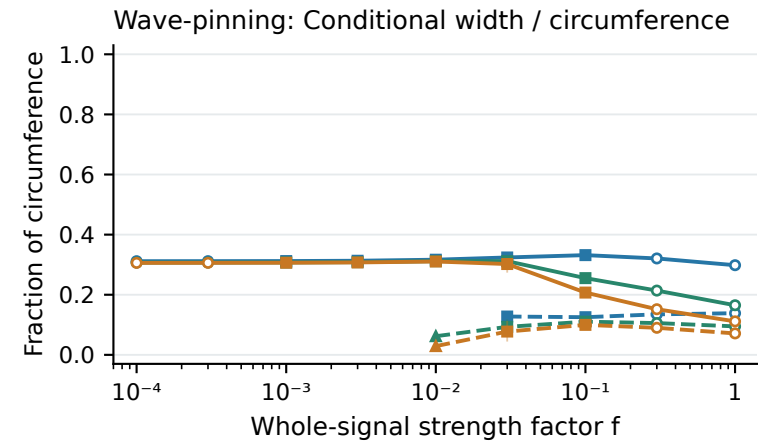
Whole observed run · chemicals0–32000; Spring0–3200. Half-window midpoint shared.

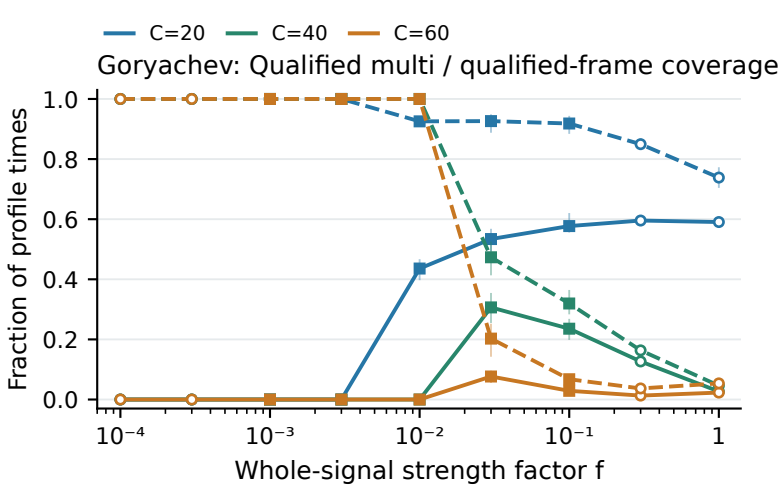
Size colors: blue smallest, teal middle, orange largest. Squares n=12; circles n=2. Whiskers: observed seed range, not confidence intervals.

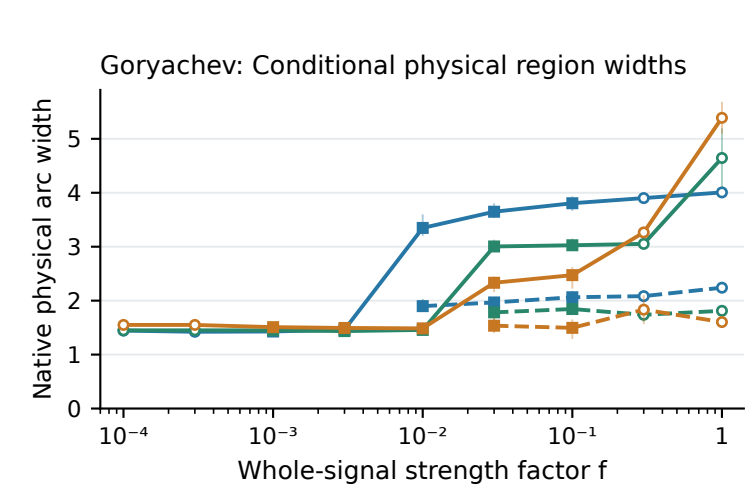
solid: qualified multi occupancy / largest width dashed: qualified-frame coverage / second-largest width

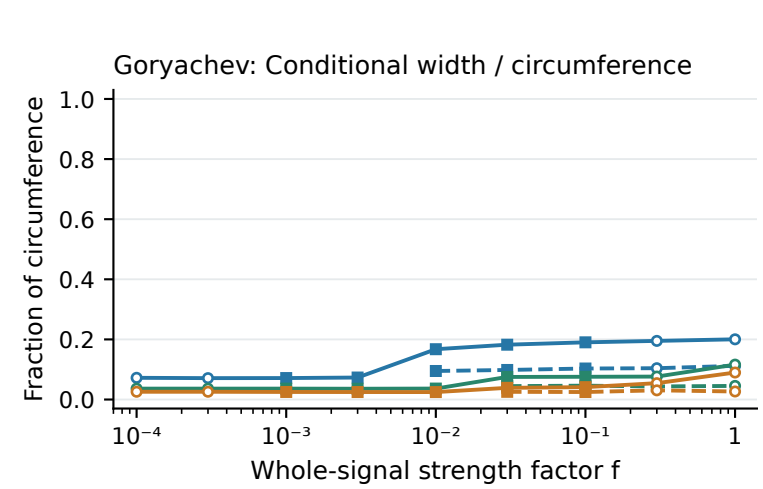


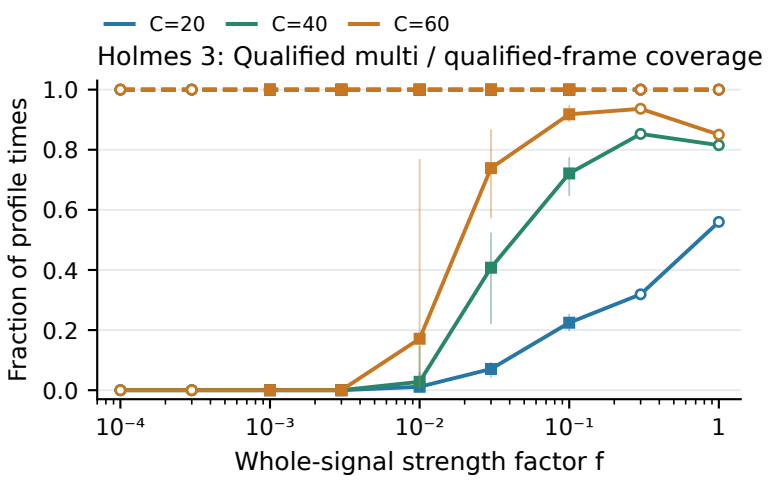


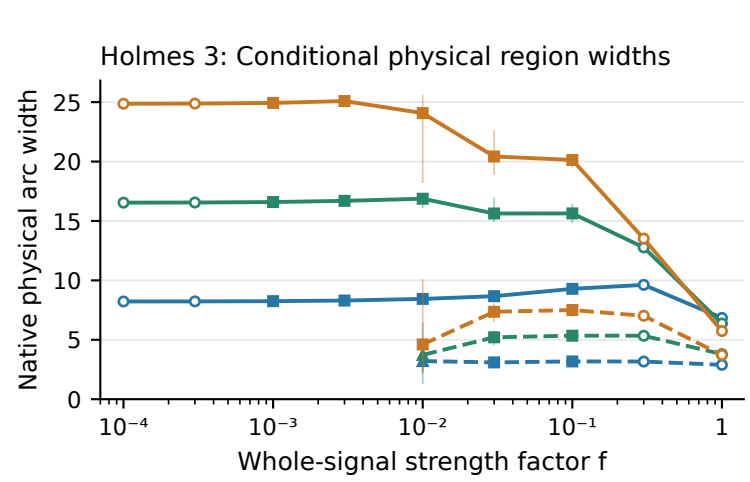


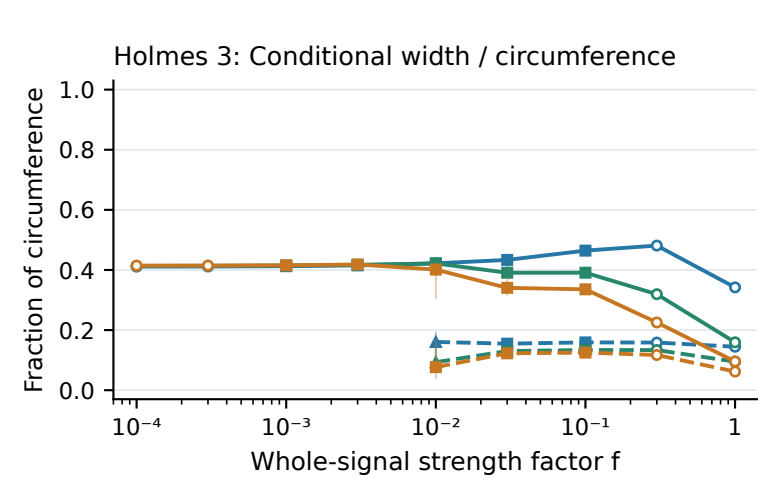


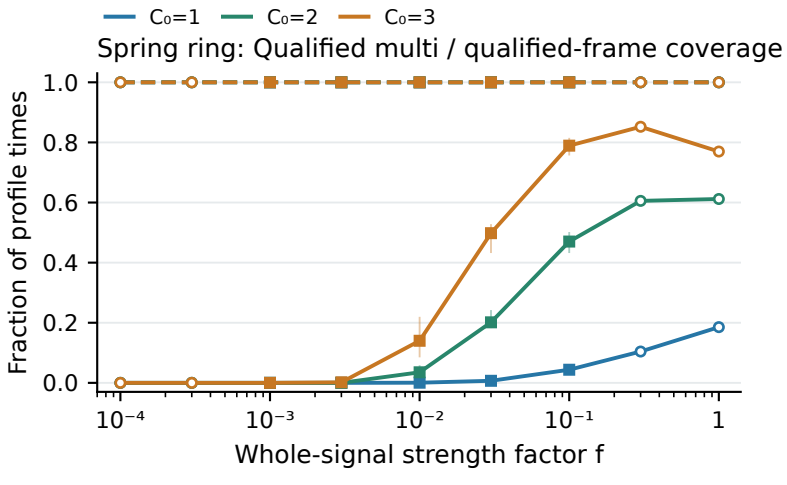


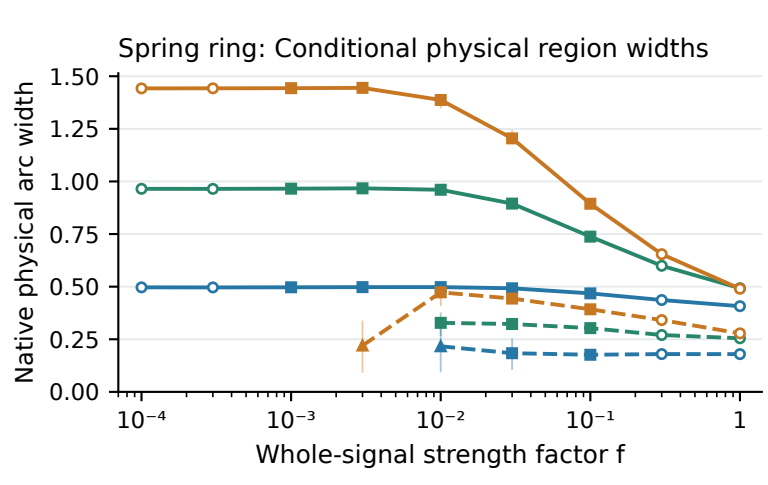


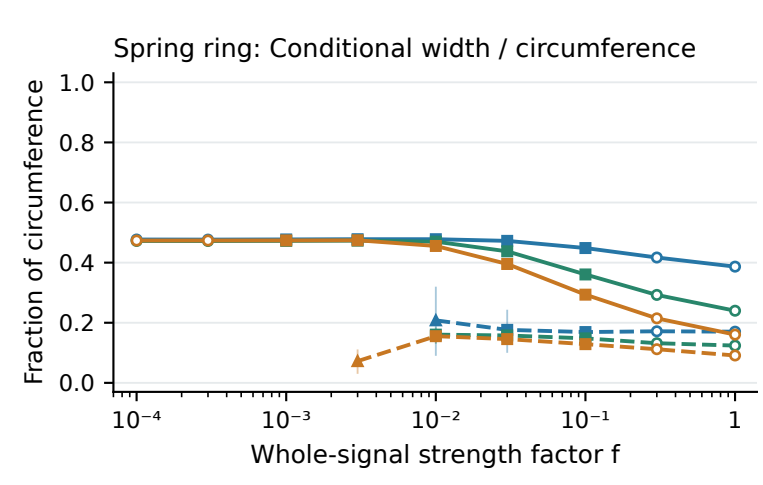












Widths join periodic seam-crossing regions: cell count/N × actual C(t). Threshold: current min+half contrast; qualification: contrast≥10% basal. Profile cadence40/4.

Largest: one per qualified frame; second: one per qualified multiregion frame. Equal available-seed means; △ fewer contributing seeds than condition n (exact counts exported).

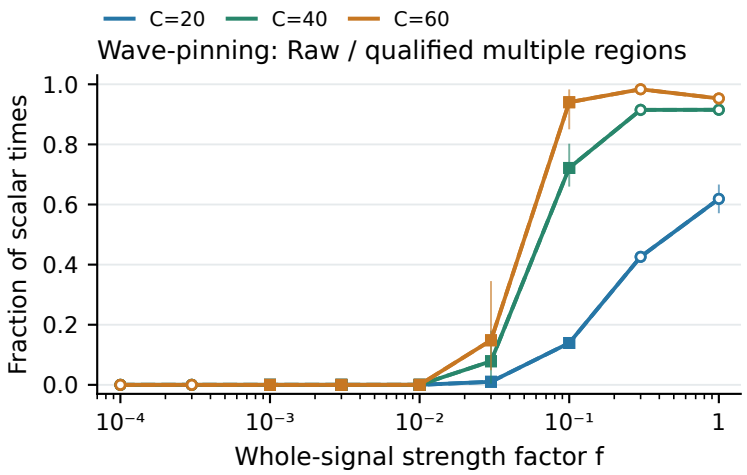
Missing conditional widths stay unavailable. Widths are descriptive geometry, outside the ten-metric numerical gate; no region births, establishment, lifetimes or travel inferred.

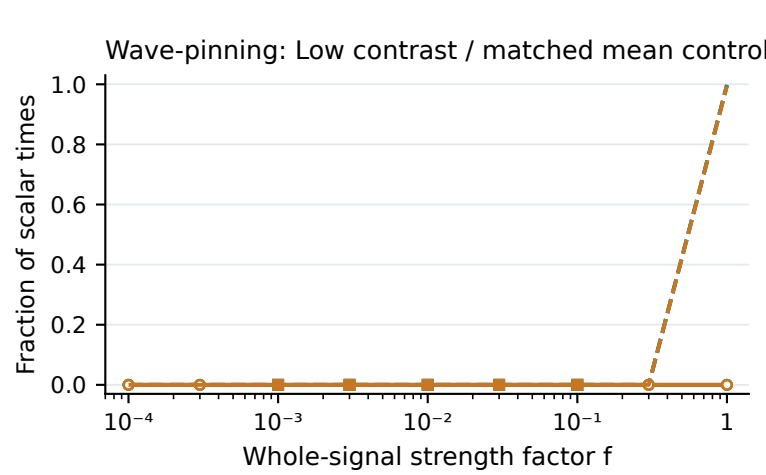
Organization, contrast and global magnitude: all sizes and strengths

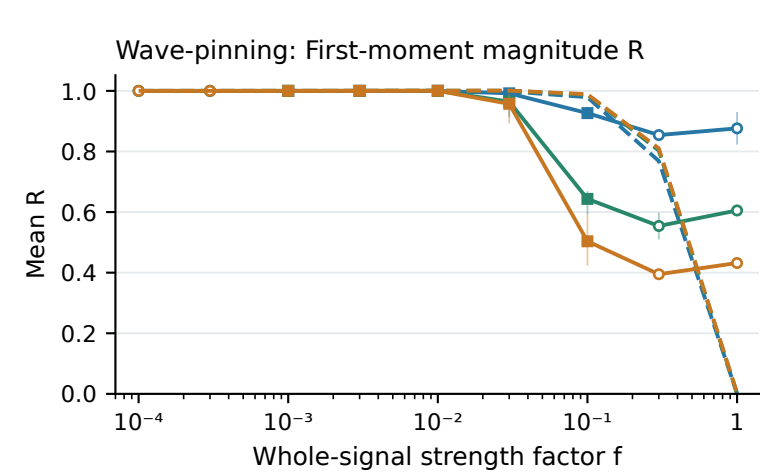
First half · chemicals0–16000; Spring0–1600. Half-window midpoint shared.

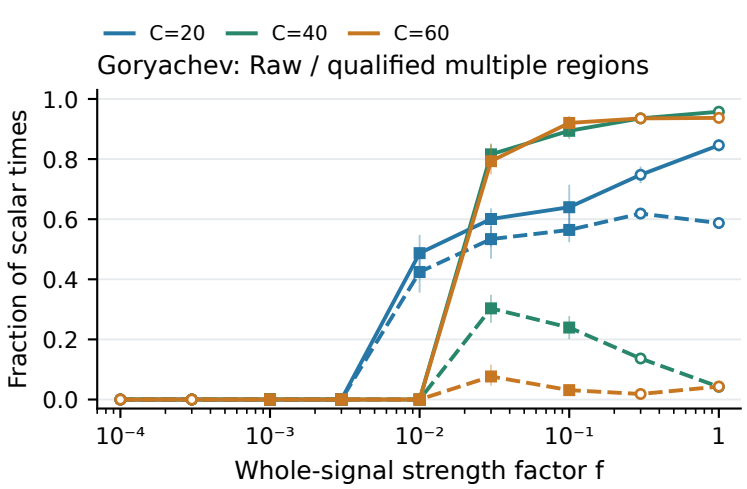
Size colors: blue smallest, teal middle, orange largest. Squares n=12; circles n=2. Whiskers: observed seed range, not confidence intervals.

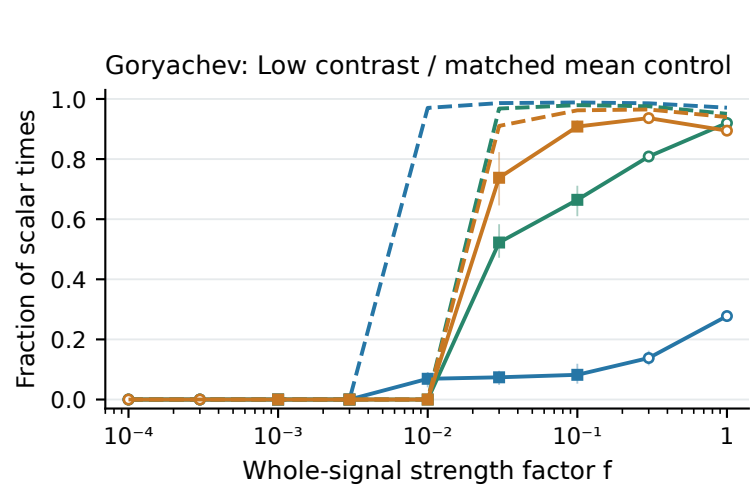
solid: raw multi / noisy low contrast / noisy R dashed: contrast-qualified multi / matched mean-control low contrast / mean-control R

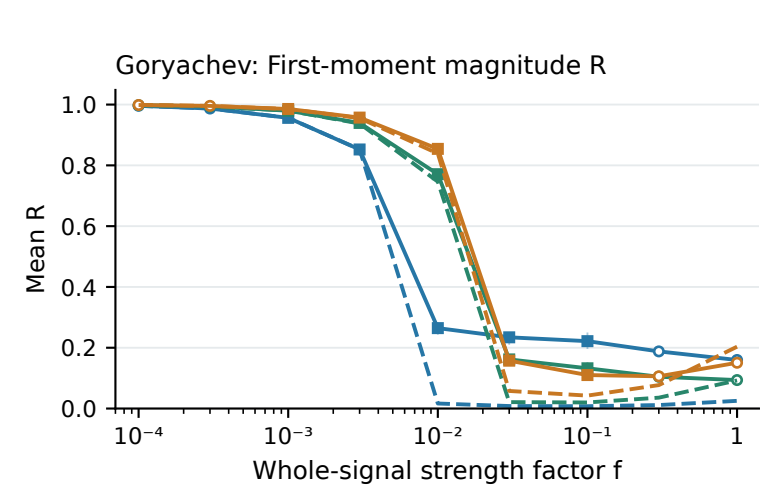


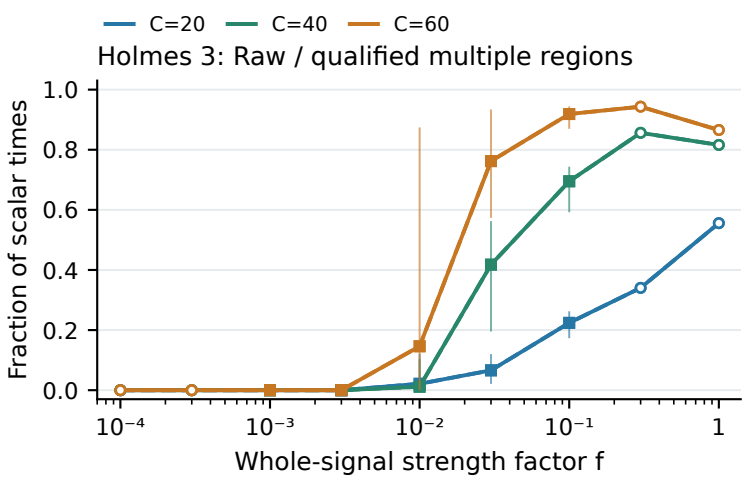


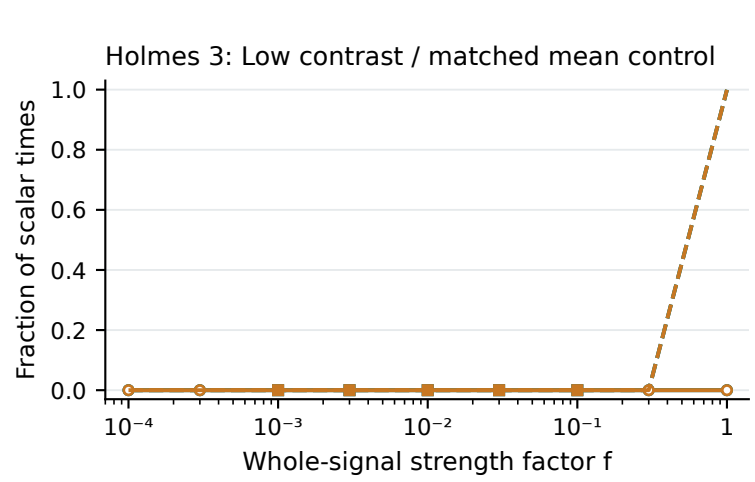


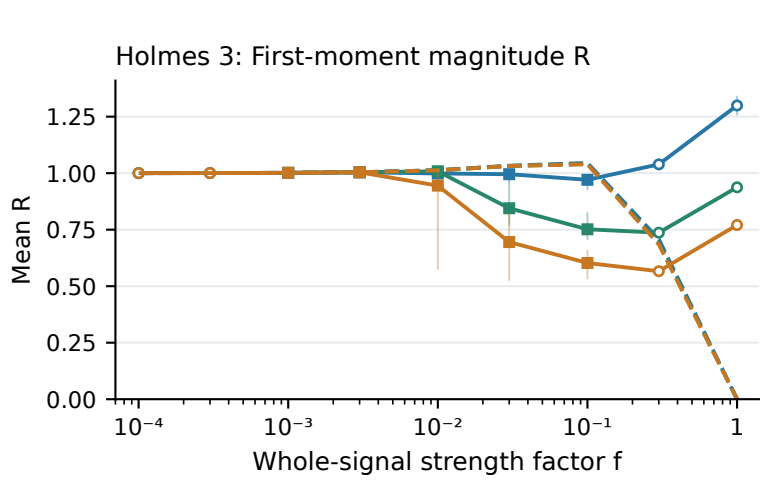


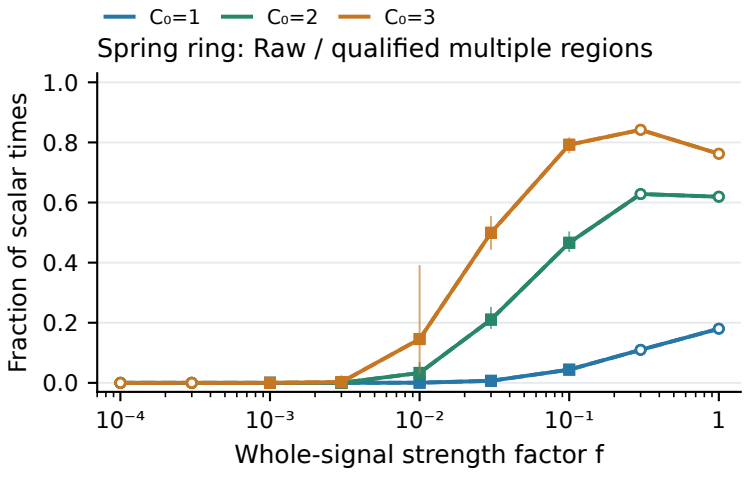


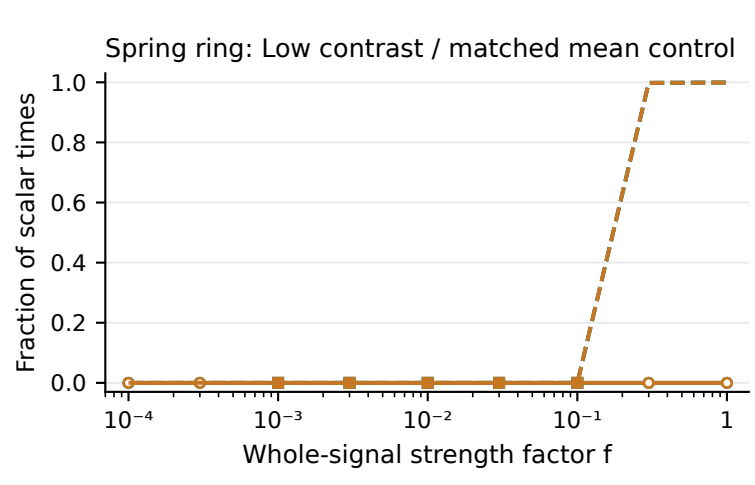


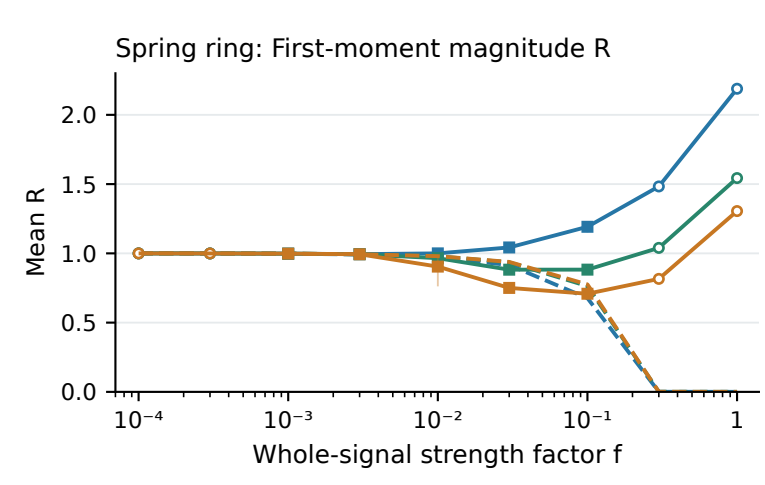












All scalar times remain in occupancy denominators; cadence5 chemicals/0.5 Spring. Threshold regions may coexist with weak or strong resultants; these are not exclusive phases.

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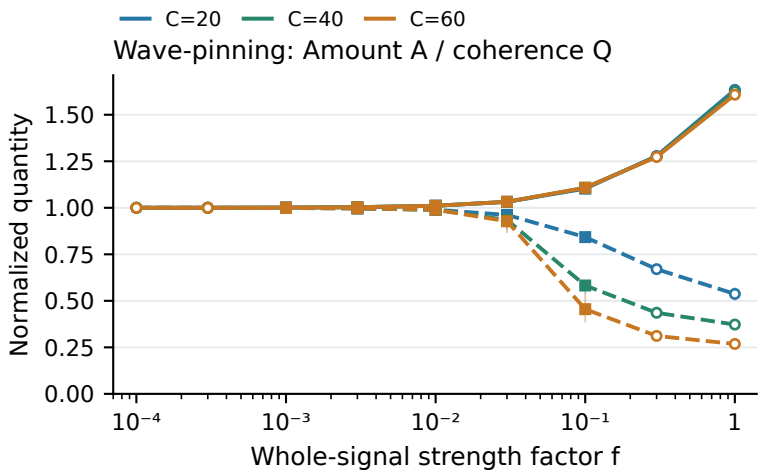
Mean controls match declared ensemble input means, without per-path normalization. Native clocks/laws/units differ; full/half differences do not certify stationarity.

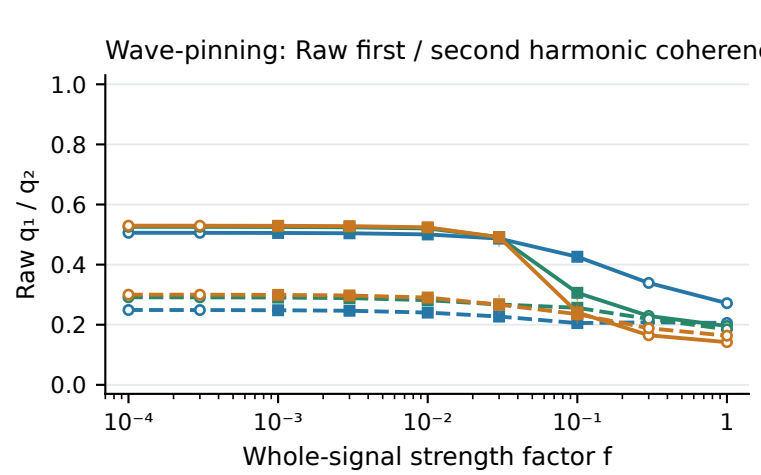
Active amount, coherence and spatial harmonics: all sizes and strengths

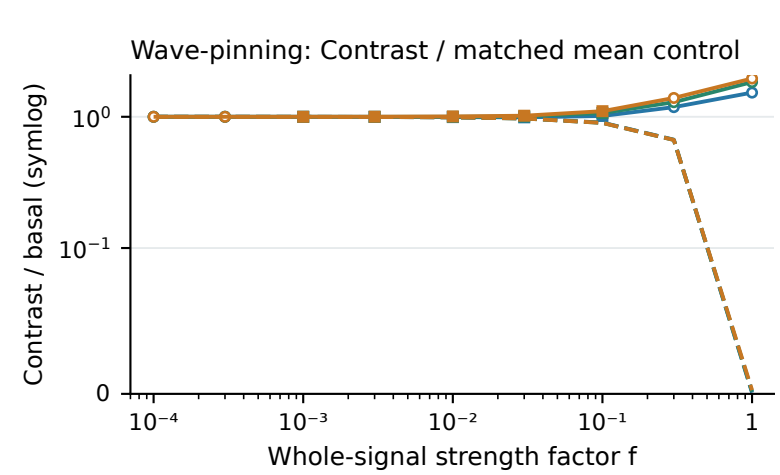
First half · chemicals0–16000; Spring0–1600. Half-window midpoint shared.

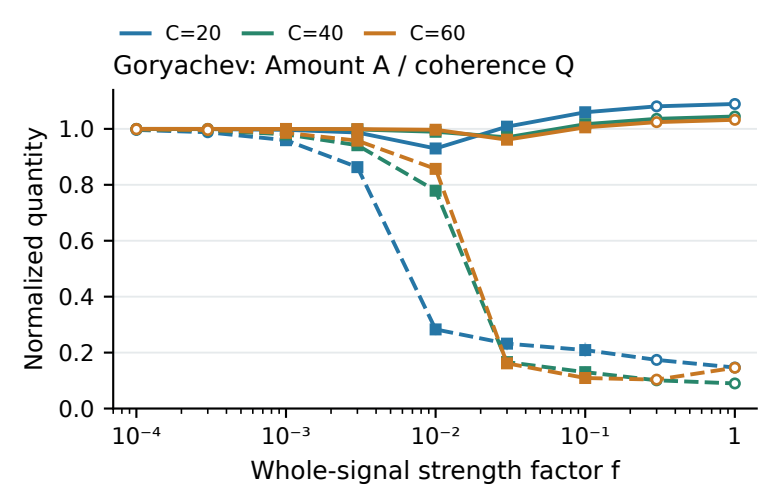
Size colors: blue smallest, teal middle, orange largest. Squares n=12; circles n=2. Whiskers: observed seed range, not confidence intervals.

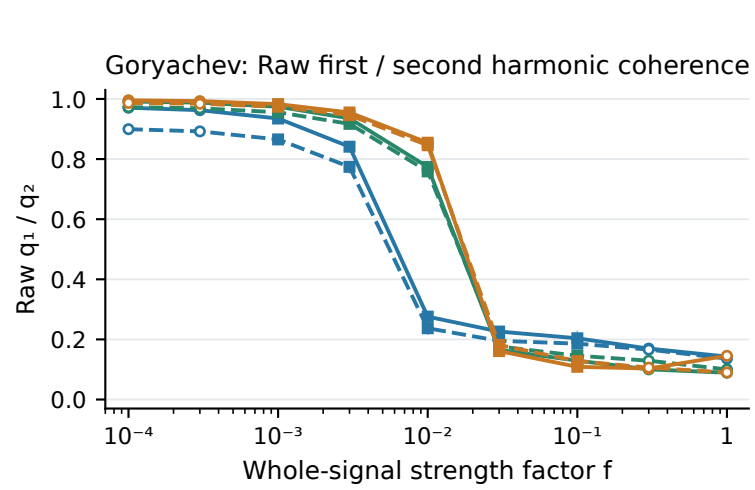
solid: active amount A / raw q_1 / noisy contrast dashed: coherence Q / q_2 / matched mean-control contrast

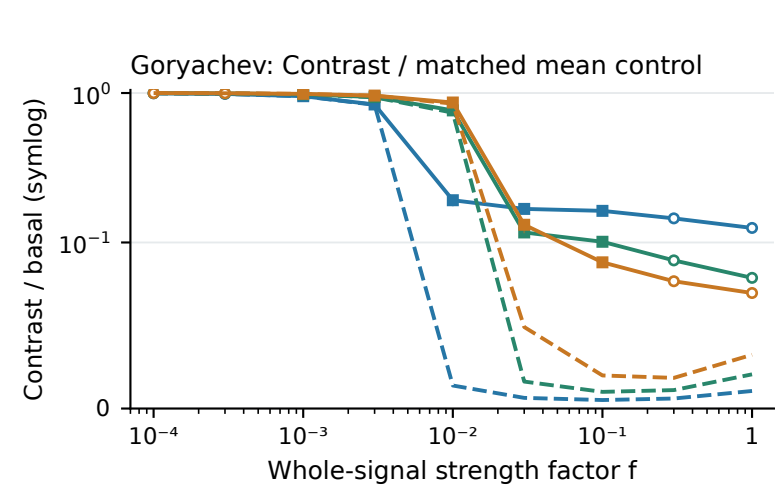


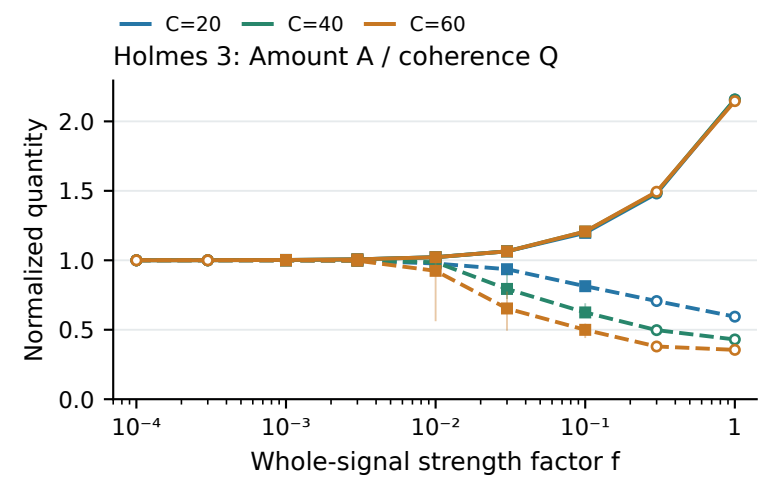


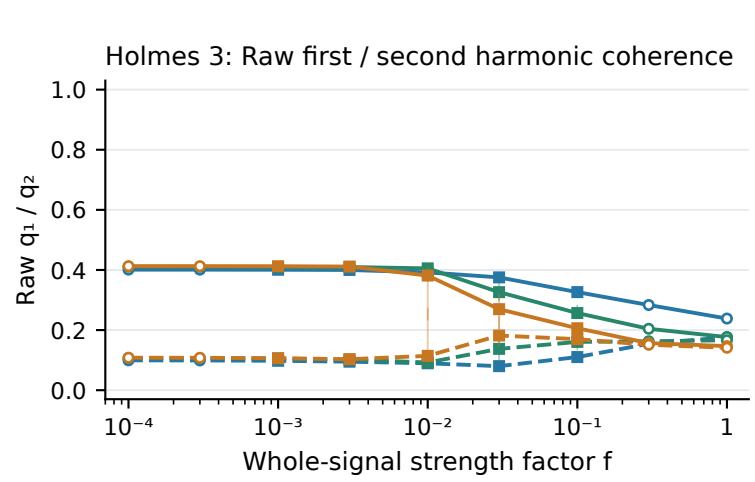


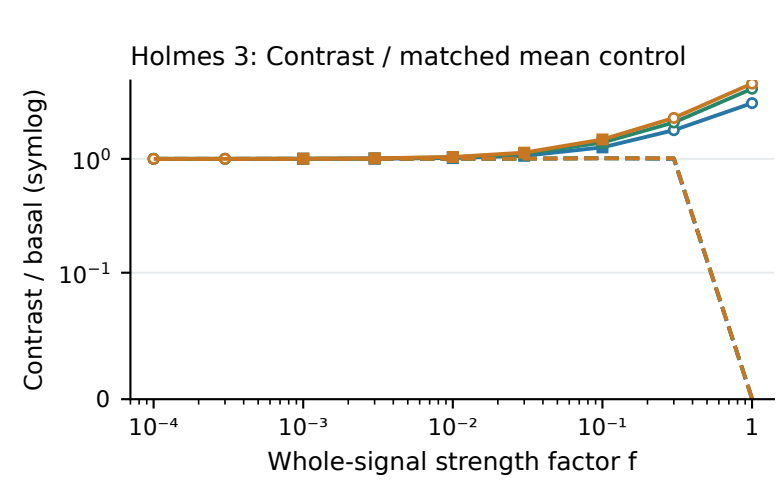


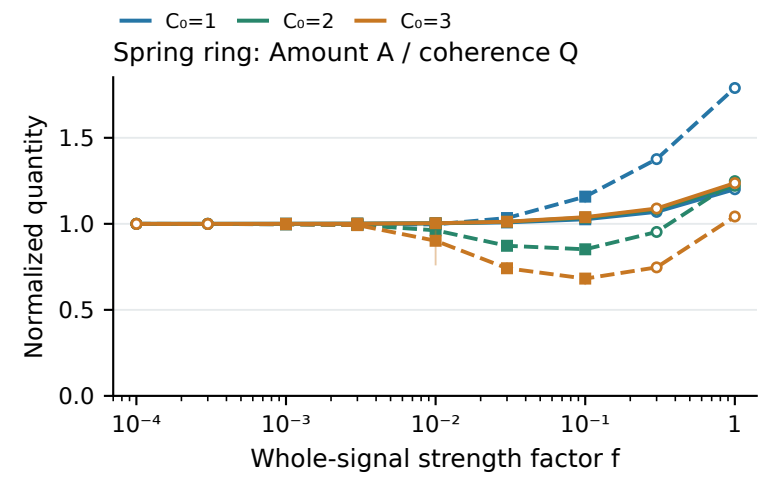


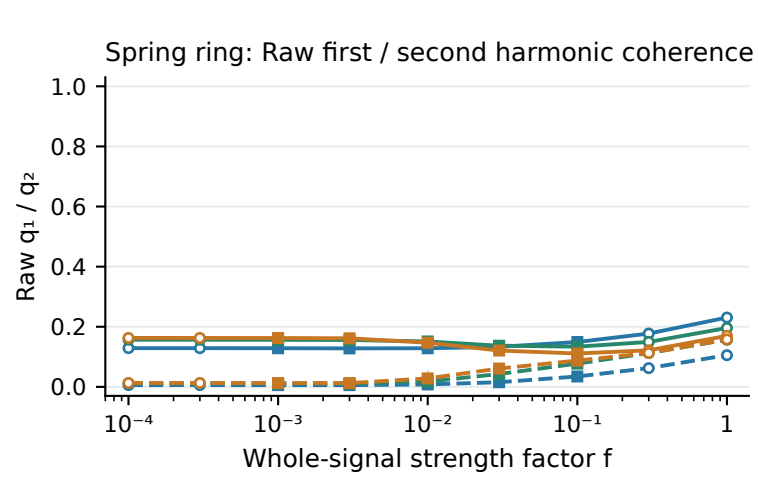


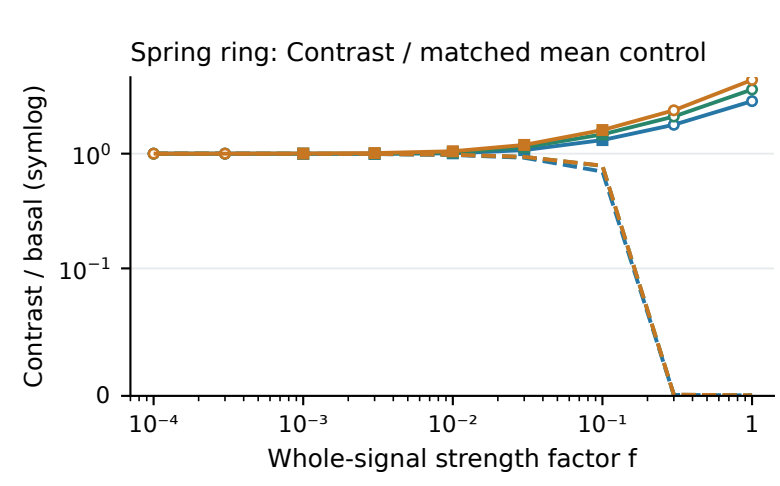












All scalar times remain in occupancy denominators; cadence5 chemicals/0.5 Spring. Threshold regions may coexist with weak or strong resultants; these are not exclusive phases.

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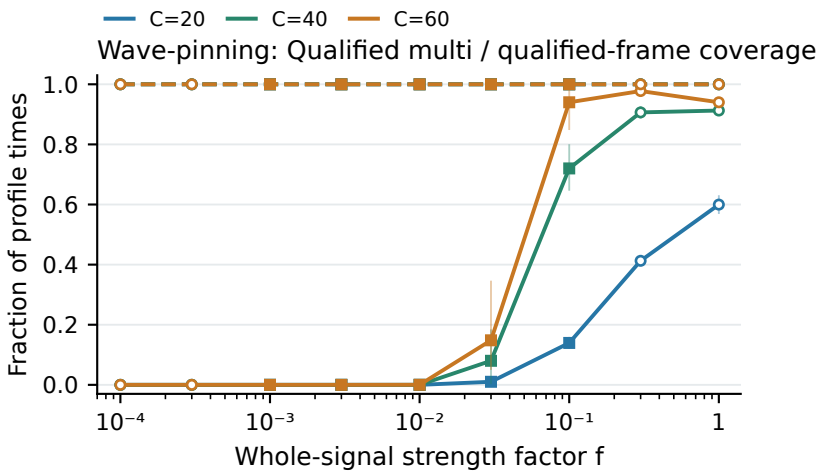
Mean controls match declared ensemble input means, without per-path normalization. Native clocks/laws/units differ; full/half differences do not certify stationarity.

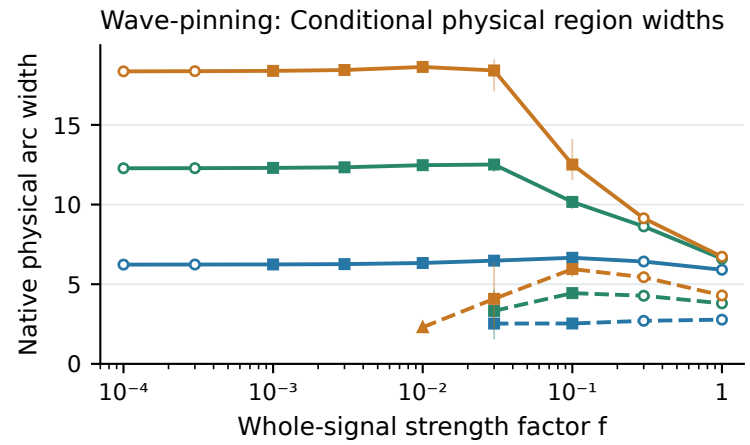
Instantaneous region widths with qualification coverage: all sizes and strengths

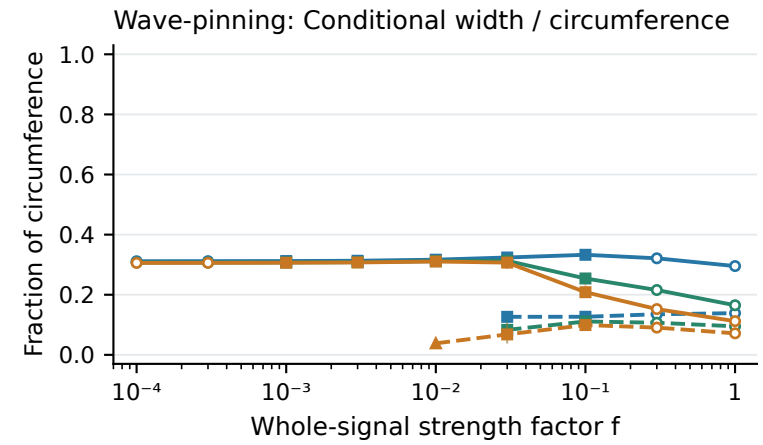
First half · chemicals0–16000; Spring0–1600. Half-window midpoint shared.

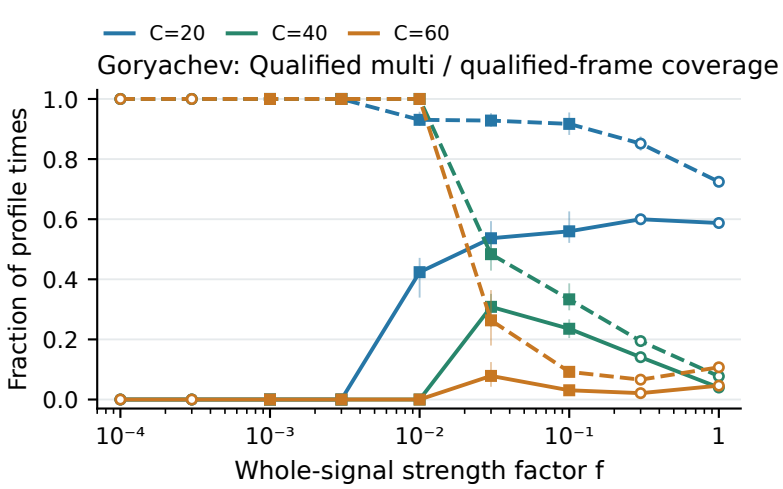
Size colors: blue smallest, teal middle, orange largest. Squares n=12; circles n=2. Whiskers: observed seed range, not confidence intervals.

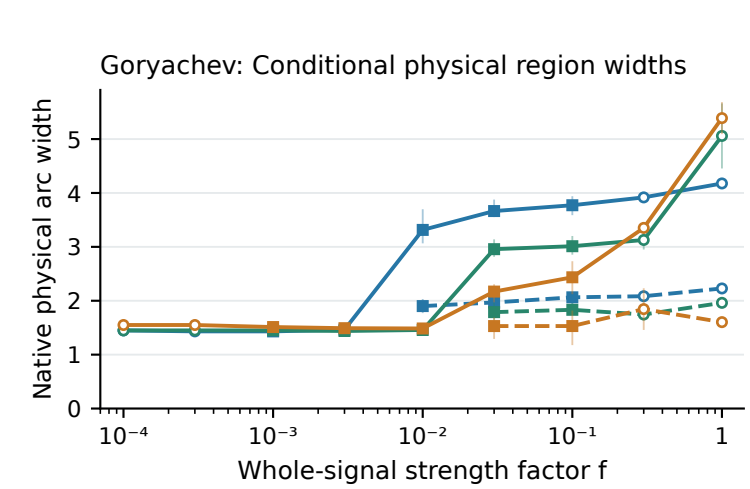
solid: qualified multi occupancy / largest width dashed: qualified-frame coverage / second-largest width

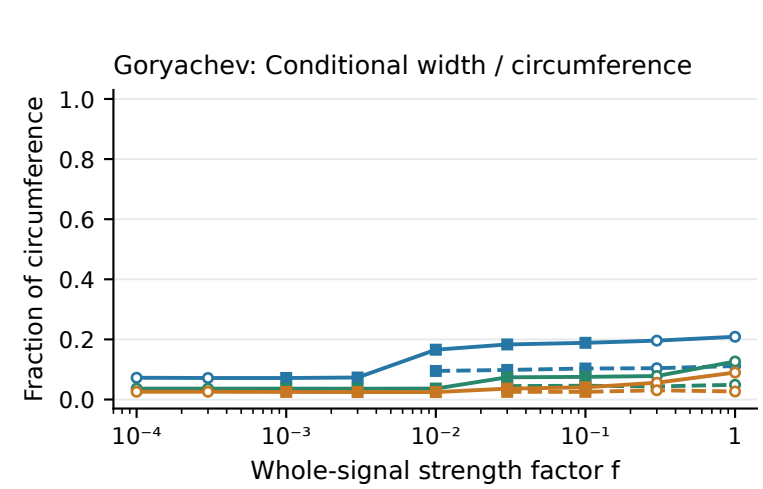


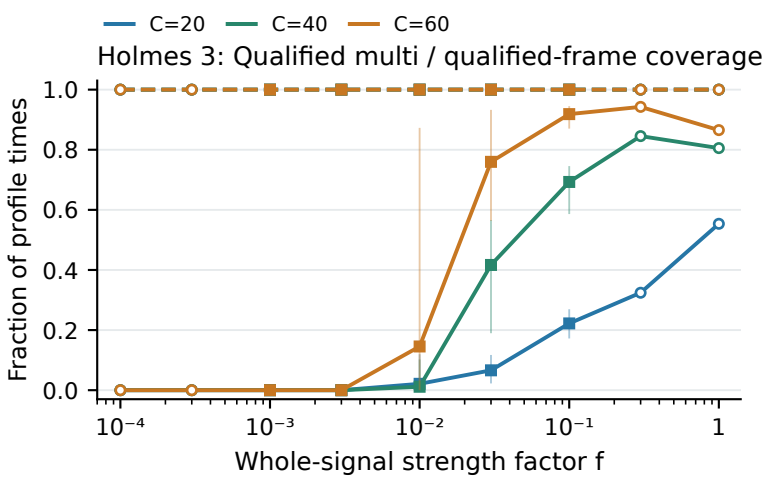


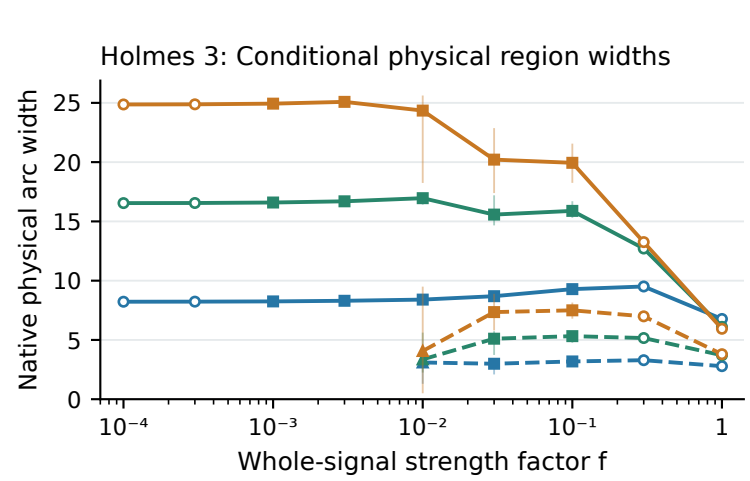


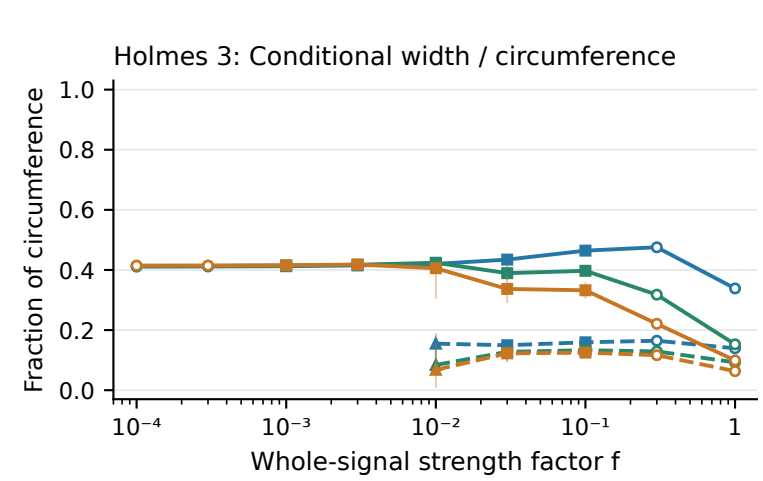


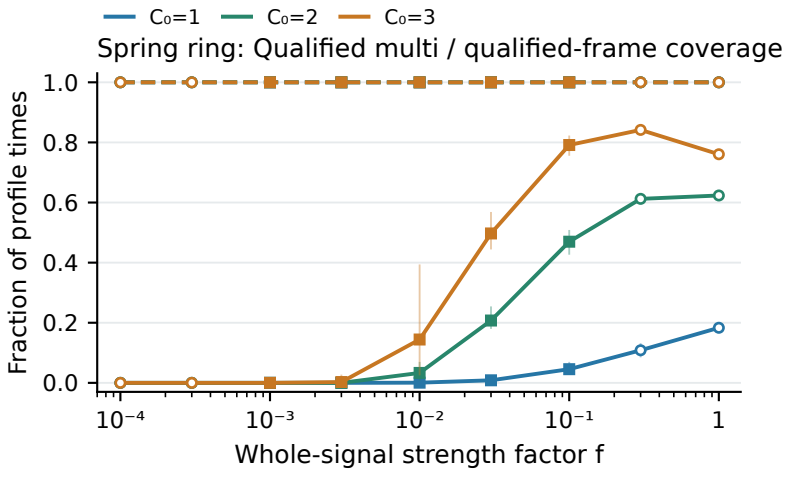


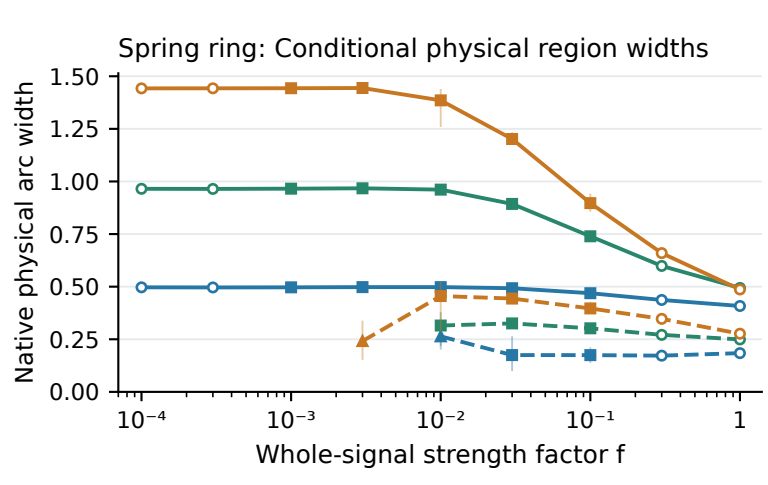


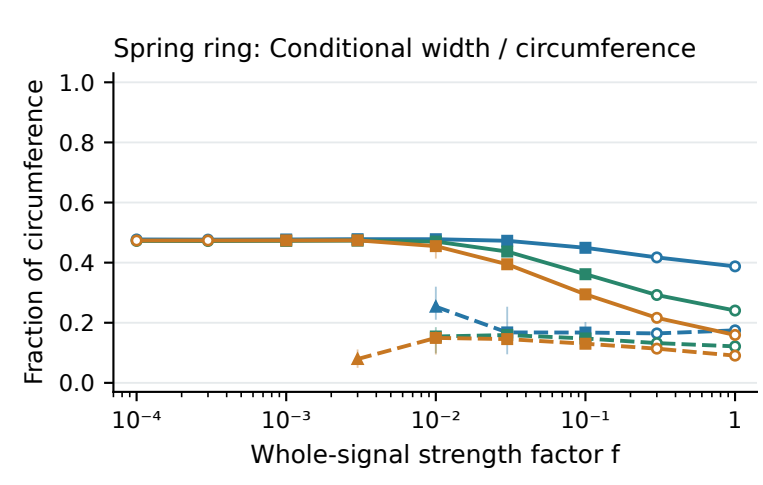












Widths join periodic seam-crossing regions: cell count/N × actual C(t). Threshold: current min+half contrast; qualification: contrast≥10% basal. Profile cadence40/4.

Largest: one per qualified frame; second: one per qualified multiregion frame. Equal available-seed means; △ fewer contributing seeds than condition n (exact counts exported).

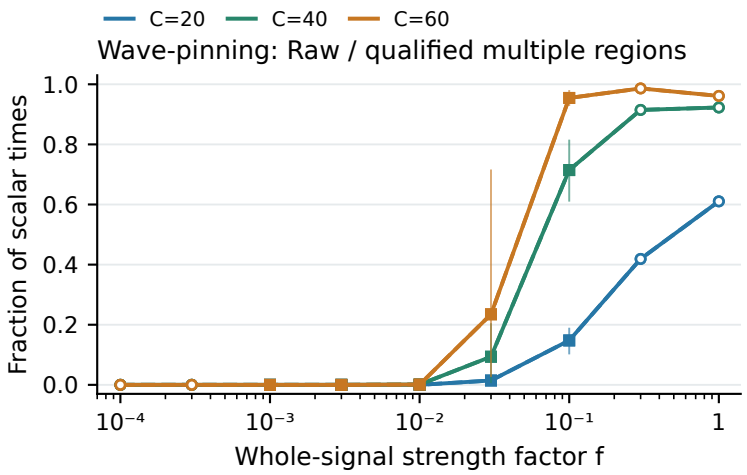
Missing conditional widths stay unavailable. Widths are descriptive geometry, outside the ten-metric numerical gate; no region births, establishment, lifetimes or travel inferred.

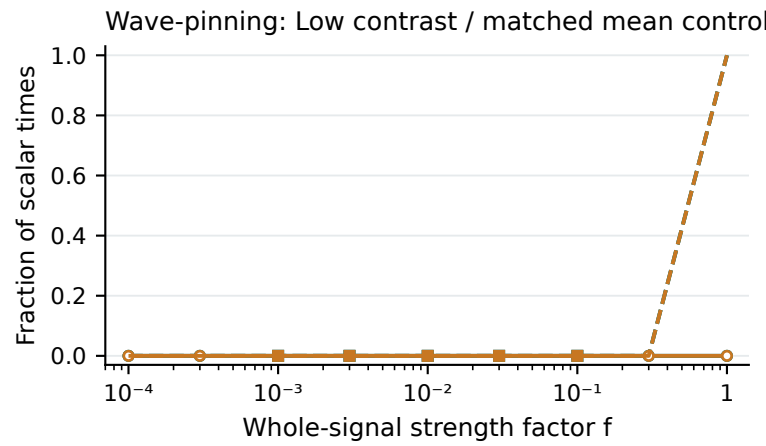
Organization, contrast and global magnitude: all sizes and strengths

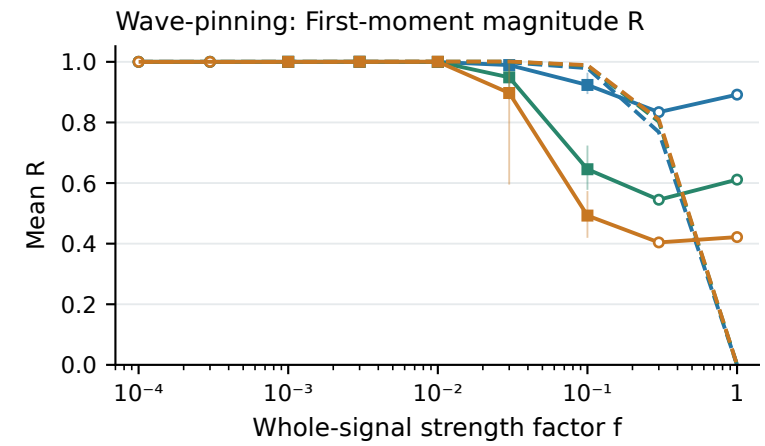
Second half · chemicals16000–32000; Spring1600–3200. Half-window midpoint shared.

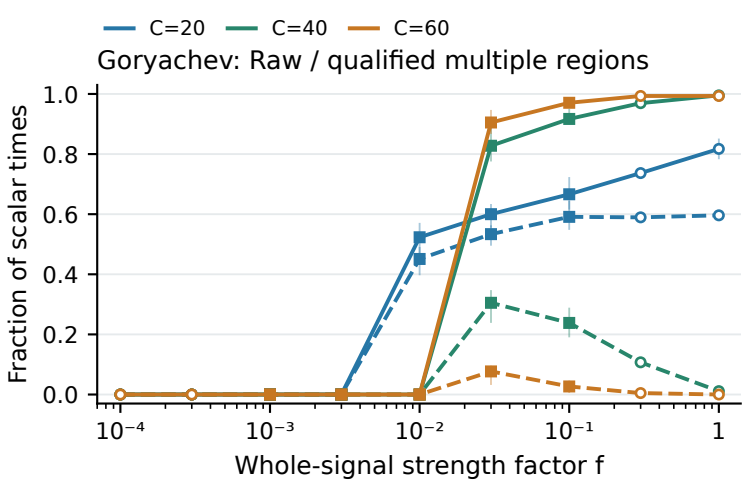
Size colors: blue smallest, teal middle, orange largest. Squares n=12; circles n=2. Whiskers: observed seed range, not confidence intervals.

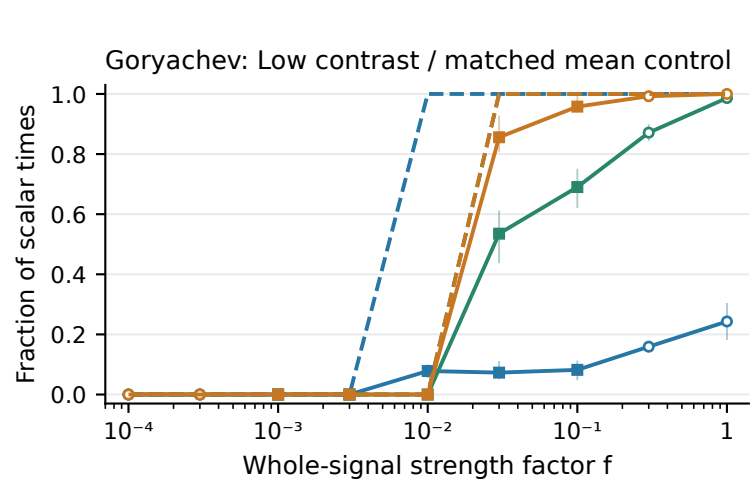
solid: raw multi / noisy low contrast / noisy R dashed: contrast-qualified multi / matched mean-control low contrast / mean-control R

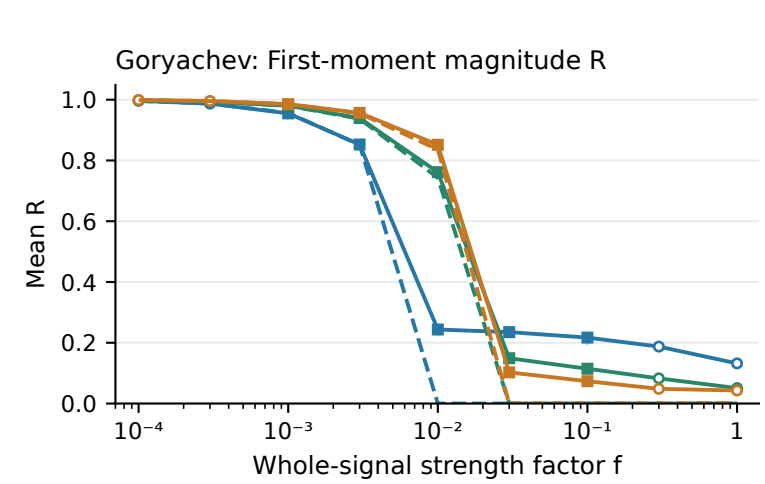


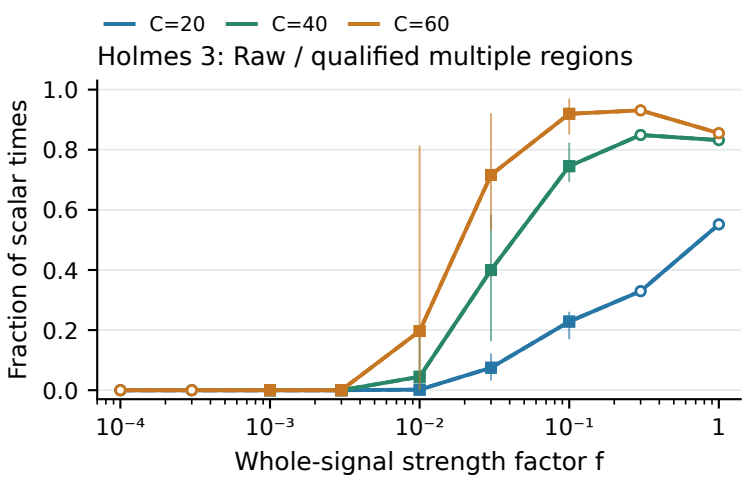


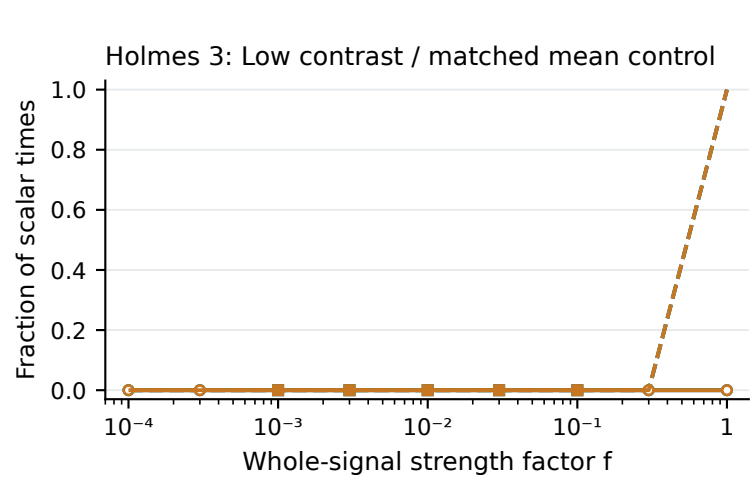


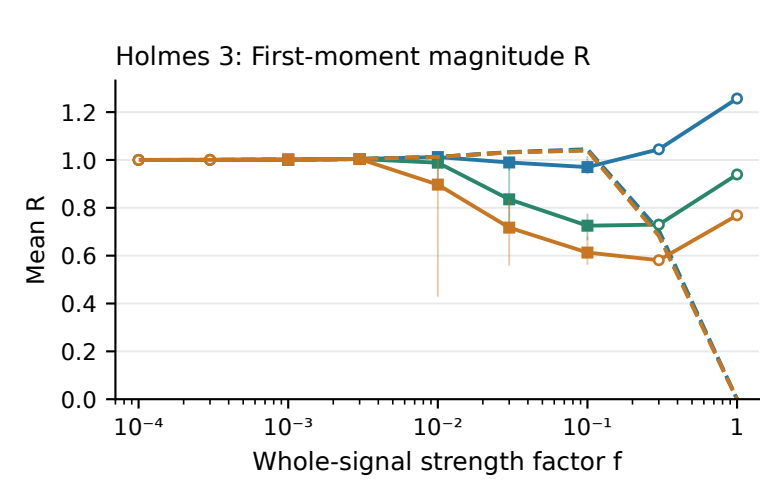


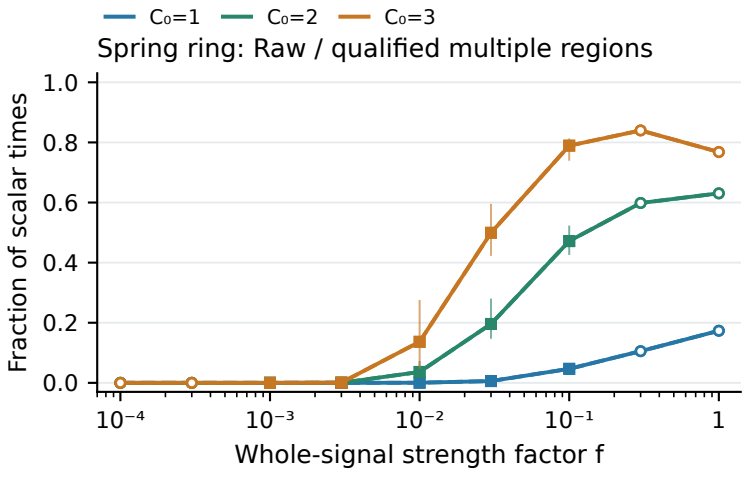


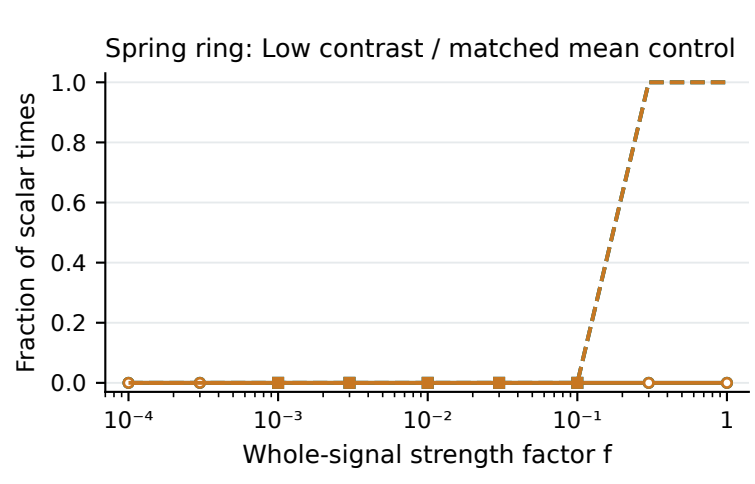


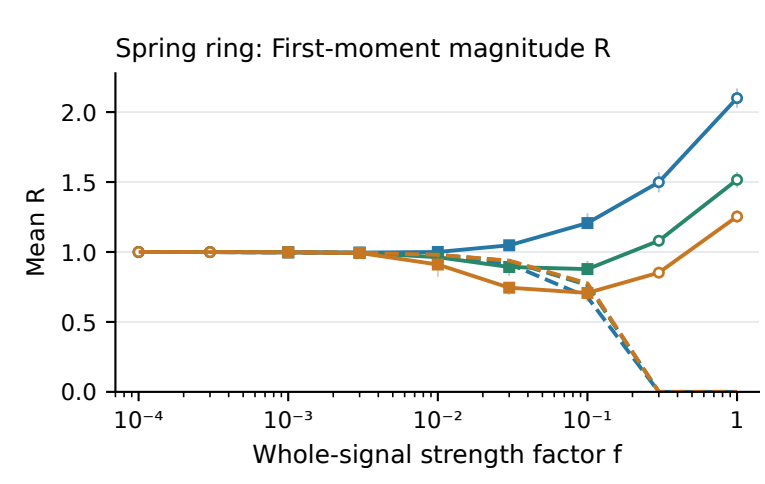












All scalar times remain in occupancy denominators; cadence5 chemicals/0.5 Spring. Threshold regions may coexist with weak or strong resultants; these are not exclusive phases.

R=AQ exactly per frame; mean R is mean of the paired product, not product of means. $q_1=|M_1|/M_0$ and $q_2=|M_2|/M_0$ retain basal coherence differences.

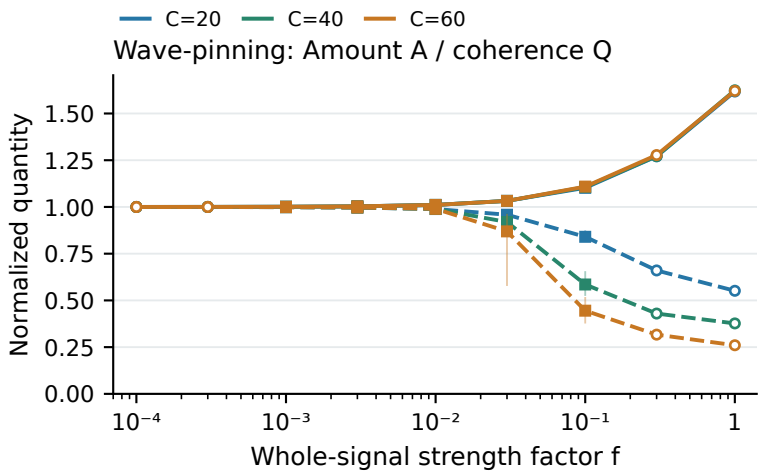
Mean controls match declared ensemble input means, without per-path normalization. Native clocks/laws/units differ; full/half differences do not certify stationarity.

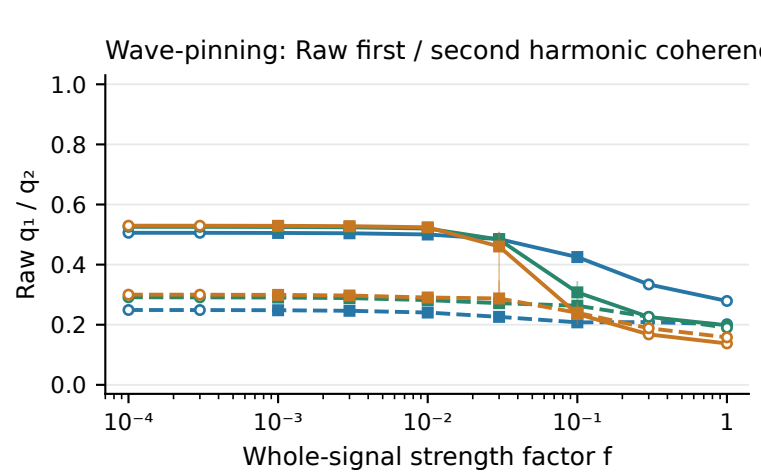
Active amount, coherence and spatial harmonics: all sizes and strengths

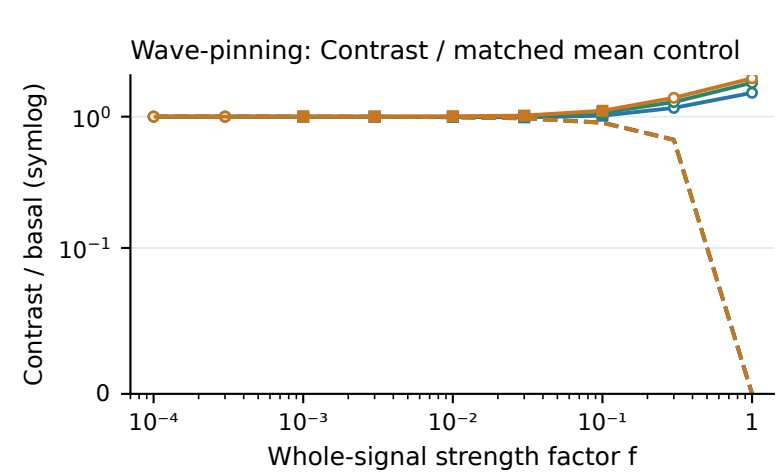
Second half · chemicals16000–32000; Spring1600–3200. Half-window midpoint shared.

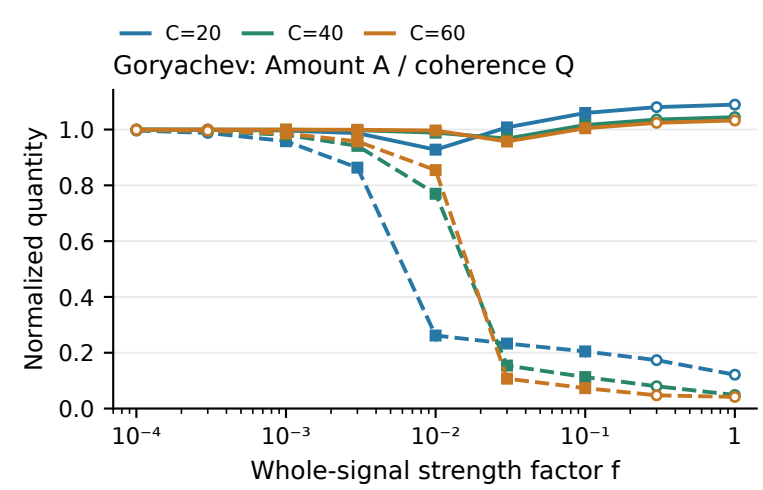
Size colors: blue smallest, teal middle, orange largest. Squares n=12; circles n=2. Whiskers: observed seed range, not confidence intervals.

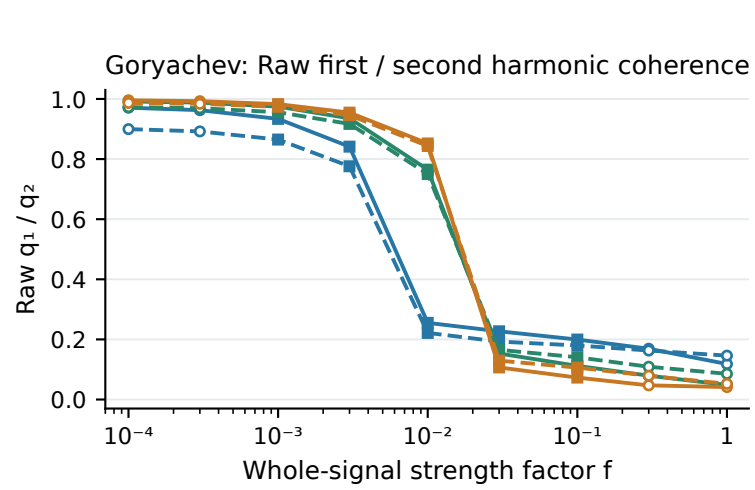
solid: active amount A / raw q_1 / noisy contrast dashed: coherence Q / q_2 / matched mean-control contrast

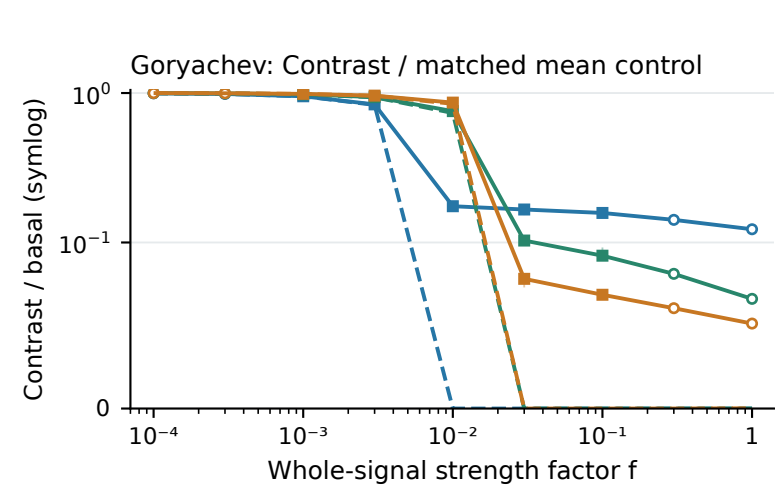


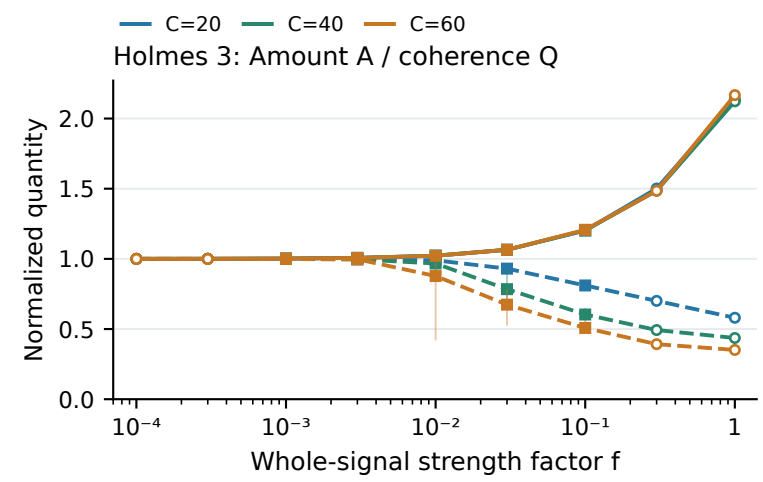


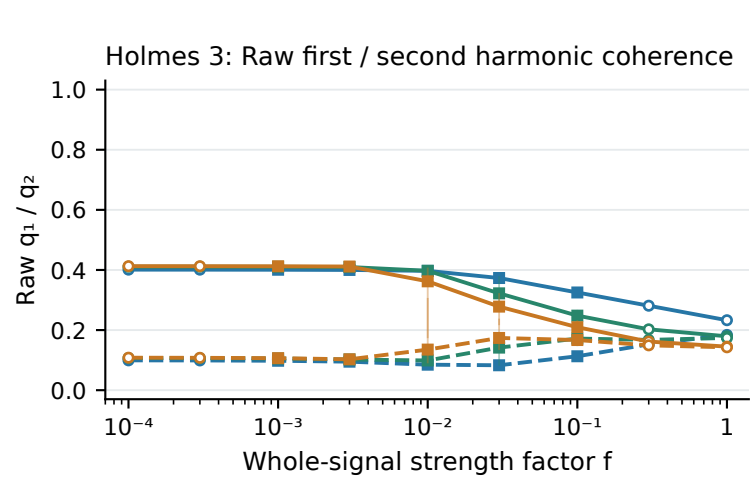


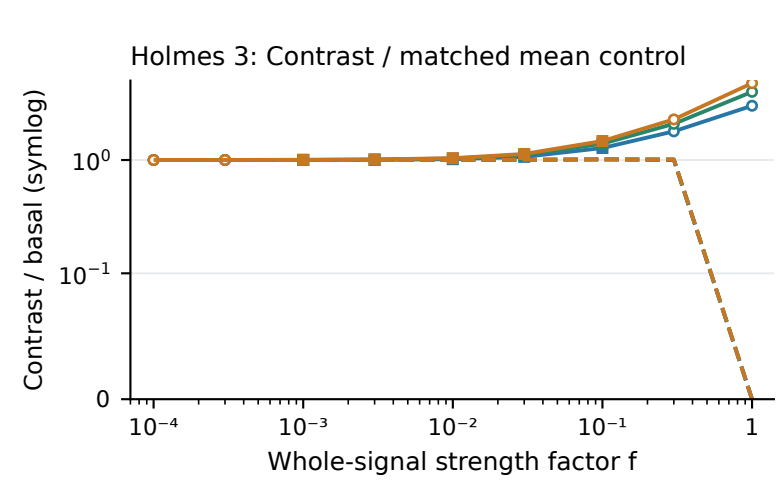


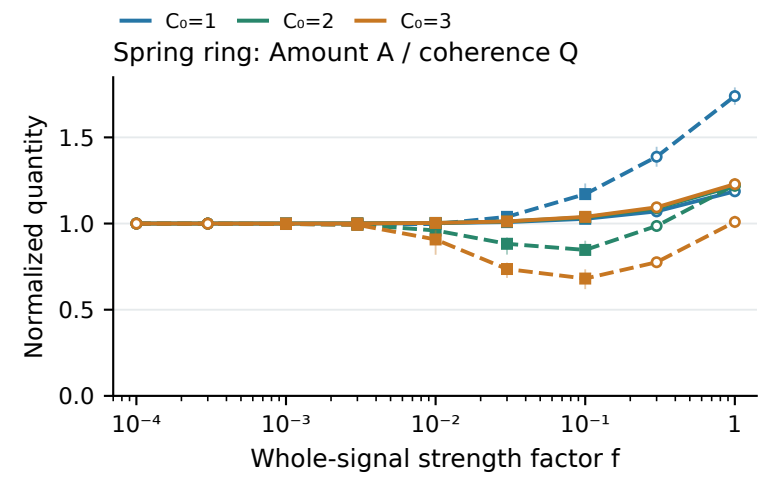


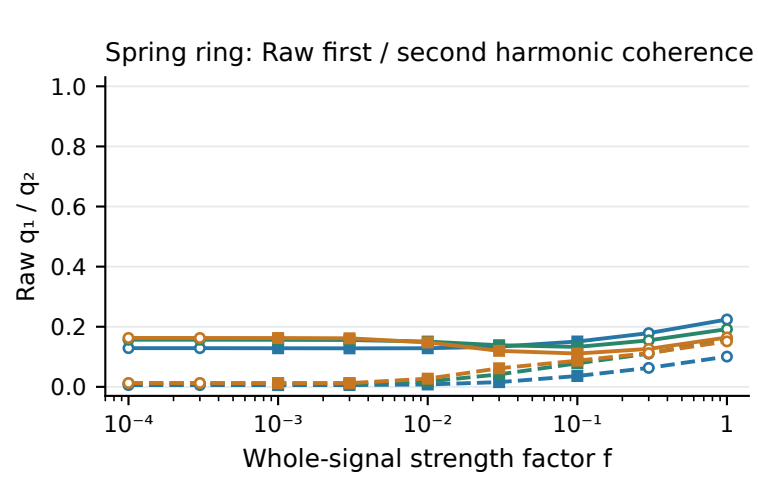


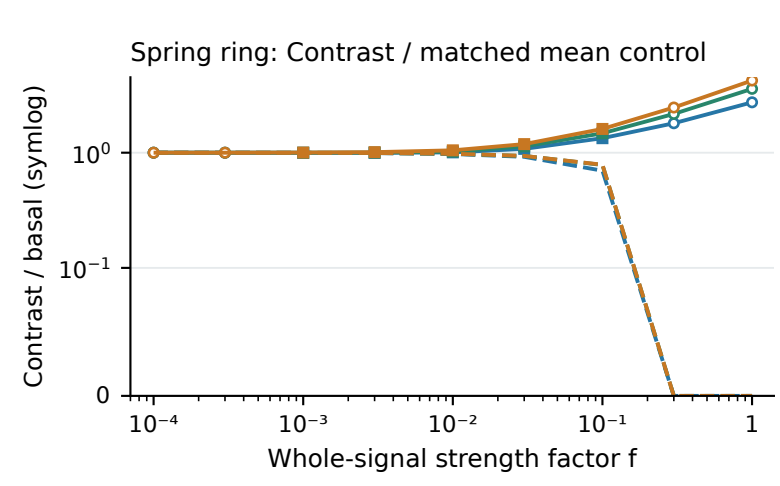












All scalar times remain in occupancy denominators; cadence5 chemicals/0.5 Spring. Threshold regions may coexist with weak or strong resultants; these are not exclusive phases.

R=AQ exactly per frame; mean R is mean of the paired product, not product of means. $q_1=|M_1|/M_0$ and $q_2=|M_2|/M_0$ retain basal coherence differences.

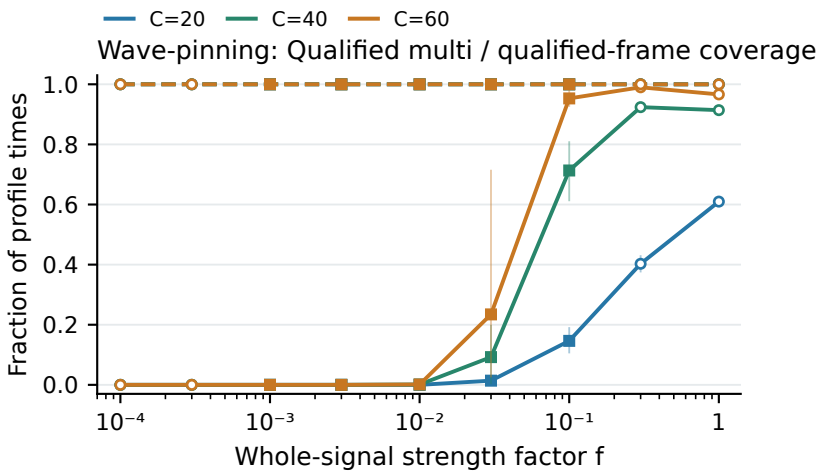
Mean controls match declared ensemble input means, without per-path normalization. Native clocks/laws/units differ; full/half differences do not certify stationarity.

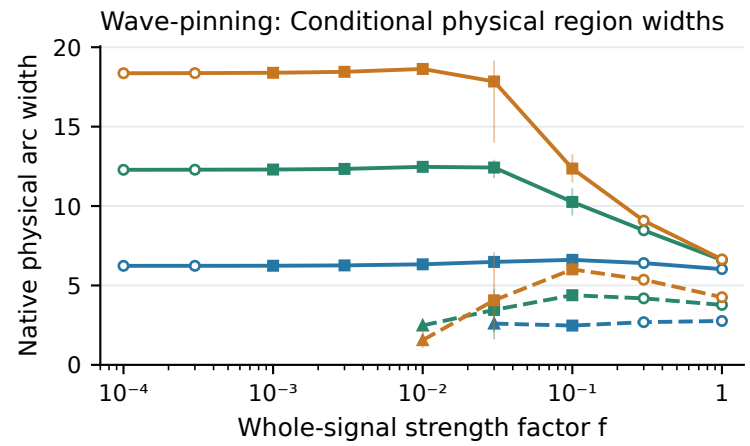
Instantaneous region widths with qualification coverage: all sizes and strengths

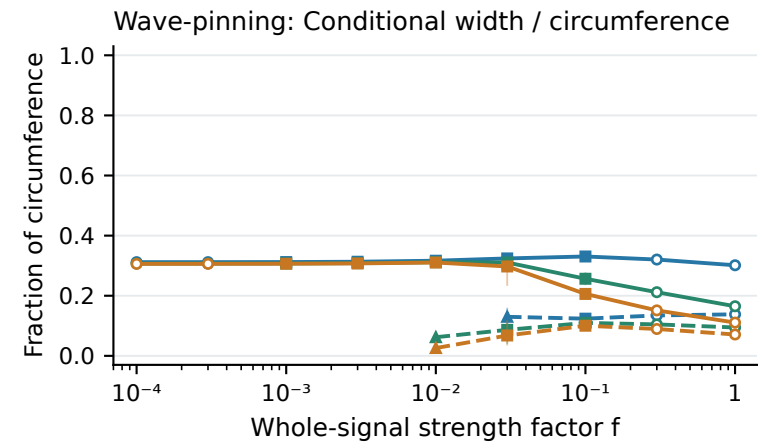
Second half · chemicals16000–32000; Spring1600–3200. Half-window midpoint shared.

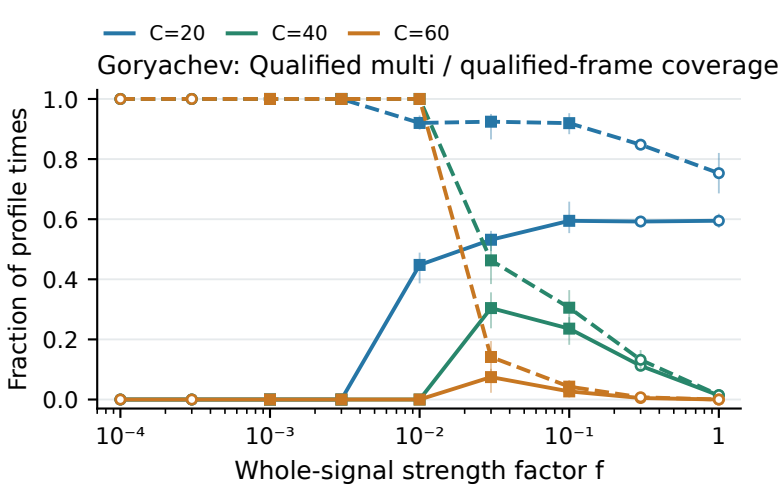
Size colors: blue smallest, teal middle, orange largest. Squares n=12; circles n=2. Whiskers: observed seed range, not confidence intervals.

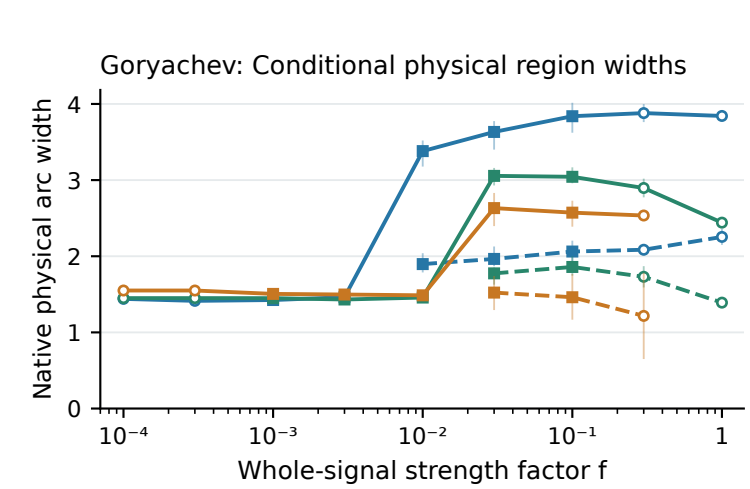
solid: qualified multi occupancy / largest width dashed: qualified-frame coverage / second-largest width

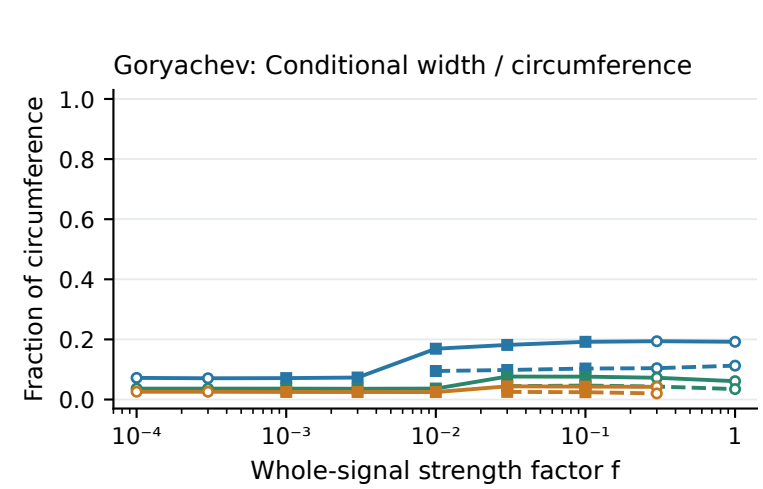


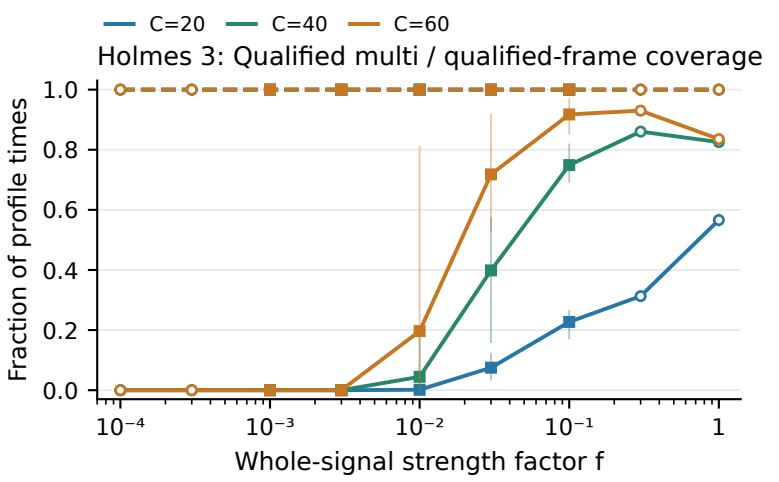


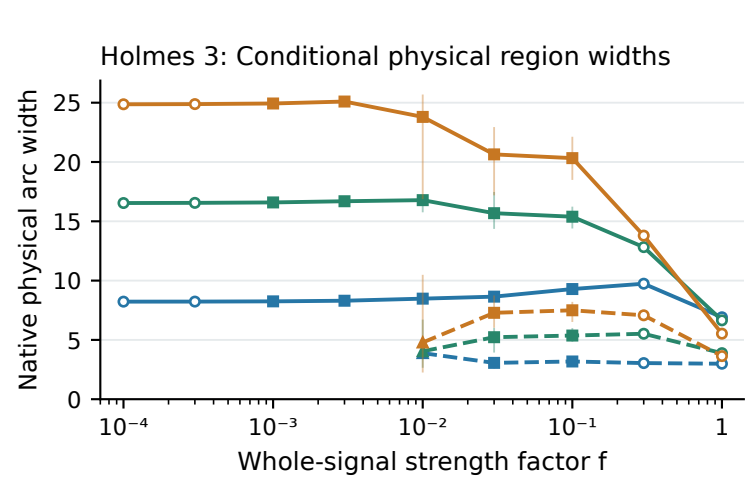


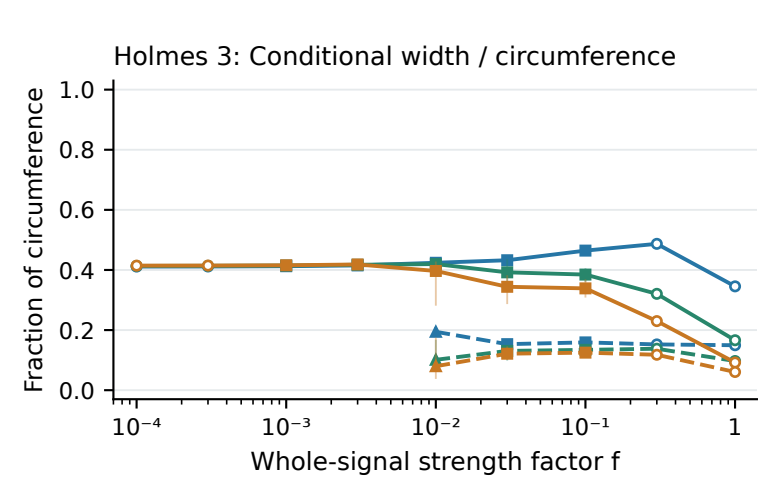


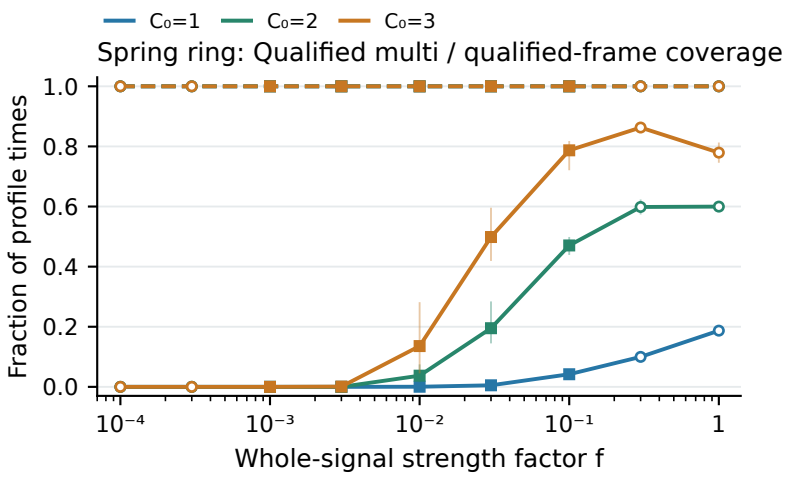


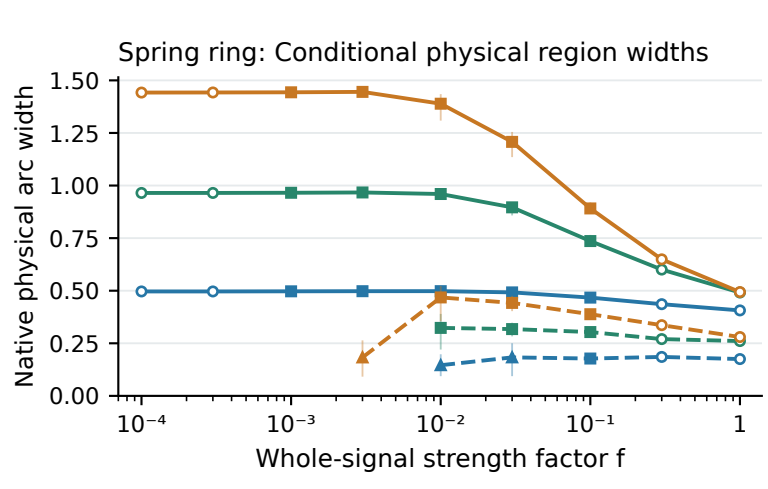


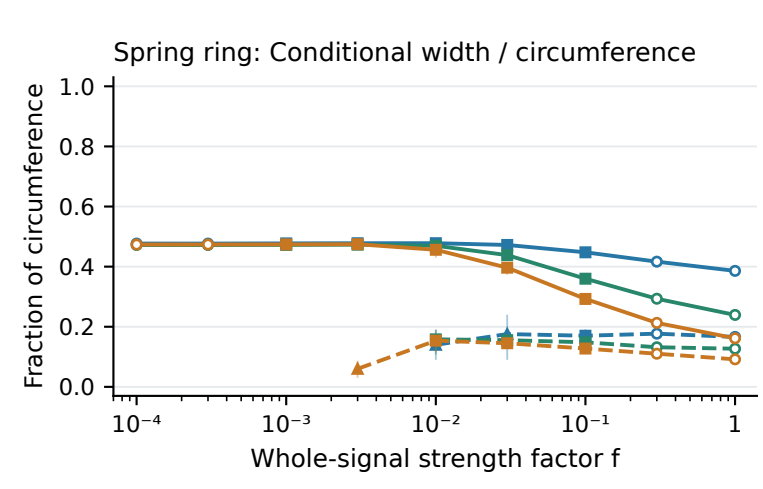












Widths join periodic seam-crossing regions: cell count/N × actual C(t). Threshold: current min+half contrast; qualification: contrast≥10% basal. Profile cadence40/4.

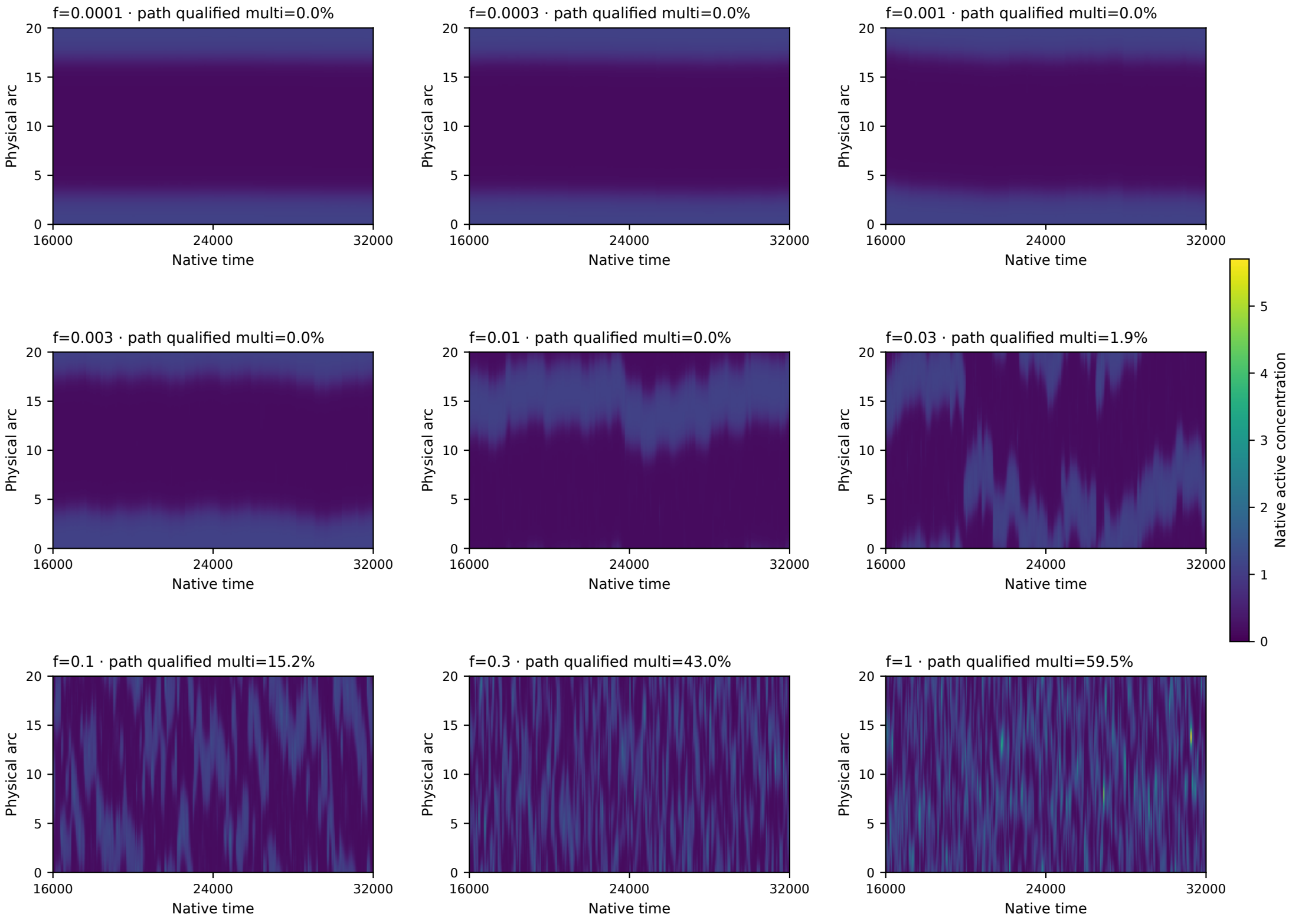
Largest: one per qualified frame; second: one per qualified multiregion frame. Equal available-seed means; △ fewer contributing seeds than condition n (exact counts exported).

Missing conditional widths stay unavailable. Widths are descriptive geometry, outside the ten-metric numerical gate; no region births, establishment, lifetimes or travel inferred.

Wave-pinning, C=20: all nine input strengths

Second-half descriptor medoid per condition, with smallest-seed tie breaking. Common zero-based native activity scale across this page.

f=0.001/0.003/0.01/0.03/0.1: n=12. Other four levels: n=2 (both tie). These examples do not estimate frequency.



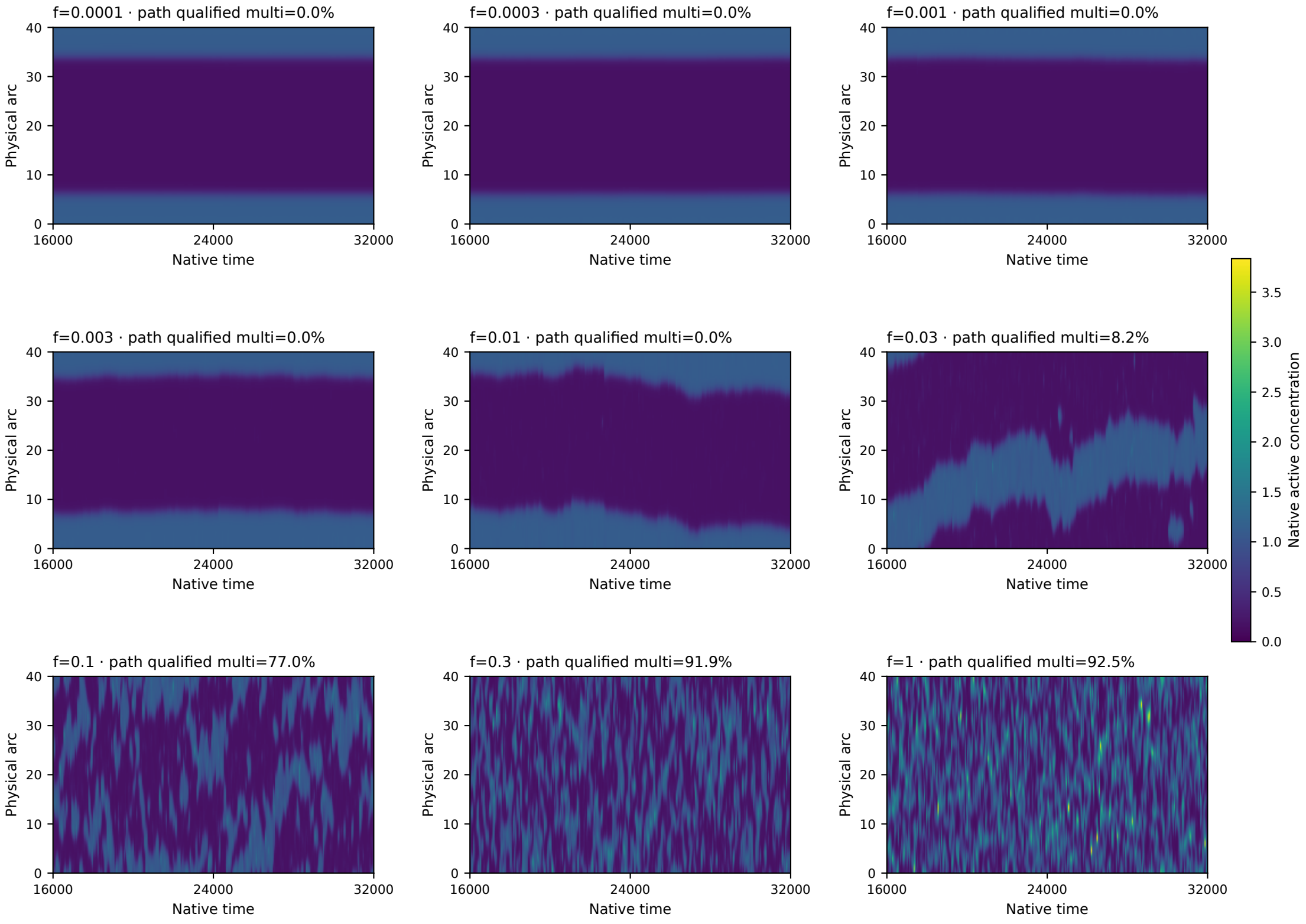
No field alignment, per-history peak normalization, birth/establishment or lifetime classifier. Multi-region counts are qualified at contrast≥10% of own basal.

The first half is discarded as an observed-window choice, without assuming stationarity. Model clocks and input laws remain native and uncalibrated.

Wave-pinning, C=40: all nine input strengths

Second-half descriptor medoid per condition, with smallest-seed tie breaking. Common zero-based native activity scale across this page.

f=0.001/0.003/0.01/0.03/0.1: n=12. Other four levels: n=2 (both tie). These examples do not estimate frequency.



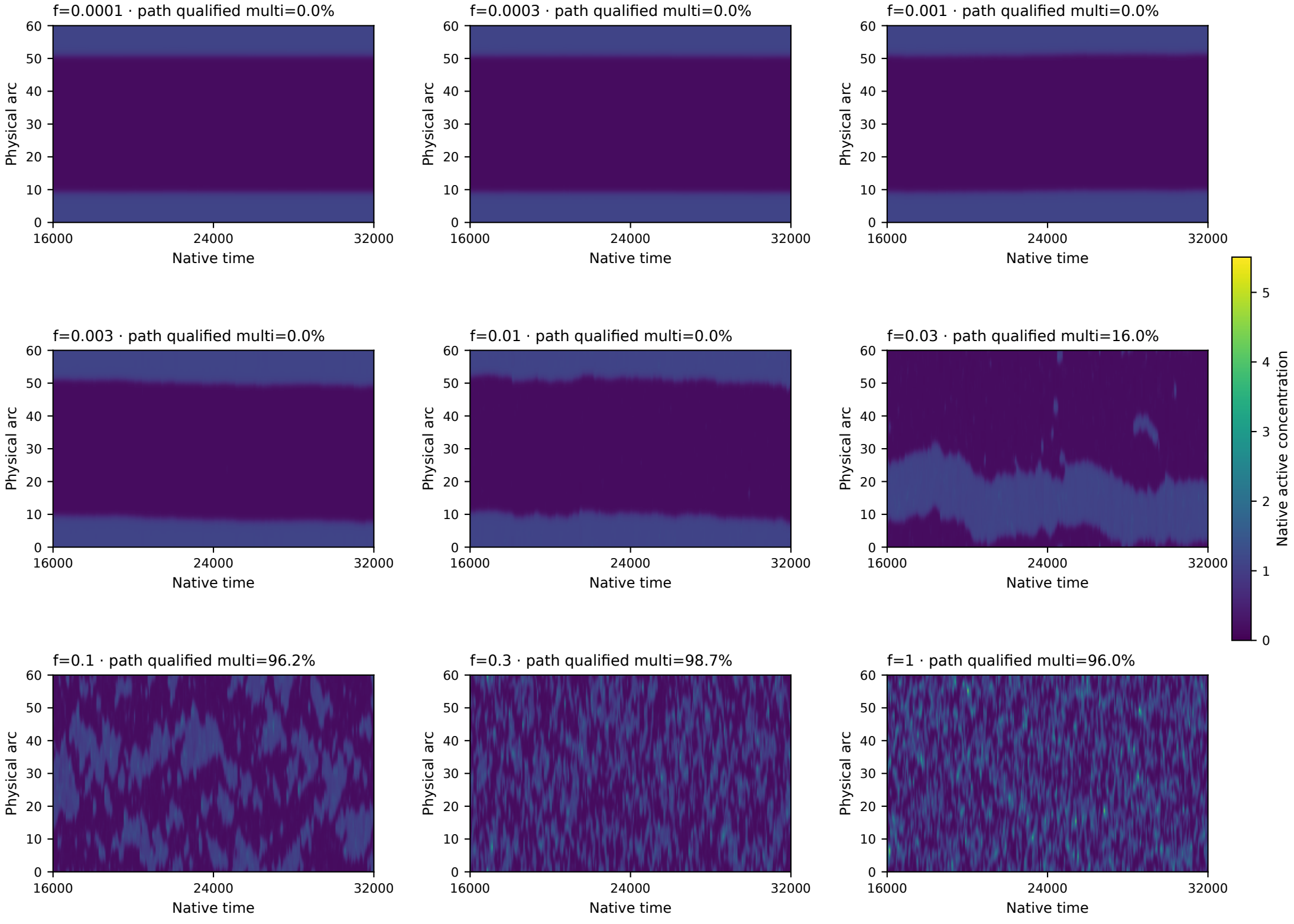
No field alignment, per-history peak normalization, birth/establishment or lifetime classifier. Multi-region counts are qualified at contrast≥10% of own basal.

The first half is discarded as an observed-window choice, without assuming stationarity. Model clocks and input laws remain native and uncalibrated.

Wave-pinning, C=60: all nine input strengths

Second-half descriptor medoid per condition, with smallest-seed tie breaking. Common zero-based native activity scale across this page.

f=0.001/0.003/0.01/0.03/0.1: n=12. Other four levels: n=2 (both tie). These examples do not estimate frequency.



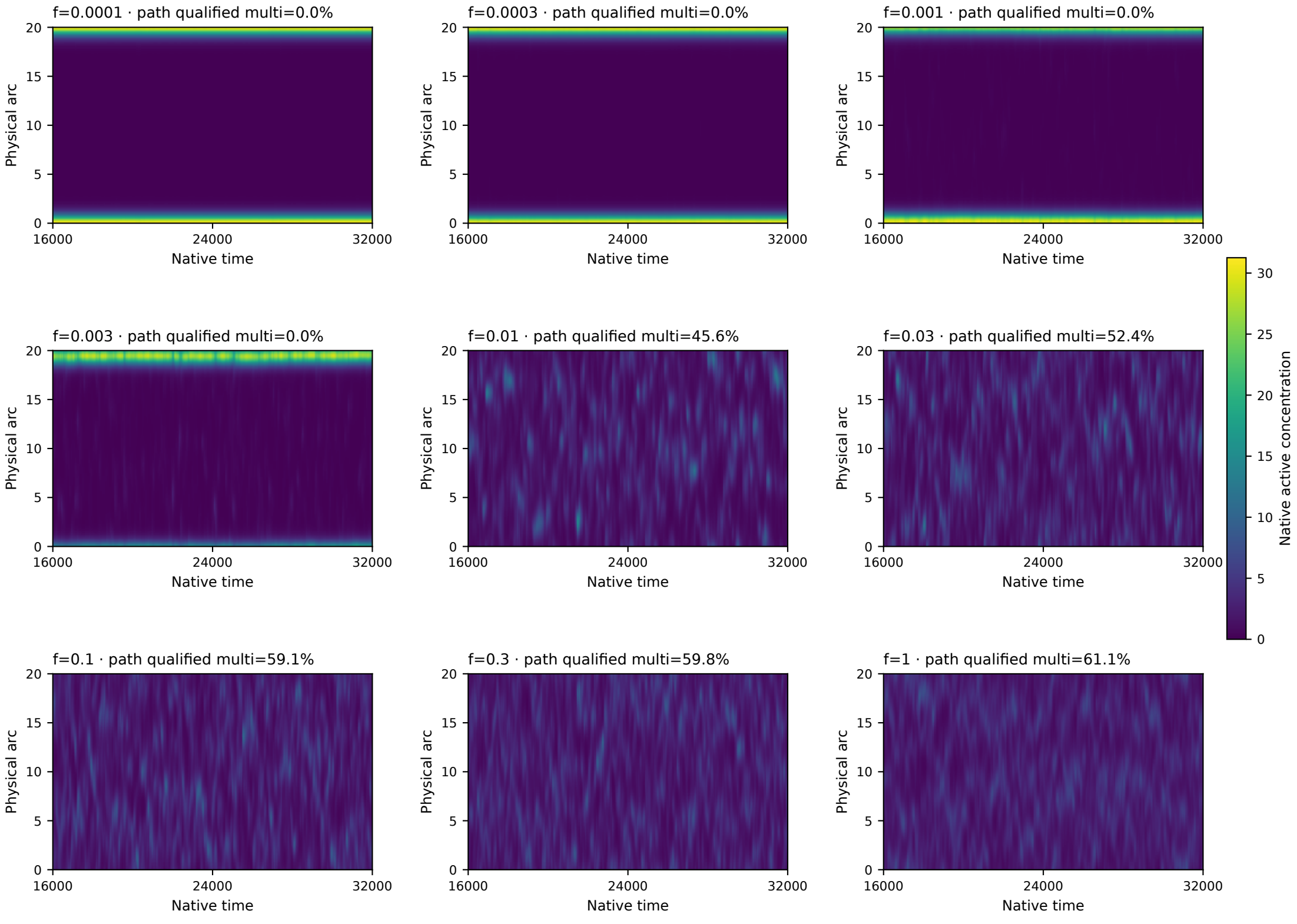
No field alignment, per-history peak normalization, birth/establishment or lifetime classifier. Multi-region counts are qualified at contrast≥10% of own basal.

The first half is discarded as an observed-window choice, without assuming stationarity. Model clocks and input laws remain native and uncalibrated.

Goryachev, C=20: all nine input strengths

Second-half descriptor medoid per condition, with smallest-seed tie breaking. Common zero-based native activity scale across this page.

f=0.001/0.003/0.01/0.03/0.1: n=12. Other four levels: n=2 (both tie). These examples do not estimate frequency.



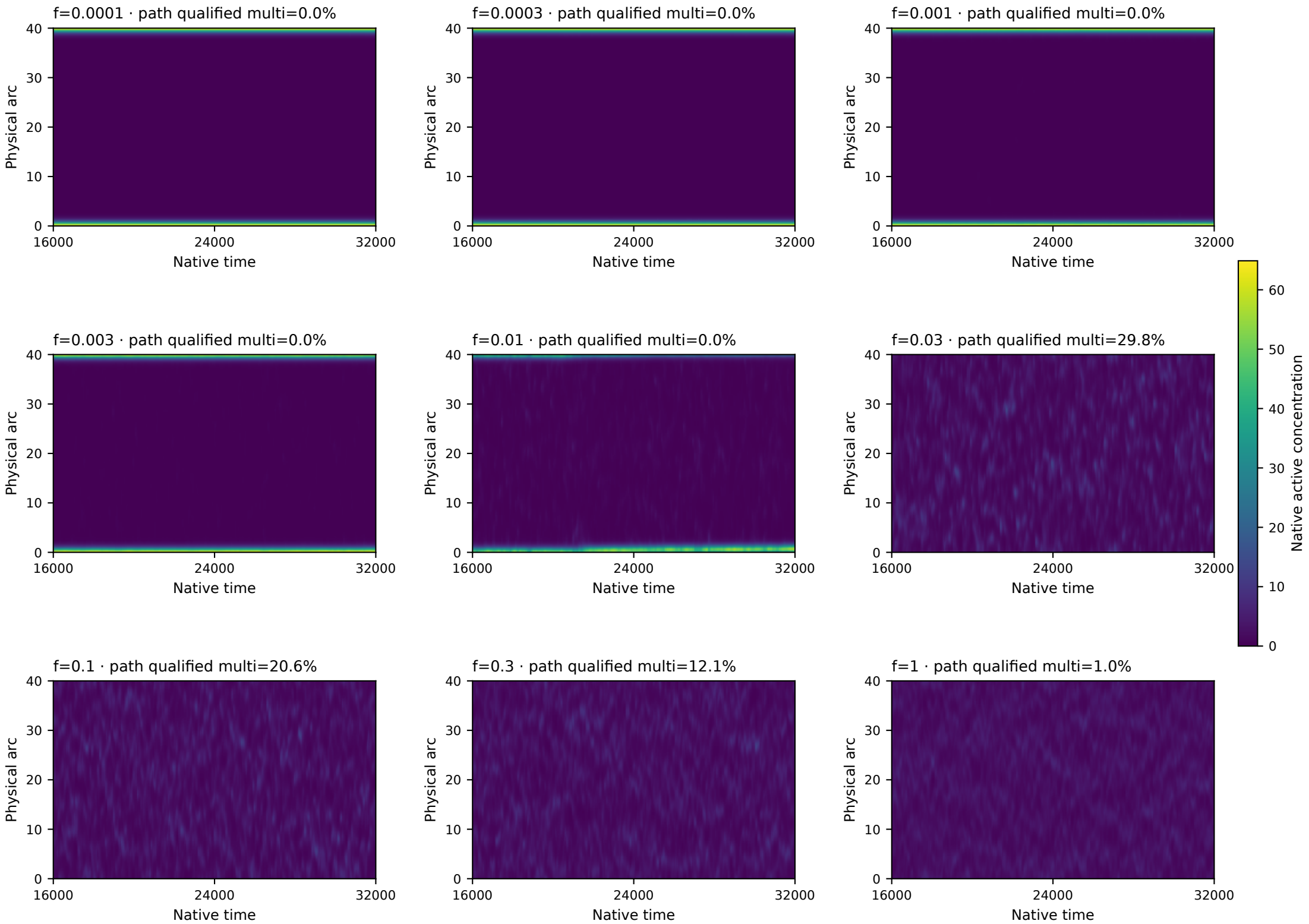
No field alignment, per-history peak normalization, birth/establishment or lifetime classifier. Multi-region counts are qualified at contrast≥10% of own basal.

The first half is discarded as an observed-window choice, without assuming stationarity. Model clocks and input laws remain native and uncalibrated.

Goryachev, C=40: all nine input strengths

Second-half descriptor medoid per condition, with smallest-seed tie breaking. Common zero-based native activity scale across this page.

f=0.001/0.003/0.01/0.03/0.1: n=12. Other four levels: n=2 (both tie). These examples do not estimate frequency.



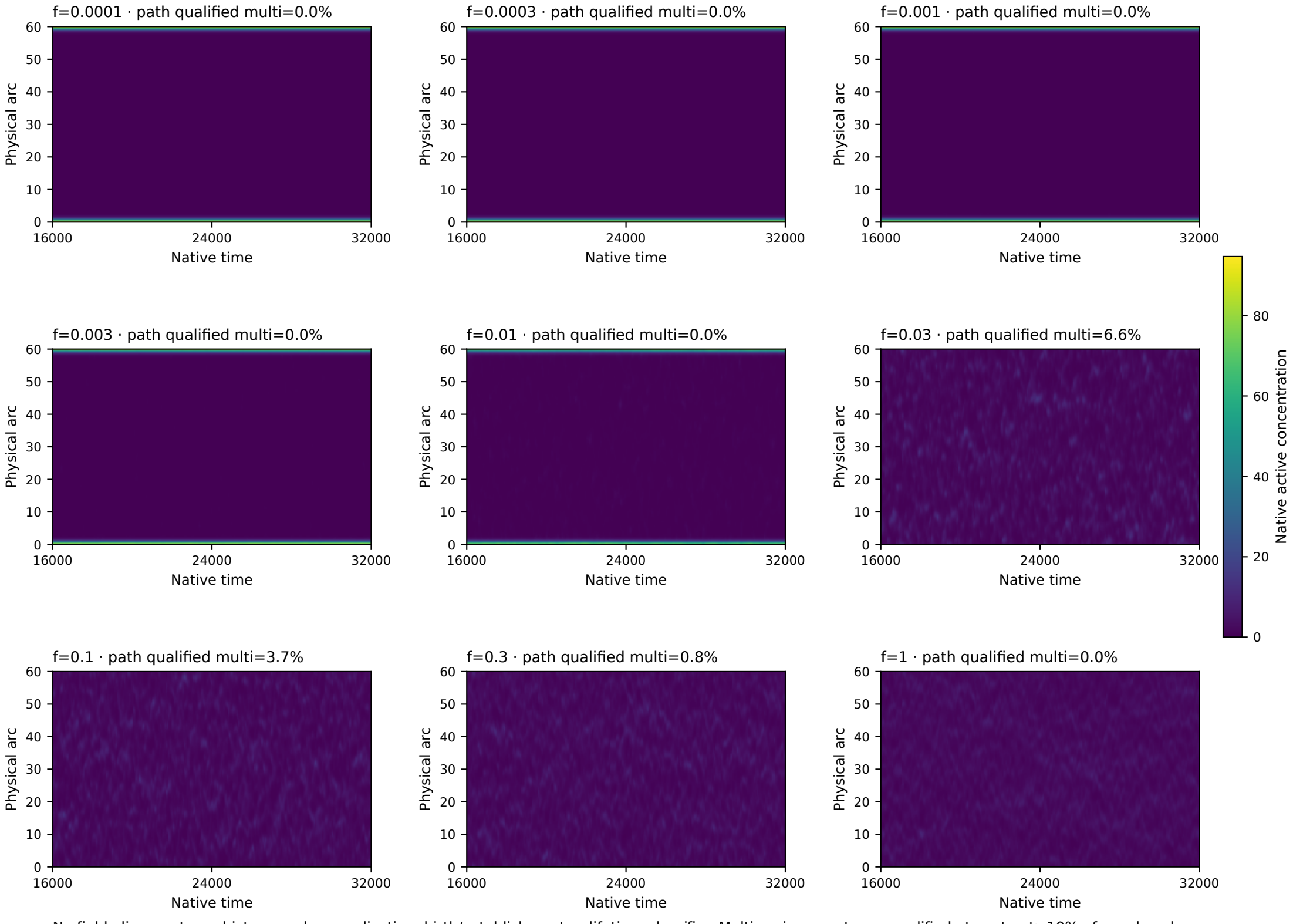
No field alignment, per-history peak normalization, birth/establishment or lifetime classifier. Multi-region counts are qualified at contrast≥10% of own basal.

The first half is discarded as an observed-window choice, without assuming stationarity. Model clocks and input laws remain native and uncalibrated.

Goryachev, C=60: all nine input strengths

Second-half descriptor medoid per condition, with smallest-seed tie breaking. Common zero-based native activity scale across this page.

f=0.001/0.003/0.01/0.03/0.1: n=12. Other four levels: n=2 (both tie). These examples do not estimate frequency.



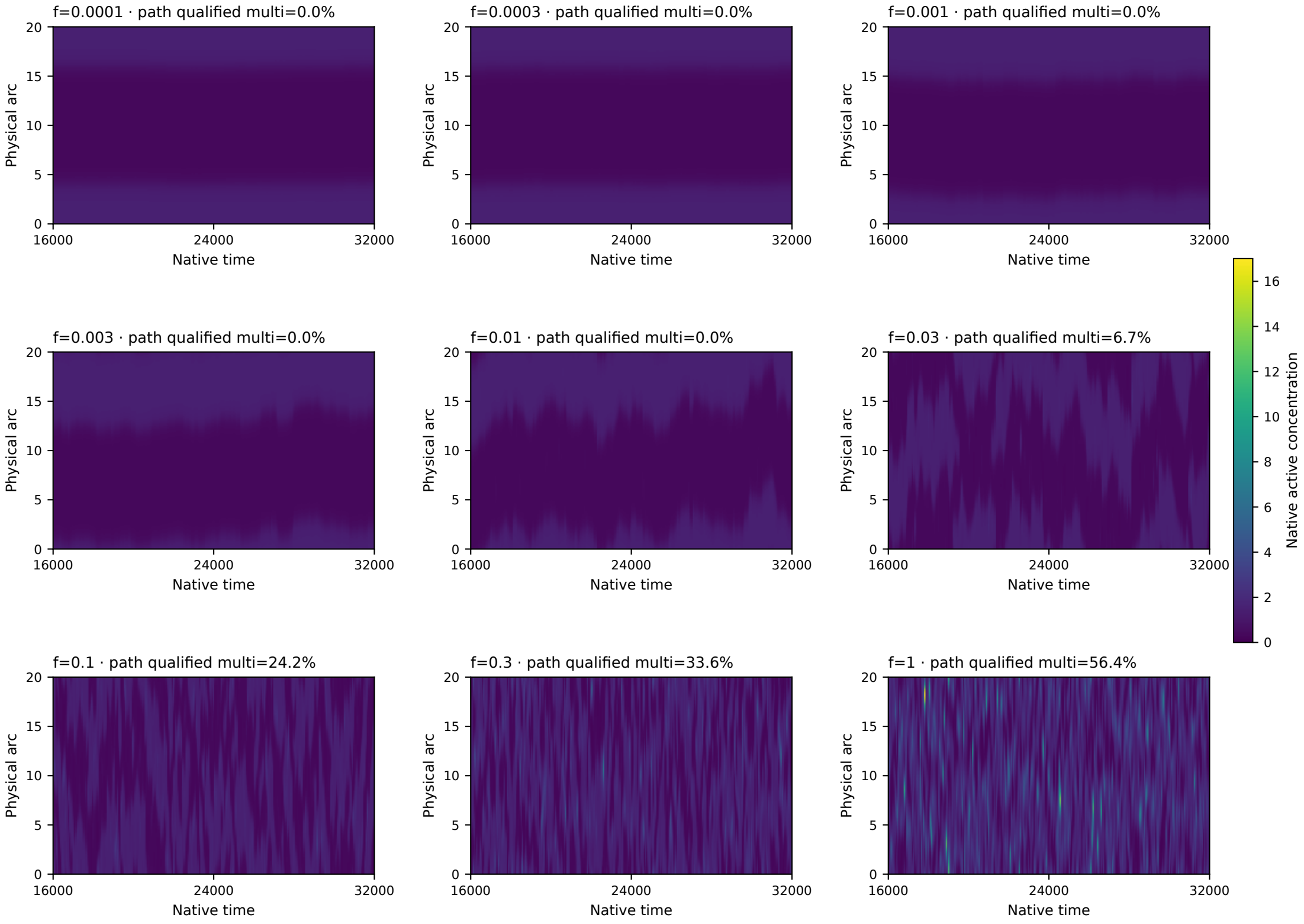
No field alignment, per-history peak normalization, birth/establishment or lifetime classifier. Multi-region counts are qualified at contrast≥10% of own basal.

The first half is discarded as an observed-window choice, without assuming stationarity. Model clocks and input laws remain native and uncalibrated.

Holmes 3, C=20: all nine input strengths

Second-half descriptor medoid per condition, with smallest-seed tie breaking. Common zero-based native activity scale across this page.

f=0.001/0.003/0.01/0.03/0.1: n=12. Other four levels: n=2 (both tie). These examples do not estimate frequency.



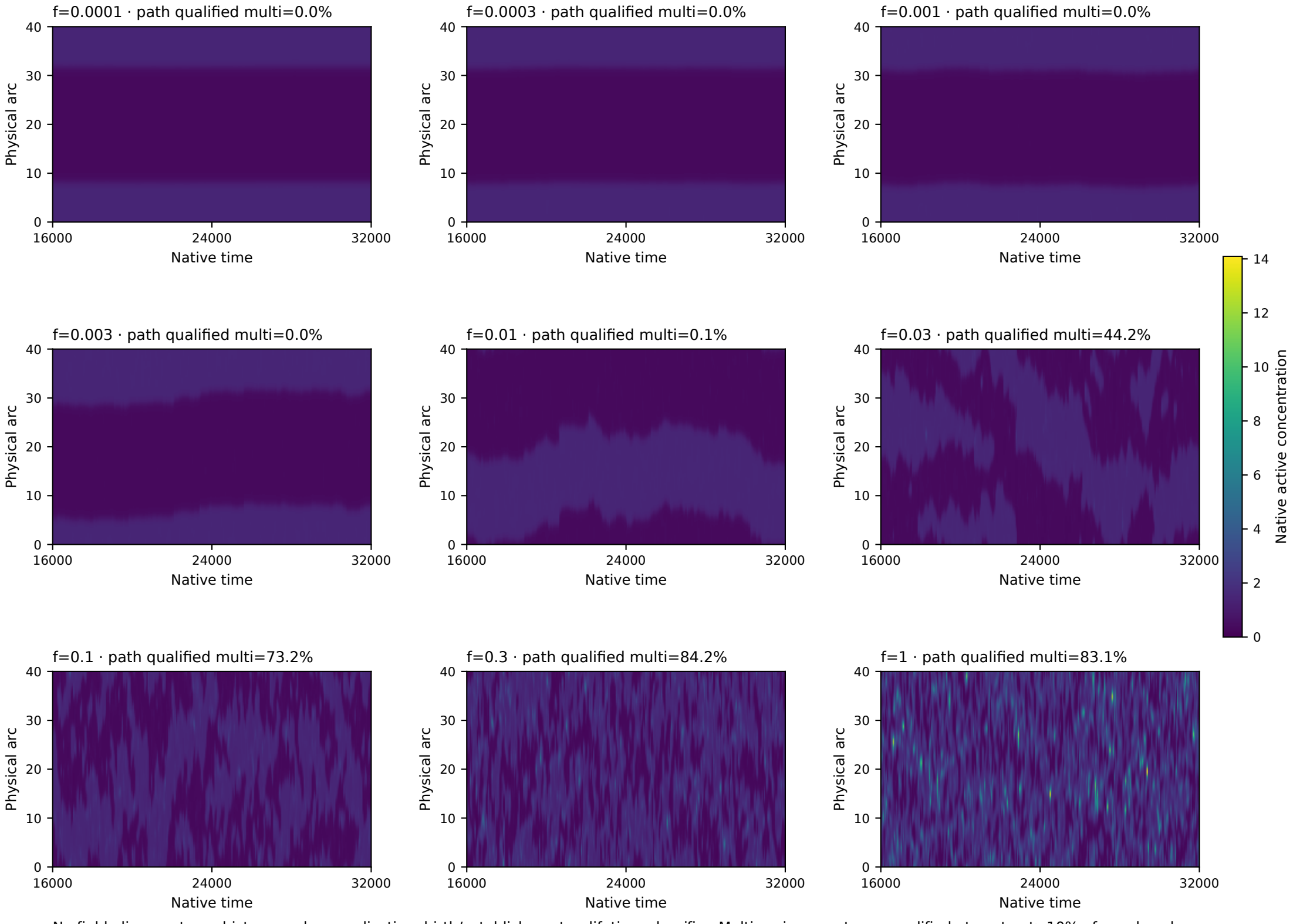
No field alignment, per-history peak normalization, birth/establishment or lifetime classifier. Multi-region counts are qualified at contrast≥10% of own basal.

The first half is discarded as an observed-window choice, without assuming stationarity. Model clocks and input laws remain native and uncalibrated.

Holmes 3, C=40: all nine input strengths

Second-half descriptor medoid per condition, with smallest-seed tie breaking. Common zero-based native activity scale across this page.

f=0.001/0.003/0.01/0.03/0.1: n=12. Other four levels: n=2 (both tie). These examples do not estimate frequency.



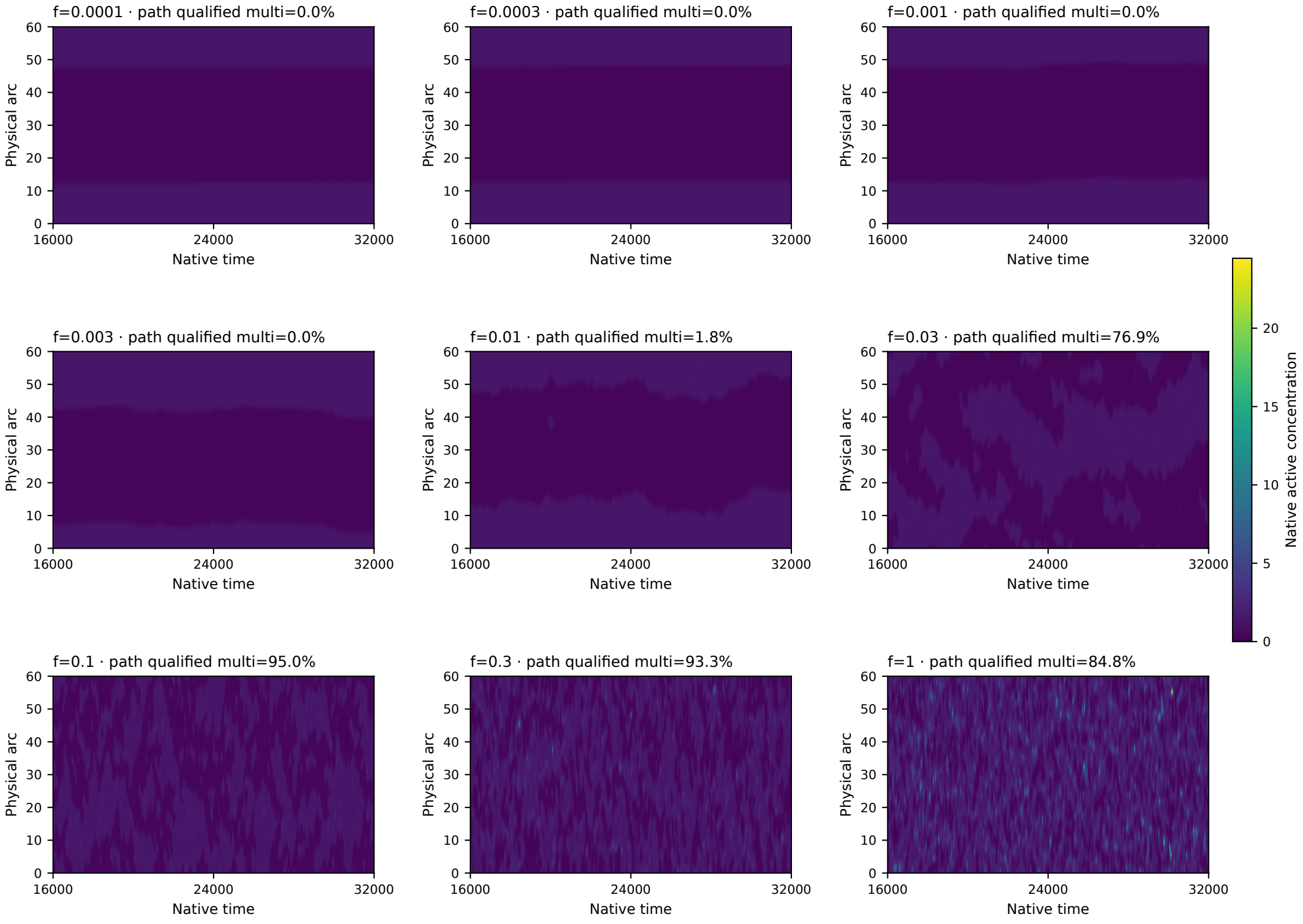
No field alignment, per-history peak normalization, birth/establishment or lifetime classifier. Multi-region counts are qualified at contrast≥10% of own basal.

The first half is discarded as an observed-window choice, without assuming stationarity. Model clocks and input laws remain native and uncalibrated.

Holmes 3, C=60: all nine input strengths

Second-half descriptor medoid per condition, with smallest-seed tie breaking. Common zero-based native activity scale across this page.

f=0.001/0.003/0.01/0.03/0.1: n=12. Other four levels: n=2 (both tie). These examples do not estimate frequency.



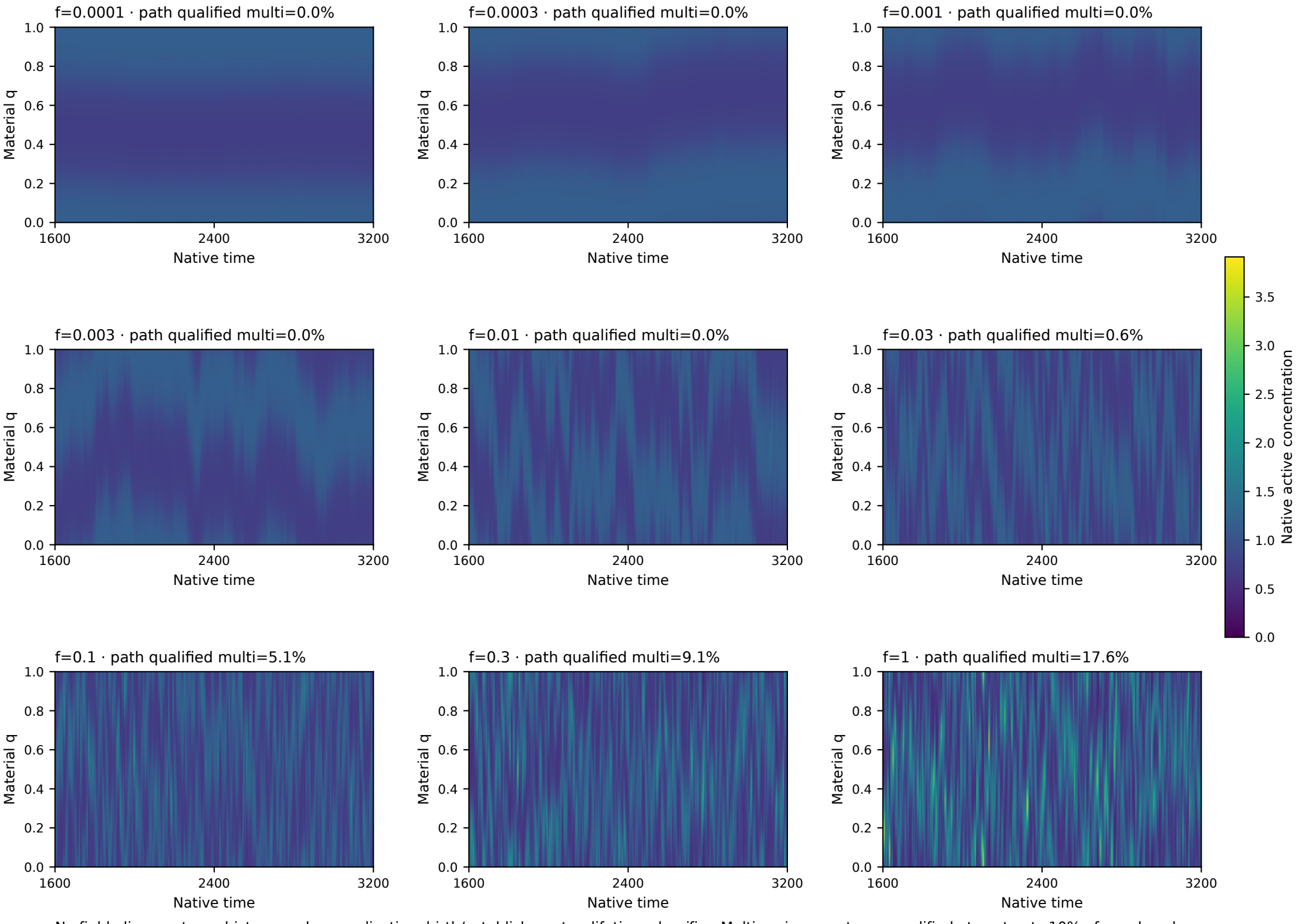
No field alignment, per-history peak normalization, birth/establishment or lifetime classifier. Multi-region counts are qualified at contrast≥10% of own basal.

The first half is discarded as an observed-window choice, without assuming stationarity. Model clocks and input laws remain native and uncalibrated.

Spring ring, C_0=1: all nine input strengths

Second-half descriptor medoid per condition, with smallest-seed tie breaking. Common zero-based native activity scale across this page.

f=0.001/0.003/0.01/0.03/0.1: n=12. Other four levels: n=2 (both tie). These examples do not estimate frequency.



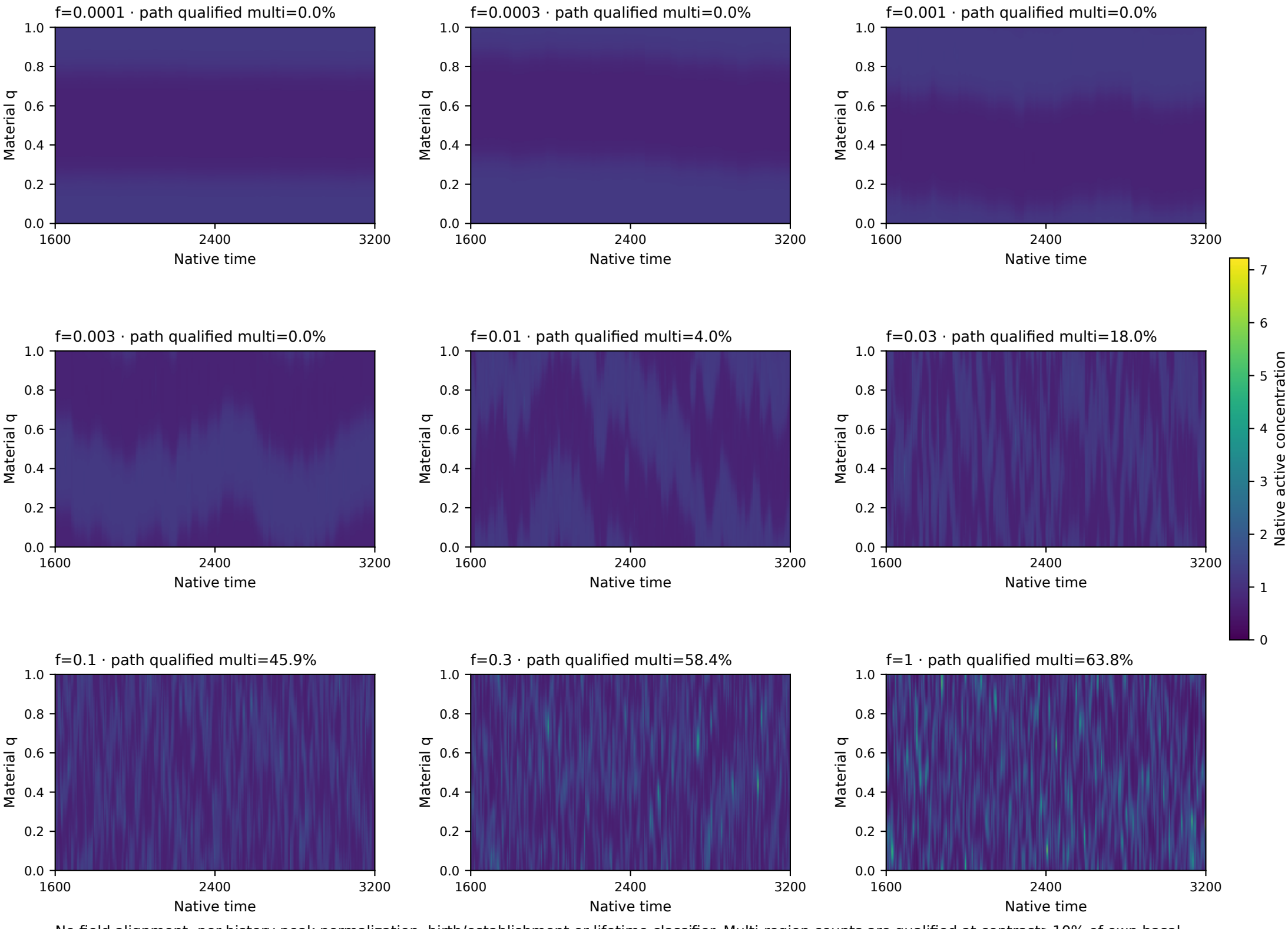
No field alignment, per-history peak normalization, birth/establishment or lifetime classifier. Multi-region counts are qualified at contrast≥10% of own basal.

The first half is discarded as an observed-window choice, without assuming stationarity. Model clocks and input laws remain native and uncalibrated.

Spring ring, C_0=2: all nine input strengths

Second-half descriptor medoid per condition, with smallest-seed tie breaking. Common zero-based native activity scale across this page.

f=0.001/0.003/0.01/0.03/0.1: n=12. Other four levels: n=2 (both tie). These examples do not estimate frequency.



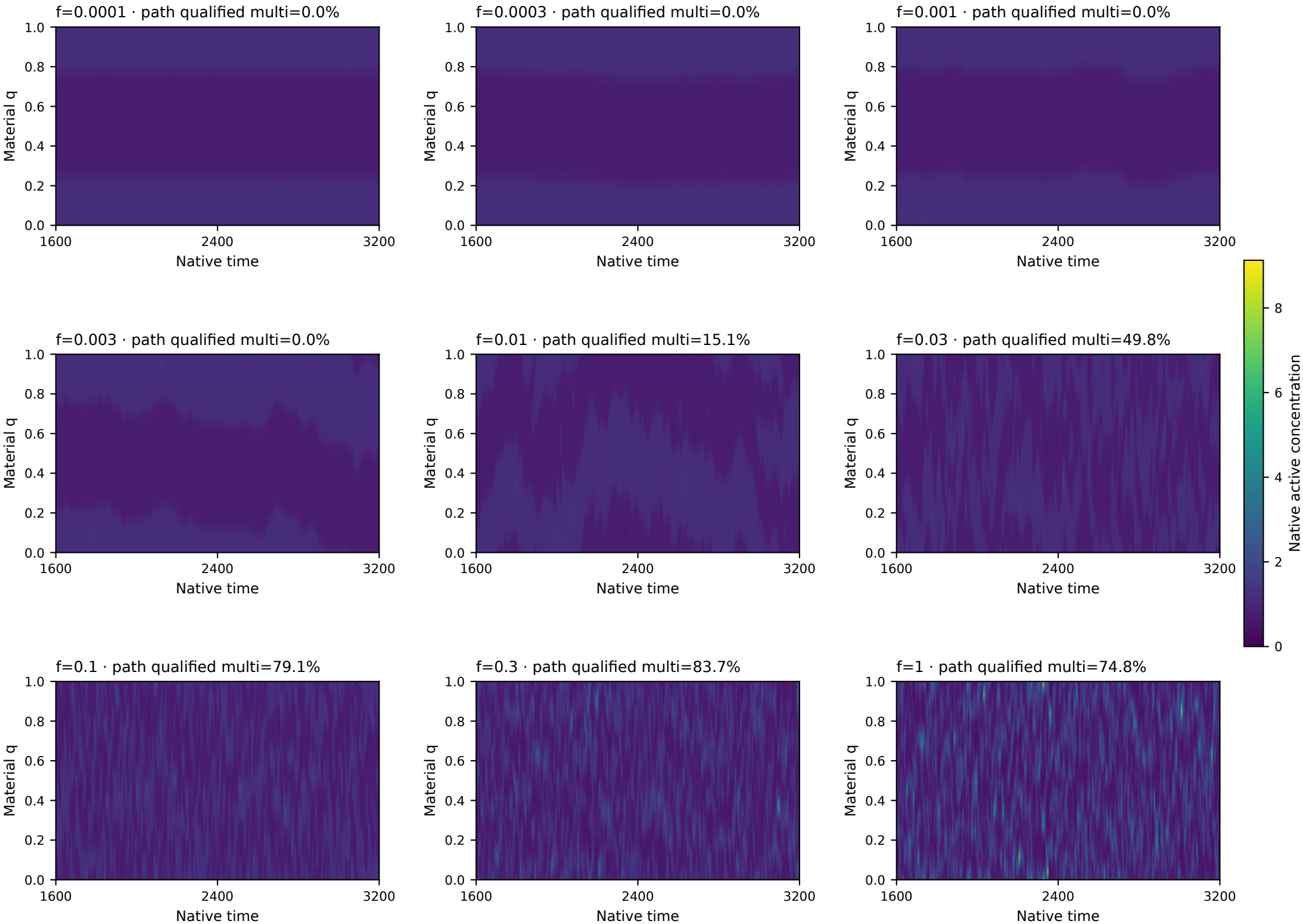
No field alignment, per-history peak normalization, birth/establishment or lifetime classifier. Multi-region counts are qualified at contrast≥10% of own basal.

The first half is discarded as an observed-window choice, without assuming stationarity. Model clocks and input laws remain native and uncalibrated.

Spring ring, C₀=3: all nine input strengths

Second-half descriptor medoid per condition, with smallest-seed tie breaking. Common zero-based native activity scale across this page.

f=0.001/0.003/0.01/0.03/0.1: n=12. Other four levels: n=2 (both tie). These examples do not estimate frequency.



No field alignment, per-history peak normalization, birth/establishment or lifetime classifier. Multi-region counts are qualified at contrast≥10% of own basal.

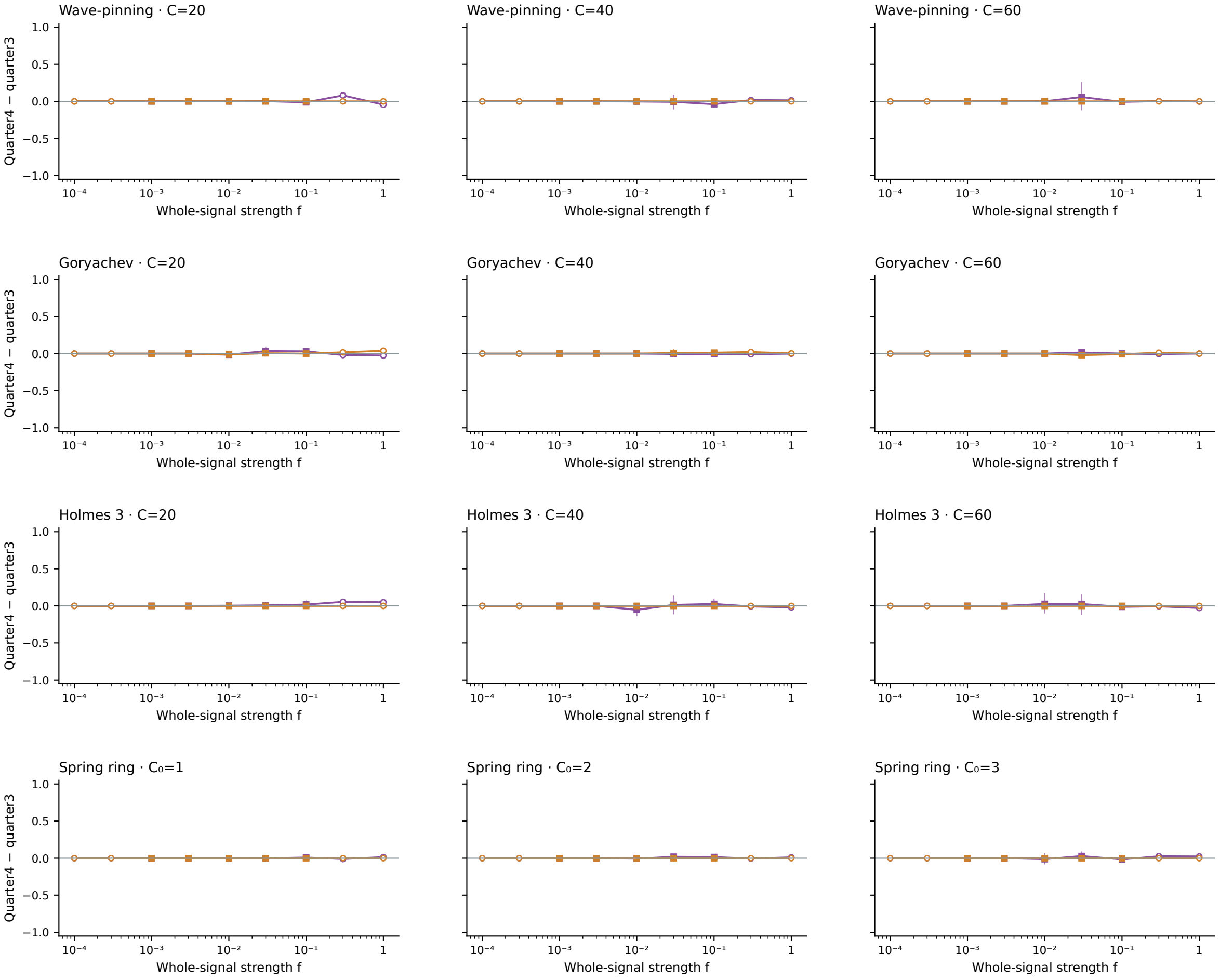
The first half is discarded as an observed-window choice, without assuming stationarity. Model clocks and input laws remain native and uncalibrated.

Late-quarter organization sensitivity

Quarter4 minus quarter3: chemicals24000–32000 minus16000–24000; Spring2400–3200 minus1600–2400.

Qualified multiple-region occupancy Low-contrast occupancy

Filled squares:n=12; open circles:n=2. Bars:95% whole-seed intervals where n≥3 and direction support≥95%; unavailable directions stay absent.



These are finite-window differences, not a stationarity pass/fail test. Intervals resample independent whole seeds with quarter pairing preserved.

Quarter direction uses shorter native lag2000/200; main lag8000/800 leaves only one pair per seed in a quarter and cannot meet the reporting support gate.

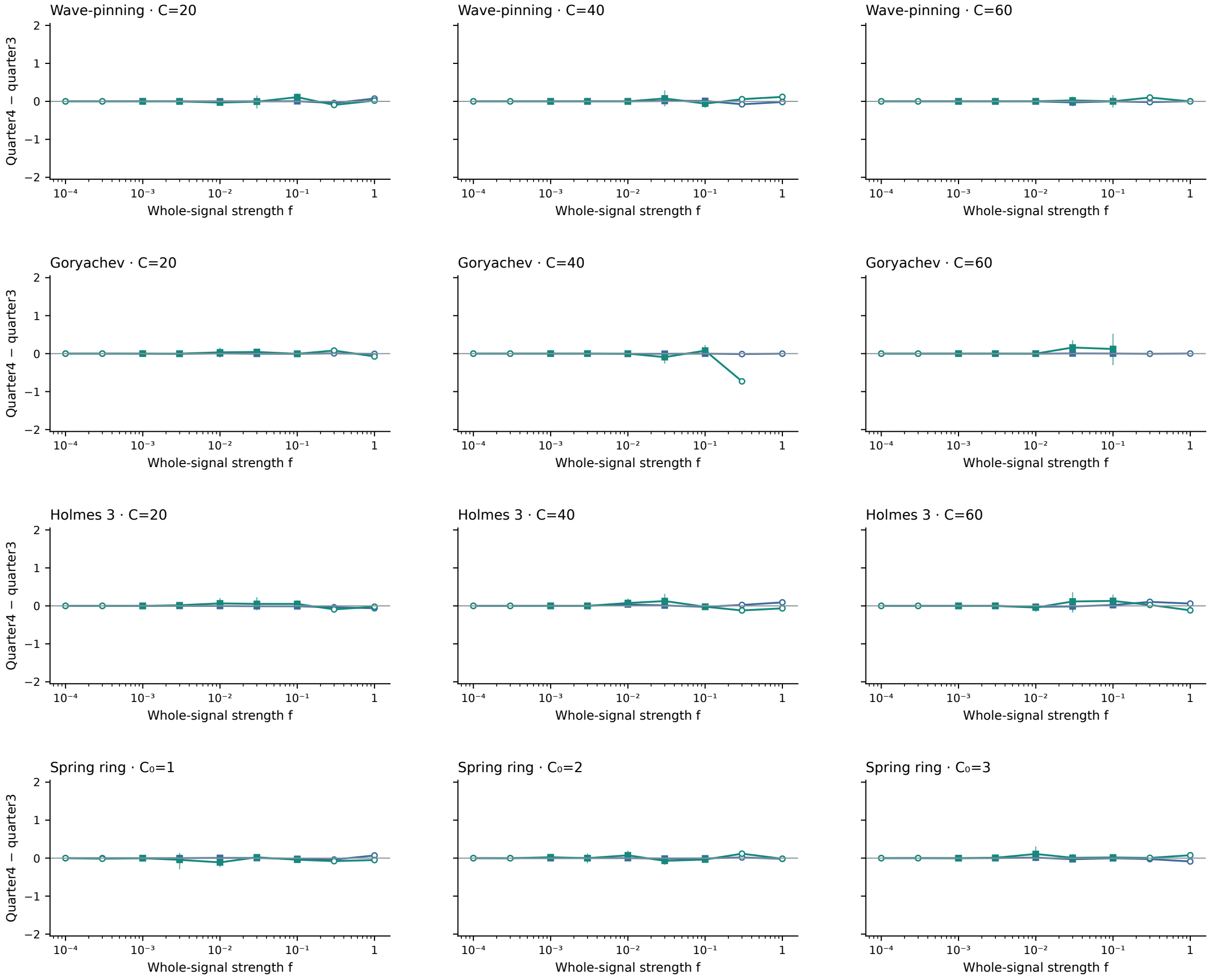
Use the exported quarter-specific pair coverage and seed counts. No continuous region lifetime or migration-velocity inference is made.

Late-quarter magnitude and direction sensitivity

Quarter4 minus quarter3: chemicals24000–32000 minus16000–24000; Spring2400–3200 minus1600–2400.

Mean R Direction cosine at native lag2000/200

Filled squares:n=12; open circles:n=2. Bars:95% whole-seed intervals where n≥3 and direction support≥95%; unavailable directions stay absent.



These are finite-window differences, not a stationarity pass/fail test. Intervals resample independent whole seeds with quarter pairing preserved.

Quarter direction uses shorter native lag2000/200; main lag8000/800 leaves only one pair per seed in a quarter and cannot meet the reporting support gate.

Use the exported quarter-specific pair coverage and seed counts. No continuous region lifetime or migration-velocity inference is made.